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Understanding variations in reported epidemiology of major lower extremity amputation in the UK: A systematic review

Journal:	*BMJ Open*
Manuscript ID	bmjopen-2021-053599
Article Type:	Original research
Date Submitted by the Author:	18-May-2021
Complete List of Authors:	Meffen, Anna; University of Leicester, Health Sciences Houghton, John; University of Leicester Department of Cardiovascular Sciences Nickinson, Andrew; University of Leicester Department of Cardiovascular Sciences Pepper, Coral; University Hospitals of Leicester NHS Trust, Library and Information Services Sayers, Rob; University of Leicester Department of Cardiovascular Sciences Gray, Laura; University of Leicester Department of Health Sciences
Keywords:	Vascular surgery < SURGERY, General diabetes < DIABETES & ENDOCRINOLOGY, EPIDEMIOLOGY

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Understanding variations in reported epidemiology of major lower extremity amputation in the UK: A systematic review

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Key words: Lower extremity amputation, systematic review, peripheral artery disease, diabetes, routinely collected health data

Word count: 3748

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Abstract

Objective

Estimate the prevalence/incidence/number of major lower extremity amputations (MLEA) in the UK, identify sources of routinely collected electronic health data used; assess time trends and regional variation and identify reasons for variation in reported incidence/prevalence of MLEA.

Methods

Medline, EMBASE, EMcare, CINAHL, The Cochrane Library, AMED, Scopus and grey literature sources were searched between 01/01/2009 and 31/12/2019. Reports which provided population based statistics, used routinely collected electronic health data, were written in English and gave a measure of MLEA in adults in the general population or those with diabetes in the UK or constituent countries were included.

Results

Twenty-four reports were included.

Reported MLEA statistics varied widely. Incidence proportion for the general population ranged from 8.8 to 51.1 per 100,000 and from 70 to 291 per 100,000 for the population with diabetes. Evidence for trends in incidence of MLEA over time was mixed, but there was no evidence of increasing incidence. Reports consistently found regional variation in England with incidence higher in the North. No studies reported prevalence.

There were considerable differences in study populations and methodology between reports including differences in database use, MLEA definition, calculation methods and multiple procedure inclusion which, together with identified inaccuracies, may account for the variation in incidence; data synthesis was not possible.

Conclusions

UK incidence and trends in MLEA remain unclear; estimates vary widely due to differences in methodology and inaccurate calculations and terminology. Reasons for regional variation also remain unexplained and prevalence un-investigated potentially due to the restrictive nature of secondary care event base data. International consensus on the definition of MLEA and medical code list is needed. Future research should recommend standards for the reporting of such outcomes and investigate the potential to use primary care data in MLEA epidemiology.

Systematic review registration

PROSPERO CRD42020165592

Meffen A, Pepper CJ, Sayers RD, et al. Epidemiology of major lower limb amputation using routinely collected electronic health data in the UK: a systematic review protocol. BMJ Open 2020;10:e037053. doi: 10.1136/bmjopen-2020-037053

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Article Summary

Strengths and limitations of this study

- This review reported the incidence proportion of major lower limb amputation in the UK providing discussion on the reasons for variation in reported statistic and use of routinely collected electronic health data.
- Both the systematic review and the protocol adhere to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines.
- A comprehensive search strategy was employed in both peer-reviewed and grey literature searches.
- Included article methods and populations were not directly comparable and so it was not possible to synthesise data.
- This study only investigated data sources in the UK thus results may have limited use outside of this population.

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Introduction

Peripheral artery disease (PAD) is the leading cause of all major lower extremity amputations (MLEA) with diabetes, smoking, increasing age, hypertension increasing risk.[1, 2] With the global rise in diabetes prevalence and an aging population, incidence of MLEA has become a key indicator of health service performance and used for international comparisons.[3-6] Monitoring incidence globally, nationally and regionally is essential to determine the success of implemented prevention services.

In the UK the incidence of MLEA is debated with variations reported.[7-9] Significant differences in regional estimates have also been reported.[8-10] The reasons for these disparities are unclear, although a recent review suggested differences in calculation methods may provide some explanation.[11]

Electronic health data are widely used for epidemiological research. Variation exists in the way individual databases collect, administer and report data and therefore differences between databases may also explain variation in reported MLEA statistics. There are many additional challenges to using such data, one notable, but improving issue is coding errors.[12-15] Including poor quality data with errors in incidence studies may affect the apparent trends in disease incidence.[15] Using different sets of MLEA codes and/or data may also account for some of the variation observed.

This systematic review aimed to ascertain the current UK incidence of MLEA; establish trends over time; report regional variation and review which routinely collected electronic health databases were used. The review also aimed to explain the reasons for variation in the reporting of MLEA epidemiology.

Methods

This review followed the Preferred Reporting Items for Systematic reviews and Meta-analyses (PRISMA) guidelines for systematic reviews.[16-18] A protocol was, published and registered in the PROSERO database (CRD42020165592).[19]

Search strategy

Electronic databases Medline, EMBASE, EMcare, CINAHL, The Cochrane Library, AMED and Scopus were searched using keywords and thesaurus terms (indexing systems) and using validated UK geographic filters where available.[20, 21] The search was developed in Medline and then adapted for other databases, Supplementary File 1.

Grey literature (non-peer-reviewed articles) was identified using www.opengrey.eu, openDoar.org, openAire.eu, base-search.net, eTHOS, OpenDisserations.org, www.gov.co.uk, www.parliament.uk, www.vvappg.com, www.digital.nhs.uk, www.QResearch.org and www.CPRD.com using the key word “amputation”/”amput*”.

Inclusion criteria

Included reports presented: population based statistics; used routinely collected electronic health data; were written in English; and gave a measure (prevalence/incidence/number) of MLEA in adults in the general population or in persons with diabetes in England, Scotland, Wales, Northern Ireland or the UK.

Studies published from 2009 onwards were included to ensure results were reflective of recent trends. This review aimed to focus on PAD and diabetes related MLEA as these may be preventable and only a small percentage of MLEA occur due to other aetiologies. [22] Analysis and any resulting

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care service improvements would therefore need to be based on vascular, diabetes and podiatric care. Studies of adults that included MLEA due to cancer and trauma were included in the study but this information is explicitly stated where available. Studies of children were excluded.

No agreed definition of 'major' lower extremity amputation exists. Therefore, we did not define MLEA and all studies that reported MLEA were included with individual study definition of MLEA extracted.

For studies that reported for both minor and major lower extremity amputation, only data for major lower extremity amputation were included.

Screening and extraction

Identified reports were imported into EndNote X9 and duplicates removed. Initially, titles and abstracts were screened for inclusion by two reviewers independently (AM and JH). Of the selected reports, two reviewers independently screened the full texts with exclusion reasons recorded (AM plus JH or AN).

Data extraction was performed by two reviewers independently using a pre-tested tabulated form. Disagreements at any stage were resolved by discussion. Where data items were not reported, data were requested from authors.

Data extracted:

- Author, title, publication date
- Data source
- Epidemiological measure of MLEA - prevalence/incidence/number including any CI, SE and variance presented
- Definition of MLEA
- Population studied (country, diabetic/non diabetic, age limits, date limits, comorbidities or reasons for amputation excluded e.g. cancer/trauma)
- Regional results
- Standardisation methods[23]
- MLEA code lists
- Reporting guidelines used

Quality assessment

The Joanna Briggs Institute (JBI) Critical Appraisal Instruments, either for Studies Reporting Prevalence Data or for Cohort Studies, were used to assess reporting quality.[24-26] Two reviewers (AM plus JH or AN) independently appraised included reports and resolved any disagreement. Additionally, use of appropriate reporting guidelines such as Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) and Reporting of Studies Conducted Using Observational Routinely-collected Data (RECORD) will be noted.

Data synthesis

Data were converted, where possible, to provide consistent statistics across reports (e.g. percentages converted to per 100,000 population). Where multiple incidence values for an individual study were reported, the most recent figure was extracted.

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Extracted data were corrected where errors in terminology or calculations were identified using the following definitions (where populations refers to the population specified by the individual study).[27, 28]

$$\text{Incidence proportion} = \frac{\text{Number of new MLEA in specified time period}}{\text{Total population at risk at the start of time period}}$$

$$\text{Incidence rate} = \frac{\text{Number of new MLEA during specified time period}}{\text{Total population time at risk during specified time period}}$$

$$\text{Point prevalence} = \frac{\text{Number of new and pre existing MLEA at specified time point}}{\text{Population total at specified time point}}$$

$$\text{Period prevalence} = \frac{\text{Number of new and pre existing MLEA over specified time period}}{\text{Average or mid interval population for specified time period}}$$

Meta-analysis of incidence/prevalence and meta-regression of time trends using a random effects model (which allows for differences in estimates due to heterogeneity) were planned with different measures pooled independently where not convertible. Where possible, this was to be repeated for subgroup analysis. Alternatively, if extracted report methods were not similar enough to justify pooling, then data were to be narratively synthesised.

Patient and public involvement

Patients or the public were not involved in the design, or conduct, or reporting, or dissemination plans of our research.

Ethics Approval

Ethics approval was not applicable for this study as it is a systematic review of published data.

Results

Search results

Titles and abstracts of 3306 reports were screened (Figure 1). Twenty-four reports were included, half of which were identified through grey literature searches.[4, 7-10, 22, 29-46] Included grey literature consisted of reports from national healthcare quality improvement programmes, government healthcare department agencies, and government associated health care audits.[22, 31-35, 39, 40, 43-46] Corresponding authors for nine reports were contacted for additional data, only one author responded.[41, 42]

Quality assessment

Overall reporting was poor to moderate with errors in key aspects such as statistic definition and calculations and poor reporting including the lack of descriptive statistics of their study population and denominator population descriptions. (Supplementary File 2). Seven reports passed only 33% of the JBI prevalence study criteria whilst 16 passed between 33% and 66%. The one report assessed by the JBI cohort study guidelines passed on roughly 70% of the criteria. Use of appropriate reporting

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guidelines such as STROBE and RECORD guidelines, where available, was unclear.[47, 48] Only one report stated the use of STROBE guidelines.[36] RECORD guidelines were published in 2015, none of the included reports published after this date stated the use of these guidelines.

Outcome measures

A variety of MLEA statistics were reported by included reports (Table 1). Where available, standardised outcome measures were presented. Fourteen (67%) reports provided a calculated statistic,[4, 7-10, 29, 31-36, 39-42] the remaining eight provided an absolute number.[22, 30, 37, 38, 43-46] Fourteen (58%) reported an incidence proportion.[7, 10, 29, 31-37, 40-42] One reported an incidence rate.[8] The statistic presented for one report was unclear.[39] None presented prevalence.

Due to heterogeneity between study designs and reporting and the inability to accordingly recalculate outcomes homogeneously owing to a lack of available data (discussed further in the results and discussion sections), data synthesis was not possible; results have been tabulated and narratively synthesised.

Incidence proportion for the general population ranged from 8.8 to 51.1 per 100,000 and from 70 to 291 per 100,000 for the population with diabetes.

Time trends

Time trends were investigated by five reports, (Figure 2).[4, 7, 36, 41, 42] Differences in populations and methods mean that a direct comparison of trends could not be made.

Although results are mixed, with some reports finding non-significant decreasing trends, there is some evidence of a decrease in MLEA in England over time for both the general population and for those with diabetes.[41]

The report for Scotland reported a statistically significant ($p<0.001$) decrease in the incidence of MLEA for those with diabetes. [36]

For the studies that reported on a frequent basis, i.e. the National Diabetes Audit (NDA) and National Vascular Registry (NVR) reports, trends in incidence of MLEA over time could not be assessed as the NDA reports differ in frequency, study length and calculation methods and the NVR reports have overlapping study periods.

Regional trends

Four reports provide regional data.[8-10, 39] Public Health England (PHE) Fingertips provided regional estimates for those with diabetes by Clinical Commissioning Group.[39] For the period of 2018/19 the regions with the highest and lowest incidence per 100,000 were Bradford City (In Northern England) (205) and Bromley (In Greater London) (22). Holman et al studied MLEA incidence rates in those with and without diabetes by commissioning area (with different boundaries to those in the PHE fingertips report) but did not provide any extractable area level data.[8] They observed a 16-fold variation in incidence rate between areas for the population without diabetes and a 10-fold variation in incidence rate between areas for those with diabetes.

The two reports providing comparable data showed regional variation was statistically significant ($p<0.001$).[9, 10] Incidence was highest in the North and lowest in the South of England (Figure 3).

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Sources of outcome measure variation

Database differences

All cases of MLEA were ascertained using secondary care data with healthcare database use differing by country. Details of these databases are summarised in Table 2. Importantly, case ascertainment of MLEA in the NVR compared to HES is low in the NVR compared to HES (48-74% between 2014-2018).[49]

Population differences

Study populations varied widely between reports with differences in country, exclusion criteria and MLEA definition and study period (Table 1).

Reports varied in how multiple procedures per person were dealt with. Seven reports made no reference to this.[4, 7, 9, 10, 29, 30, 37] Four used the highest level of amputation.[8, 36, 41, 42] Reports varied in how contralateral amputation during the same admission were counted. One report included the first procedure.[38] The NDA reports implied each person was only included once no matter how many operations were performed although they do not specify which (first or highest) was included.[31-35, 40] The NVR 2015 report looked at unilateral amputations, no other information is given.[43] The NVR 2016-2019 reports excluded bilateral amputations and those associated with a bypass.[22, 44-46] Repeated amputations were not mentioned. None of the reports included revision specific procedures.

Study periods ranged from one to 17 years between the years 1996-2018 (Figure 4). No two reports with comparable populations covered the same period. Reports providing whole study estimates of incidence over a number of years generally reported higher incidence than those that reported for yearly incidence, as would be expected. This can be seen in the NDA reports in Table 1.

Coding differences

Sixteen reports provided code lists (Supplementary File 3, Table 1), all of which ascertained cases in HES or HES linked CPRD and used OPCS-4 codes apart from one which used ICD-10-PCS codes.[4, 7-10, 29, 31-35, 38-42] There were inconsistencies between the definition of MLEA used and the code lists and one report included an ICD-10-PCS code which does not exist whilst not including an expected code.[4]

For those that reported for the population with diabetes only, differences were found in the case ascertainment methods of those with diabetes (Supplementary file 3, Table 2).

Outcome measure calculation differences

Statistics were inconsistently reported. Some reports used ‘prevalence rate’ or ‘period prevalence’ when actually stating an incidence proportion.[7, 10, 29, 31-35] Five repots used the terms ‘incidence’ and ‘amputation rate’ when reporting incidence proportion.[4, 9, 36, 41, 42] One report did not clearly define the statistic used[39].

Six reports used standardisation methods and only presented standardised results, these reports did not present enough information to be able to calculate crude (unstandardised) outcome measures.[7, 10, 29, 31, 39, 42] These reports varied in method and variables used to perform standardisation. Three reports standardised by age only;[7, 10, 29] two by age and sex[31, 42] and one by age and ethnicity.[39] Only one included report provided both standardised and crude outcome meaasures.[31] In this case, the effect of age and sex standardisation to reflect the national population was a greater than threefold increase in incidence proportion (crude: 87/100,000, standardised: 290/100,000).

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Denominators were poorly reported with few reports providing the inclusion/exclusion criteria. One report stated the population country as the UK, however, the cases were ascertained using HES which only covers England and the denominator population was taken from Office for National Statistics population statistics for England and Wales.[4] Another report misreported incidence proportion as number of MLEAs per 100,000 population, the calculated figures were actually reported per 10,000 population.[9] As the population data were available to do so, correct recalculated statistics were presented in this review.

Discussion

Principle findings

Reported values of incidence proportion for the general population and population with diabetes varied with approximately a six and fourfold difference respectively.

Despite mixed reports of a decreasing trend in MLEA in England, it is encouraging that there is no evidence for an increasing trend considering the increase in prevalence of type 2 diabetes.[49] Some reports found evidence of an increase in the incidence of minor lower limb amputations which may partly explain this.[7] Although reports agree that incidence is higher in the North of England compared with the South, this difference remains unexplained by demographic (including deprivation) or risk factors.[8-10] One suggested explanation of regional variation in incidence is the difference in care provided, for example, areas where multidisciplinary preventative care services are available may have reduced regional incidence compared to other areas.[8, 50]

Two main sources of variation were found in reported MLEA incidence between reports. Firstly, there were considerable differences in the methodology used across reports including differences in country, study period, case and population (denominator) exclusions, MLEA definition (and coding), data source, standardisation, and statistic. For example, with Moxey et al 2010 and Ahmad et al 2014a, the slightly different study period and differences in age exclusions, standardisation methods, denominator populations and MLEA codes contributed to a twofold difference in reported incidence proportion. Secondly, there were reporting quality issues including inconsistent terminology and poorly reported denominators. It was not possible to further determine reasons for or the effect of the variation as reports did not include enough information to be able to clarify this. Adherence to reporting guidelines would have aided in clarifying variation sources and effects, however, adherence was low within included reports. Specifically, reports often failed to include; both crude (unstandardised and/or adjusted) and standardised whole study population outcome estimates; raw numerator and denominator values; both case and population (denominator) inclusion/exclusion criteria; referencing and detailing standardisation methods and data source. In addition to including details listed in RECORD guidelines, it would be of particular use in this case for reports to provide a breakdown of how many times an individual is included in the analysis to aid in interpretation and understanding of multiple procedure incidence.

Database use has an effect on MLEA epidemiology reporting. All studies ascertained cases using secondary care databases such as HES and the NVR which contain event based data and so only events within the study period can be analysed. Historical MLEA performed outside of the study period cannot be gained from these examples of event based databases. This explains the finding that prevalence of MLEA has not been investigated. It is also not possible to obtain population (/control) data for analysis from these databases, this would limit the ability to explain the regional variation found in some reports which is currently unexplained by the available population demographic (including deprivation) and risk factor data sourced externally to secondary care

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databases. Another option would be to use primary care data, which contains the whole medical history of each registered patient and so prevalence could potentially be calculated and more detailed control data obtained for use in analysis. However, current primary care databases in the UK do not contain data on the whole population and are therefore only a population sample.[51-53] Another example of the effect database choice has on MLEA reporting is seen when considering the low case ascertainment of the NVR compared to HES. Although none of the reports using NVR calculated incidence proportion, using this database may underestimate MLEA incidence and could infer a potentially false increasing trends as case ascertainment improves. The lack of established linkages to the NVR and limited amount of patient data compared to HES may mean it could also be of limited use in further analysis.

Some reports ascertained cases in secondary care data and linked this to patient data in primary care records.[31-35, 38, 40] This obtains additional case patient information but does not aid in calculating prevalence or gaining population/control data.

Comparison with other studies

A systematic review by Davies et al found many similar issues regarding denominator populations, MLEA coding and definition and general method differences.[11] They state that "prevalence of MLEA ranges from 0.7 to 332.4 per 100,000 in the diabetic population and 3.0 to 76.1 per 100,000 in the general population" a 480 and 25 fold difference respectively. This review differed in inclusion criteria by excluding reports that included 'minor' amputations; including only studies that use routinely collected electronic health data; excluding studies only reporting regional data and by excluding subgroup data from the main review outcome. These review method differences resulted in a considerable reduction in the magnitude of variation. However, the magnitude is still large and this review found additional sources of variation within the inclusion/exclusion criteria of reports; in multiple procedure counting; in standardisation techniques and data source. This review also included grey literature, investigated regional variation and discussed the effect of database use on MLEA epidemiology reporting.

As lower extremity amputation is a key health care quality indicator globally, national incidence of lower extremity amputation is an increasing topic of research in many countries.[54-57] Systematic reviews and reports on the national and international incidence of lower extremity amputation have found similar problems with report comparisons as found in this review, i.e. population differences, lower extremity amputation definitions, database characteristics, and calculations methods.[4, 5, 54-57]

Implications for clinical practice and further research

Being unable to accurately estimate the UK incidence of MLEA has important implications for clinicians and policy makers who will be unable to gauge trends and allocate resources appropriately. This will impact on national service configuration of vascular, diabetes and rehabilitation services. Additionally, unexplained regional differences in incidence mean that any resources allocated to specific areas may not be directly targeted at the cause. It also leads to inequalities in healthcare access and outcomes; inefficiency in resource allocation and uncertainty of the effect of preventative measures.

As MLEA definitions varied and were often not consistent with medical code lists, to aid in comparisons of future research, an internationally agreed definition of MLEA, alongside a corresponding medical coding list, should be agreed upon. Additionally, the appropriate reporting guidelines (STROBE, and more specifically, RECORD) should be strictly adhered to. Reports (particularly peer-reviewed) will, by nature, differ in research question, but by adhering to reporting

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guidelines, the data should be able to be unpicked so that sources of variation and their effect are more easily identified.

The use of primary care databases for ascertaining case and population data should be explored further to establish whether it is possible to estimate MLEA prevalence and explain the reasons for regional variation.

Strengths and limitations

The main strength of this review is the extensive search strategy including a comprehensive grey literature search. Although not peer-reviewed, the included grey literature reports were found to be of no lesser quality, on the whole, than peer-review research and so were analysed alongside peer-review reports. Other strengths include, the clarification of epidemiological calculations and population definitions, the adherence to the PRISMA reporting guidelines, and the peer reviewed protocol.[19]

This systematic review was limited to reports on MLEA; so reports that did not distinguish between major and minor amputations were excluded. Only reports published since January 2009 were included. Outcome measures from these reports prior to 2009 were included in the narrative synthesis so may not be reflective of current trends. Some reports published prior to 2009 with data available from 1996 may be available but excluded due to publication date. Although UK grey literature sources were searched thoroughly, international sources (e.g. World Health Organisation (WHO) websites etc.) were not and thus there may be additional international reports fitting the criteria that were missed.

Furthermore, this review was limited by the lack of reports with comparable methods, which led to the inability to synthesise data. It was also limited by data availability. Notably, the majority of the data is from England, some reports included cases from Wales and Northern Ireland, however, none focused on these countries specifically. Additionally, the event based nature of the databases used by the included reports meant that determining and defining multiple amputations per patient and therefore also prevalence was problematic. This limitation is not restricted to the UK and has been a finding of reports globally.[4, 5, 54-57]

Ranges in reported incidence were sensitive to the inclusion of particular reports, for example, removing the report that analysed only PAD related MLEA (excluding many who experienced MLEA related to diabetes who had no PAD associated diagnoses) would reduce this to a twofold difference in the general population.[4] Removing the report where the application of standardisation methods inflated incidence proportion by over three times the crude value would result in a reduction to a twofold fold difference in reported MLEA incidence proportion in the population with diabetes.[31]

Conclusions

The UK incidence and trends in MLEA remain unclear with estimates varying widely. There has been no research into the prevalence of MLEA. Encouragingly, there is no evidence of an increase in MLEA over time despite the national increase in diabetes diagnoses. Variation in reporting is due to differences in populations, methods and MLEA definitions, inaccurate calculations and terminology. There is evidence for regional variation in incidence, however, the reasons behind this are unexplained. This review highlights the need for improved MLEA epidemiology reporting with adherence to RECORD reporting guidelines. Importantly, an internationally agreed MLEA definition and code list is required. It also highlights the restrictive event based nature of secondary care data on analysis and the potential for further research of MLEA epidemiology using primary care data

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which could provide a prevalence estimate and the reasons behind the regional variation in MLEA in the UK.

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Acknowledgements

We would like to thank George Davies and the George Davies Charitable Trust for the charitable donation which help fund this work. This project was supported by the National Institute for Health Research (NIHR) Applied Research Collaboration East Midlands (ARC EM). The views expressed are those of the author(s) and not necessarily those of the NIHR or the Department of Health and Social Care.

Footnotes

Author Contributions: AM, RS, LG conceived and designed the study. CP, AM created and performed the search. AM, JH, AN screened, extracted data and undertook quality assessment. AM analysed the data, supervised by LG, RS. AM drafted and revised the manuscript which was reviewed by LG, all authors. Authors approved the final draft. The corresponding author attests that all listed authors meet authorship criteria and that no others meeting the criteria have been omitted.

Funding statement: AM, JH, AN are funded through the George Davies Charitable Trust (Registered Charity Number 1024818). RS is part funded through this Trust. The funders had no role in the design and conduct of the study, the collection, analysis, and interpretation of the data, or the preparation, review, or approval of the manuscript

Competing interests: All authors have completed the ICMJE uniform disclosure form at www.icmje.org/coi_disclosure.pdf and declare: RS reports grants from George Davies Charitable Trust during the conduct of the study and is Chair of the Vascular CRG for NHS England. All other authors declare no competing interests, or activities that could appear to have influenced the submitted work.

Data Sharing Statement: Statistical code, and dataset available from the corresponding author.

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Tables

Table 1: Table showing the method details and outcomes extracted from the included articles

Article	Country	Study period	**Case data source**	Exclusions	MLEA definition	Statistic	Standardisation	**Statistic value**
General population								
Ahmad, N. 2014a[29]	England	2003 – 2008*	**HES**	Age<50 & >84	Proximal to the ankle	Incidence proportion	Age	**26**
Ahmad, N. 2014b[10]	England	2003 – 2008*	**HES**	Age<50 & >84	Proximal to the ankle	Incidence proportion	Age	**26.3**
Ahmad, N. 2016[7]	England	2003 – 2012*	**HES**	Age<50 & >84	Proximal to the ankle	Incidence proportion	Age	**22.9**†
Ambler, GK. 2019[30]	UK	2014 – 2016	**NVR**	None	Proximal to the ankle	Number	N/A	**9549**‡
Behrendt, CA. 2018[4]	England	2014	**HES**	Trauma, cancer	Proximal to the ankle	Incidence proportion	None	**8.8**§
Moxey, PW. 2010[9]	England	2003 – 2007*	**HES**	None	BKA/TKA/AKA	Incidence proportion	None	**51.1**
Moxey, PW. 2012[37]	England	2002 – 2005*	**HES**	Age<50 & >84, trauma, cancer	BKA/TKA/AKA	Number	N/A	**14168**‡
Nickinson, A. 2019[38]	England	2000 – 2016	**HES linked CPRD**	Age<17, trauma, cancer	Proximal to the ankle	Number	N/A	**3260**‡
NVR report 2015[43]	UK	2014	**NVR**	Bilateral	BKA/AKA	Number	N/A	**2465**‡
NVR report 2016[44]	UK	2014 – 2015	**NVR**	Trauma, bilateral MLEA	BKA/AKA	Number	N/A	**5318**‡
NVR report 2017[45]	UK	2014 – 2016	**NVR**	Trauma, bilateral MLEA	BKA/AKA	Number	N/A	**8866**‡

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NVR report 2018[46]	UK	2015 – 2017	**NVR**	Trauma, bilateral MLEA	BKA/AKA	Number	N/A	**9293**‡
NVR report 2019[22]	UK	2016 – 2018	**NVR**	Trauma, bilateral MLEA	BKA/AKA	Number	N/A	**9508**‡
Vamos, EP. 2010a[41]	England	2004 – 2008*	**HES**	Trauma	Through/ proximal to the ankle	Incidence proportion	None	**6.9**†
Vamos, EP. 2010b[42]	England	1996 – 2005*	**HES**	Trauma	Through/ proximal to the ankle	Incidence proportion	Age, sex	**4.9**†
Population with diabetes								
Holman, N. 2012[8]	England	2007 – 2009*	**HES**	Age <17	Proximal to the ankle	Incidence rate	None	**2.51**¶
Kennon, B. 2012[36]	Scotland	2004 – 2008	**SMR01**	Trauma, cancer	Through/ proximal to the ankle	Incidence proportion	None	**139**
NDA 2009-2010[40]	England	2009*	**HES**	None	'Major Amputation' not further defined	Incidence proportion	None	**70**
NDA 2010-2011[31]	England & Wales	2010*	**HES, PEDW**	None	BKA/TKA/AKA	Incidence proportion	Age, sex	**291**
NDA 2011-2012[32]	England & Wales	2010 – 2011*	**HES, PEDW**	None	BKA/TKA/AKA	Incidence proportion	None	**170**
NDA 2012-2013[33]	England & Wales	2012*	**HES, PEDW**	None	BKA/TKA/AKA	Incidence proportion	None	**70**
NDA 2015-2016[35]	England & Wales	2015*	**HES, PEDW**	None	'Major Amputation' not further defined	Incidence proportion	None	**75**
NDA 2017-2018[34]	England & Wales	2017*	**HES, PEDW**	None	'Major Amputation' not further defined	Incidence proportion	None	**73**
PHE fingertips 2018[39]	England	2015 – 2017*	**HES**	Age <18	Proximal to the ankle	Incidence proportion (/rate)	Age, ethnicity	**82**

Statistic value is presented as incidence proportion per 100,000 population for the entire study period unless otherwise stated. If both standardised and crude values were available, the standardised value has been presented. AKA = above knee amputation, BKA = below knee amputation, CPRD = Clinical Practice Research Datalink, HES = Hospital Episode Statistics, NDA = National Diabetes Audit, NVR = National Vascular Registry, PEDW = Patient Episode Database for Wales, PHE = Public Health England, SMR01 = Scottish Morbidity Records, TKA = through knee amputation.

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* Study period conducted over the financial year (April-March) rather than the calendar year
† Data presented for the most recent year in study period
‡ Data represented as total number
§ Incidence proportion of peripheral artery disease related amputations within the general population
Incidence proportion of people without diabetes rather than the general population (which includes those with diabetes)
¶ Incidence rate per 1000 person years

Table 2: Characteristics of databases used to ascertain cases of MLEA by included reports

Database	Coverage	Linkage	Coding	Variables
Hospital Episode Statistics (HES)[58, 59] • England secondary care data • Records work done so hospital can get paid	Whole England population	• Linkage using NHS number, postcode, date of birth, sex & postcode • Well established linkages: ○ Mortality data ○ Cancer registry ○ National longitudinal studies ○ Primary care data	• ICD-10 for diagnoses • OPCS-4 for procedures	• No information on prescriptions • No information on primary • care • Over 400 variables including: ○ Episode (event) information ○ Diagnoses ○ Patient characteristics
Patient Episode Database for Wales (PEDW)[60, 61] • Wales secondary care data • Records work done so hospital can get paid	Whole Wales population	• Linkage using NHS number, postcode, date of birth, sex & postcode • Well established linkages: ○ Mortality data ○ Cancer registry ○ National longitudinal studies ○ Primary care data	• ICD-10 for diagnoses • OPCS-4 for procedures	• No information on prescriptions • No information on primary care • Over 400 variables including: ○ Episode (event) information ○ Diagnoses ○ Patient characteristics
Scottish Morbidity Record (SMR01)[62, 63] • Scotland secondary care data • Records work done so hospital can get paid	Whole Scotland population	• Scottish patient IDs include date of birth for ease of linkage • Well established linkages: ○ Mortality data ○ Cancer registry	• ICD-10 for diagnoses • OPCS-4 for procedures	• No information on prescriptions • No information on primary care • Similar amount of variables to HES and PEDW with extras geographical information

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		○ National longitudinal studies Primary care data		including: ○ Census geographical information ○ Electoral ward and parliamentary constituency
National Vascular Registry (NVR)[64] • UK data on 5 main vascular procedures (inc major and minor lower limb amputation)	UK data but procedure recording only mandatory for hospitals in England Contains 48% to 74% of MLEA cases over the years 2014-2018 compared to HES[49]	• No established linkages	• No Internationally classified code use • Surgeon recorded procedure	• Limited tick box of pre-operative medications • No information on primary care • Detailed procedure information • Comordbities recorded by limited tick boxes • Minimal patient characteristics

ICD-10 = International Classification of Diseases version 10, MLEA = Major lower extremity, NHS = National Health Service, amputation, OPCS-4 = Operation Procedure Codes version 4.

Figure and supplement captions

Figures

Figure 1: PRISMA flow diagram of the screening process.

Figure 2: Reported time trends in incidence proportion of MLEA for (A) the general population in England and (B) the population with diabetes (England & Scotland).

Figure 3: Reported regional incidence proportion of MLEA by (A) Moxey et al 2010 and (B) Ahmad et al 2014b.

Figure 4: Included article study periods by population type.

Supplementary files

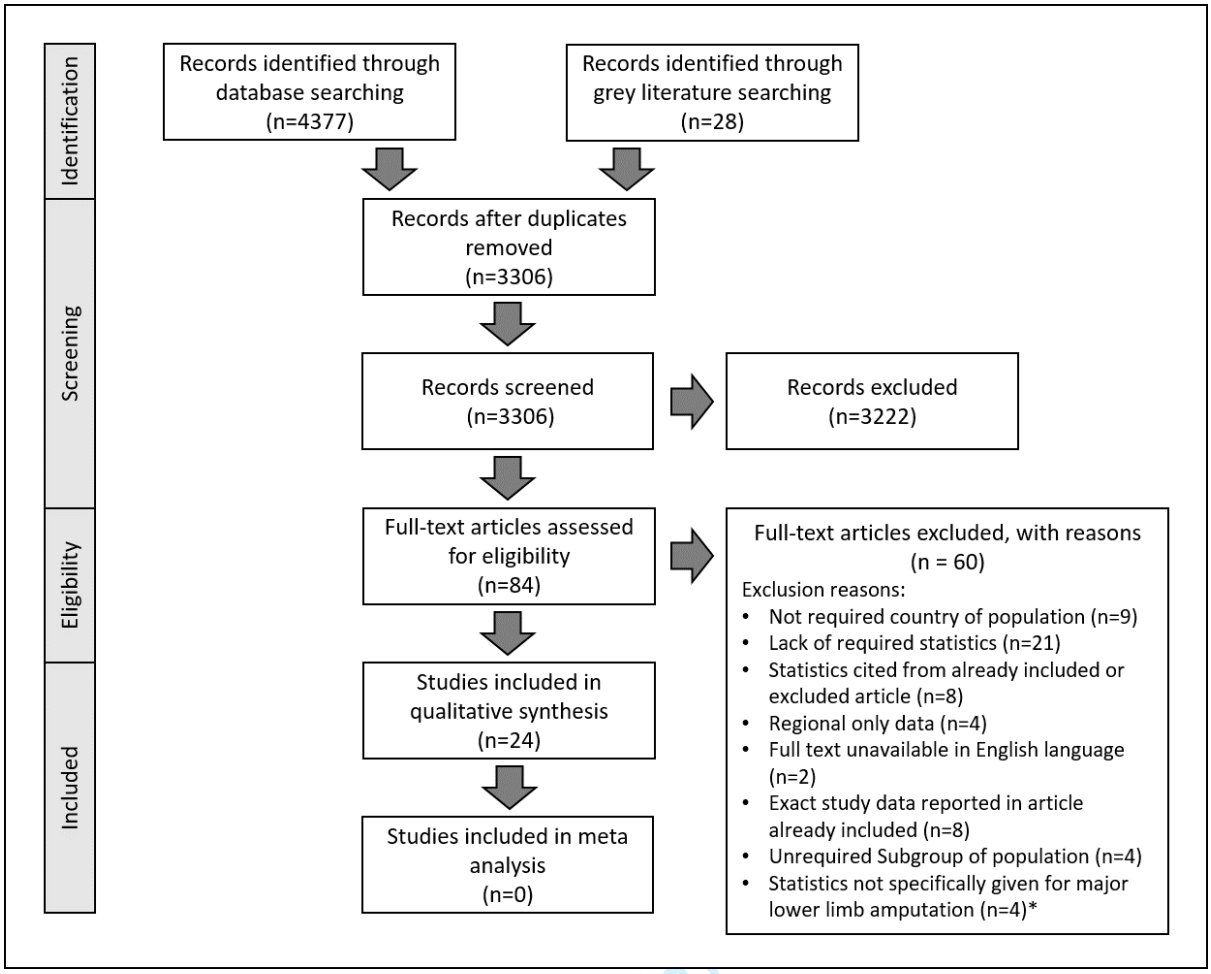
Supplementary file 1: Database search strategy

Supplementary file 2: Critical appraisal Table 1: Outcome of JBI critical appraisal for studies reporting prevalence data, Table 2: Outcome of JBI critical appraisal for cohort studies

Supplementary file 3: Table 1 - MLEA codes, Table 2 - Diabetes definitions

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Figure 1: PRISMA flow diagram of the screening process

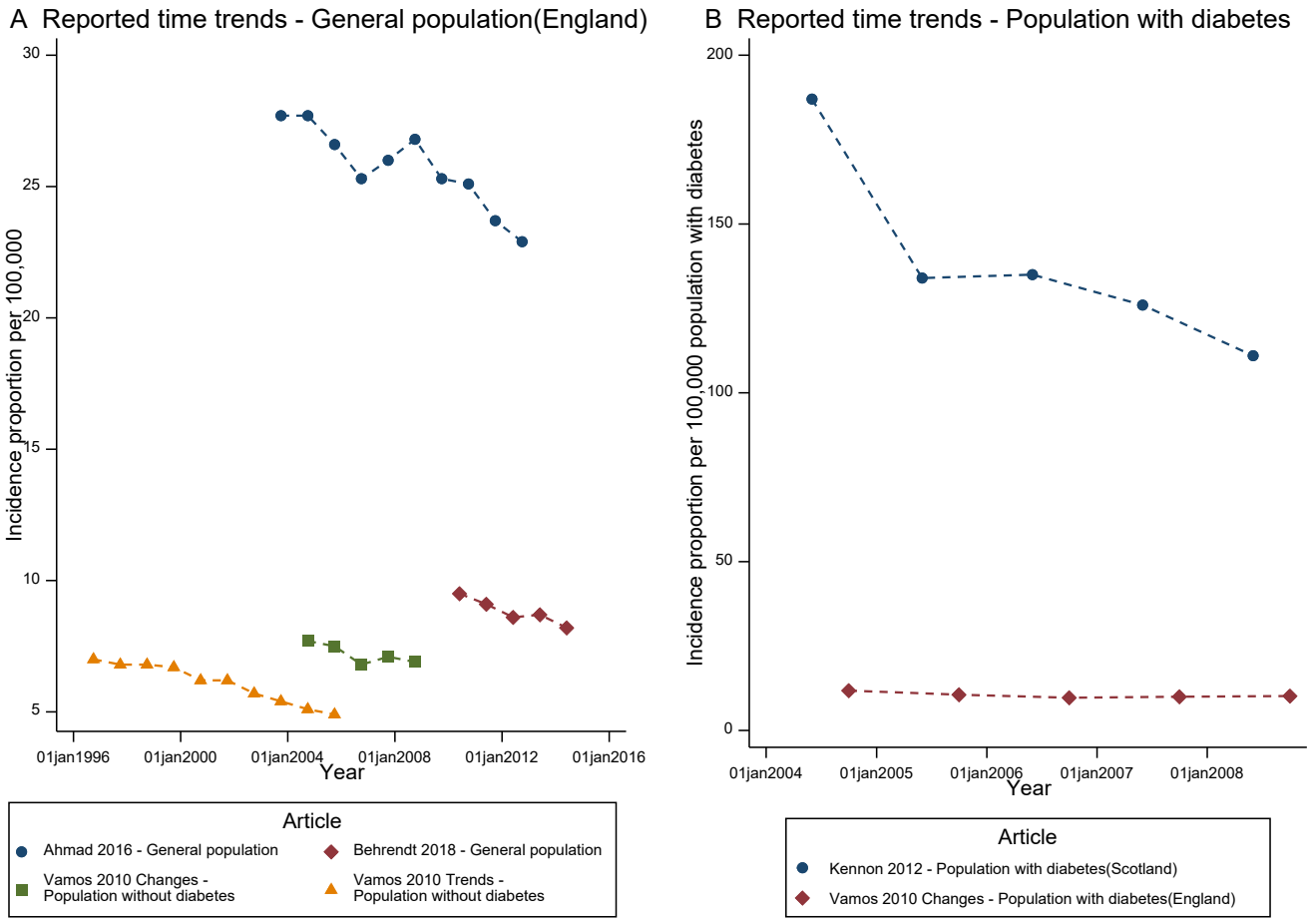


*Articles excluded for this reason [1-4]

1. Carinci F, Massi Benedetti M, Klazinga NS, Uccioli L. Lower extremity amputation rates in people with diabetes as an indicator of health systems performance. A critical appraisal of the data collection 2000-2011 by the Organization for Economic Cooperation and Development (OECD). Acta diabetologica. 2016;53(5):825-32.
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4. Hoffstad O, Mitra N, Walsh J, Margolis DJ. Diabetes, Lower-Extremity amputation, and death. Diabetes Care. 2015;38(10):1852-7.

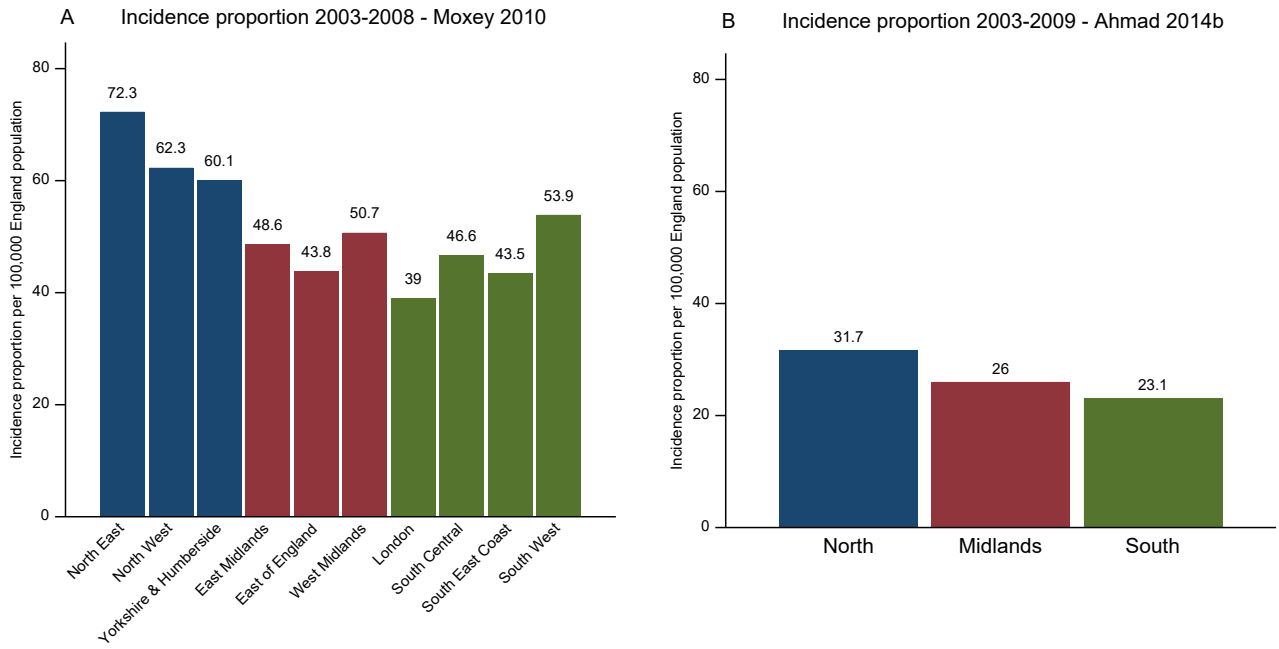
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Figure 2: Reported time trends in incidence proportion of MLEA for (A) the general population in England and (B) the population with diabetes (England & Scotland).



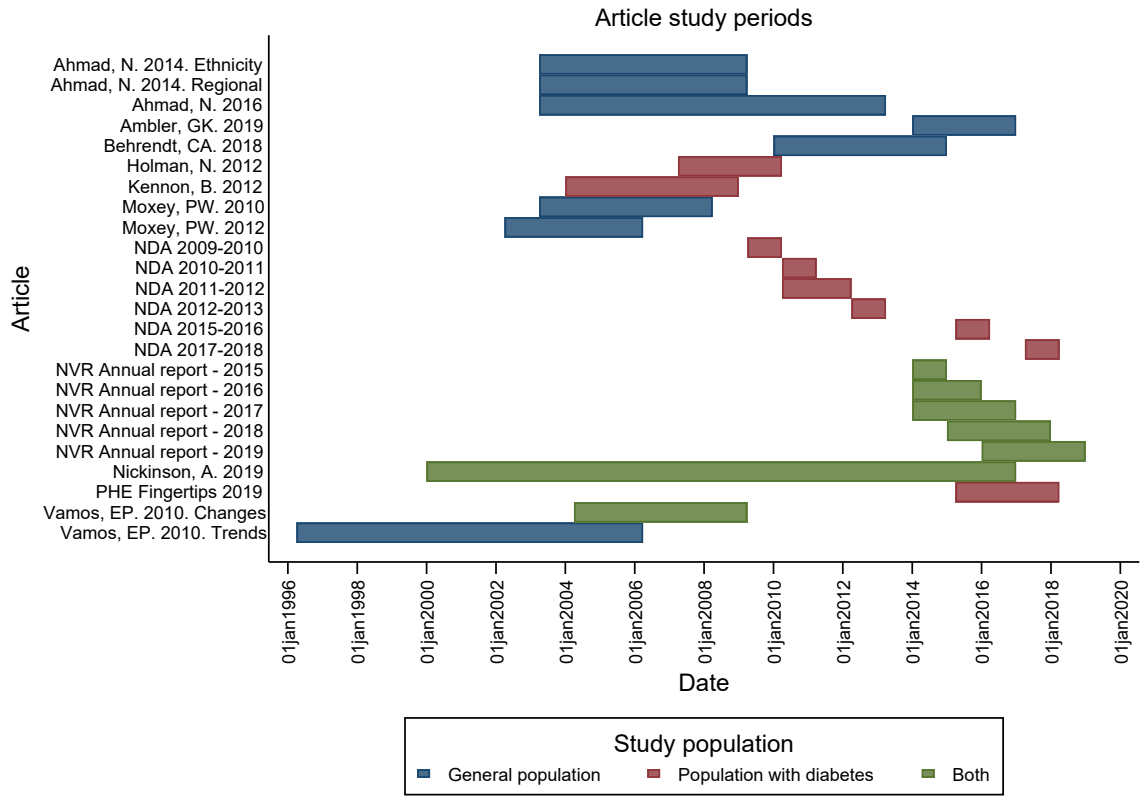
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Figure 3: Reported regional incidence proportion of MLEA by (A) Moxey et al 2010 and (B) Ahmad et al 2014b



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Figure 4: Included article study periods by population type



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Supplementary file 1 – Database search strategy

Database: Medline

Database: Ovid MEDLINE(R) and Epub Ahead of Print, In-Process & Other Non-Indexed Citations and Daily <1946 to December 12, 2019>

Search Strategy:

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1 Amputees/ (3376)

2 exp Amputation/ (20930)

3 amput*.tw. (42609)

4 disarticulat*.tw. (1606)

5 prosthe*.tw. (118947)

6 (artificial adj (limb* or leg*)).tw. (536)

7 Artificial Limbs/ (6665)

8 LEA.tw. (2347)

9 1 or 2 or 3 or 4 or 5 or 6 or 7 or 8 (166117)

10 "lower limb*".tw. (47509)

11 "lower extremit*".tw. (52565)

12 leg*.tw. (293270)

13 ((above or below or through) adj3 knee*).tw. (7081)

14 "trans-tibial".tw. (358)

15 transtibial.tw. (1733)

16 knee*.tw. (141203)

17 hip*.tw. (325565)

18 thigh*.tw. (28640)

19 "trans-femoral".tw. (350)

20 transfemoral.tw. (5026)

21 "trans-pelvic".tw. (6)

22 transpelvic.tw. (82)

23 hip/su (2871)

24 knee/su (2650)

25 leg/su (5107)

26 thigh/su (1447)

27 lower extremity/su (1856)

28 10 or 11 or 12 or 13 or 14 or 15 or 16 or 17 or 18 or 19 or 20 or 21 or 22 or 23 or 24 or 25 or 26 or 27 (818087)

29 vascular*.tw. (589135)

30 dysvascular*.tw. (239)

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31 exp Vascular Surgical Procedures/ (246355)

32 exp Vascular Diseases/ (1626902)

33 (peripheral adj3 ("arter* disease*" or "arter* disorder*" or arteriopath*)).tw. (14144)

34 "limb* threat*".tw. (1439)

35 ((leg* or limb* or peripheral) adj3 isch?emi*).tw. (14235)

36 PVD.tw. (2376)

37 PAOD.tw. (758)

38 (peripheral adj3 ("vascul* disorder*" or "vascul* disease*")).tw. (10062)

39 CLI.tw. (2465)

40 (ulcer* adj3 (arter* or vascular*)).tw. (1442)

41 Ischemia/ (49164)

42 ALI.tw. (6610)

43 ulcer*.tw. (198843)

44 diabet*.tw. (603319)

45 atherosclero*.tw. (144792)

46 arteriosclero*.tw. (15481)

47 claudic*.tw. (10402)

48 ((arter* or vascular) adj3 (occlus* or reocclus* or "re-occlus*" or steno* or restenos* or obstruct* or lesio* or block* or harden* or stiffen* or obliter* or calcificat* or restrict* or narrow* or block* or insufficien* or sclerosis)).tw. (135311)

49 (limb* adj3 (occlus* or reocclus* or "re-occlus*" or steno* or restenos* or obstruct* or lesio* or block* or harden* or stiffen* or obliter* or calcificat* or restrict* or narrow* or block* or insufficien* or sclerosis)).tw. (3792)

50 (leg* adj3 (occlus* or reocclus* or "re-occlus*" or steno* or restenos* or obstruct* or lesio* or block* or harden* or stiffen* or obliter* or calcificat* or restrict* or narrow* or block* or insufficien* or sclerosis)).tw. (3356)

51 (peripheral adj3 (occlus* or reocclus* or "re-occlus*" or steno* or restenos* or obstruct* or lesio* or block* or harden* or stiffen* or obliter* or calcificat* or restrict* or narrow* or block* or insufficien* or sclerosis)).tw. (16324)

52 ("lower extremit*" adj3 (occlus* or reocclus* or "re-occlus*" or steno* or restenos* or obstruct* or lesio* or block* or harden* or stiffen* or obliter* or calcificat* or restrict* or narrow* or block* or insufficien* or sclerosis)).tw. (2057)

53 Gangrene/ (8076)

54 gangren*.tw. (16525)

55 29 or 30 or 31 or 32 or 33 or 34 or 35 or 36 or 37 or 38 or 39 or 40 or 41 or 42 or 43 or 44 or 45 or 46 or 47 or 48 or 49 or 50 or 51 or 52 or 53 or 54 (2865492)

56 Epidemiology/ (12267)

57 incidence/ (252140)

58 prevalence/ (279362)

59 exp Epidemiologic Methods/ (6001928)

60 exp epidemiologic studies/ (2404737)

61 Population Surveillance/ (57836)

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62 epidemiolog*.mp. (1828006)

63 inciden*.mp. (951016)

64 prevalen*.mp. (796766)

65 frequenc*.mp. (942710)

66 (rate* adj3 (occur* or population*)).mp. (34809)

67 56 or 57 or 58 or 59 or 60 or 61 or 62 or 63 or 64 or 65 or 66 (7703017)

68 exp Great Britain/ (358725)

69 (national health service* or nhs*).ti,ab,in. (182826)

70 (english not ((published or publication* or translat* or written or language* or speak* or literature or citation*) adj5 english)).ti,ab. (93210)

71 (gb or "g.b." or britain* or (british* not "british columbia") or uk or "u.k." or united kingdom* or (england* not "new england") or northern ireland* or northern irish* or scotland* or scottish* or ((wales or "south wales") not "new south wales") or welsh*).ti,ab,jw,in. (1990860)

72 (bath or "bath's" or ((birmingham not alabama*) or ("birmingham's" not alabama*) or bradford or "bradford's" or brighton or "brighton's" or bristol or "bristol's" or carlisle* or "carlisle's" or (cambridge not (massachusetts* or boston* or harvard*)) or ("cambridge's" not (massachusetts* or boston* or harvard*)) or (canterbury not zealand*) or ("canterbury's" not zealand*) or chelmsford or "chelmsford's" or chester or "chester's" or chichester or "chichester's" or coventry or "coventry's" or derby or "derby's" or (durham not (carolina* or nc)) or ("durham's" not (carolina* or nc)) or ely or "ely's" or exeter or "exeter's" or gloucester or "gloucester's" or hereford or "hereford's" or hull or "hull's" or lancaster or "lancaster's" or leeds* or leicester or "leicester's" or (lincoln not nebraska*) or ("lincoln's" not nebraska*) or (liverpool not (new south wales* or nsw)) or ("liverpool's" not (new south wales* or nsw)) or ((london not (ontario* or ont or toronto*)) or ("london's" not (ontario* or ont or toronto*)) or manchester or "manchester's" or (newcastle not (new south wales* or nsw)) or ("newcastle's" not (new south wales* or nsw)) or norwich or "norwich's" or nottingham or "nottingham's" or oxford or "oxford's" or peterborough or "peterborough's" or plymouth or "plymouth's" or portsmouth or "portsmouth's" or preston or "preston's" or ripon or "ripon's" or salford or "salford's" or salisbury or "salisbury's" or sheffield or "sheffield's" or southampton or "southampton's" or st albans or stoke or "stoke's" or sunderland or "sunderland's" or truro or "truro's" or wakefield or "wakefield's" or wells or westminster or "westminster's" or winchester or "winchester's" or wolverhampton or "wolverhampton's" or (worcester not (massachusetts* or boston* or harvard*)) or ("worcester's" not (massachusetts* or boston* or harvard*)) or (york not ("new york*" or ny or ontario* or ont or toronto*)) or ("york's" not ("new york*" or ny or ontario* or ont or toronto*))))).ti,ab,in. (1341900)

73 (bangor or "bangor's" or cardiff or "cardiff's" or newport or "newport's" or st asaph or "st asaph's" or st davids or swansea or "swansea's").ti,ab,in. (52438)

74 (aberdeen or "aberdeen's" or dundee or "dundee's" or edinburgh or "edinburgh's" or glasgow or "glasgow's" or inverness or (perth not australia*) or ("perth's" not australia*) or stirling or "stirling's").ti,ab,in. (199885)

75 (armagh or "armagh's" or belfast or "belfast's" or lisburn or "lisburn's" or londonderry or "londonderry's" or derry or "derry's" or newry or "newry's").ti,ab,in. (24719)

76 68 or 69 or 70 or 71 or 72 or 73 or 74 or 75 (2563344)

77 (exp africa/ or exp americas/ or exp antarctic regions/ or exp arctic regions/ or exp asia/ or exp oceania/) not (exp great britain/ or europe/) (2783662)

78 76 not 77 (2422301)

79 9 and 28 and 55 and 67 and 78 (704)

80 exp animals/ not humans.sh. (4651026)

81 79 not 80 (701)

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82 limit 81 to yr="2009-2020" (341)

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- **Lines 1-9** are amputation terms
- **Lines 10-28** are lower extremity terms
- **Lines 29-55** are vascular terms (including diabetes)

The aim of combining these groups is to pick up only major lower limb amputation for vascular conditions.

Filters used:

- **Lines 68-74 is the UK filter** for Medline[22].
  This has been adapted for EMBASE[23].
  Validated UK filters for the other databases were not available.
- **Lines 76-77 is a human filter**, taken from the end of the Cochrane Handbook filter for RCTs[32].
- **Lines 56-67 is a study type filter**[33].

Database: EMBASE

Search Strategy:

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1 Amputee/ (933)

2 exp Amputation/ (44320)

3 amput*.tw. (51765)

4 disarticulat*.tw. (1845)

5 prosthe*.tw. (137059)

6 (artificial adj (limb* or leg*)).tw. (555)

7 exp "limb prosthesis"/ (6486)

8 LEA.tw. (2728)

9 "lower limb*".tw. (69278)

10 "lower extremit*".tw. (70925)

11 leg*.tw. (367108)

12 ((above or below or through) adj3 knee*).tw. (9584)

13 "trans-tibial".tw. (445)

14 transtibial.tw. (2073)

15 knee*.tw. (182897)

16 hip*.tw. (416631)

17 thigh*.tw. (39584)

18 "trans-femoral".tw. (840)

19 transfemoral.tw. (8600)

20 "trans-pelvic".tw. (13)

21 transpelvic.tw. (88)

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22 hip/su (1527)

23 exp leg/su (6845)

24 "lower limb"/ (15078)

25 vascular*.tw. (798977)

26 dysvascular*.tw. (270)

27 exp "Vascular Surgery"/ (442769)

28 exp "Vascular Disease"/ (2359700)

29 (peripheral adj3 ("arter* disease*" or "arter* disorder*" or arteriopath*)).tw. (22290)

30 "limb* threat*".tw. (1723)

31 ((leg* or limb* or peripheral) adj3 isch?emi*).tw. (20995)

32 PVD.tw. (4002)

33 PAOD.tw. (1230)

34 (peripheral adj3 ("vascul* disorder*" or "vascul* disease*")).tw. (14328)

35 CLI.tw. (4204)

36 (ulcer* adj3 (arter* or vascular*)).tw. (2062)

37 Ischemia/ (77155)

38 "Peripheral Ischemia"/ (934)

39 "Leg Ischemia"/ (6107)

40 ALI.tw. (11904)

41 ulcer*.tw. (254945)

42 diabet*.tw. (893260)

43 atherosclero*.tw. (199411)

44 arteriosclero*.tw. (15582)

45 claudic*.tw. (14191)

46 ((arter* or vascular) adj3 (occlus* or reocclus* or "re-occlus*" or steno* or restenos* or obstruct* or lesio* or block* or harden* or stiffen* or obliter* or calcificat* or restrict* or narrow* or block* or insufficien* or sclerosis)).tw. (184353)

47 (limb* adj3 (occlus* or reocclus* or "re-occlus*" or steno* or restenos* or obstruct* or lesio* or block* or harden* or stiffen* or obliter* or calcificat* or restrict* or narrow* or block* or insufficien* or sclerosis)).tw. (5473)

48 (leg* adj3 (occlus* or reocclus* or "re-occlus*" or steno* or restenos* or obstruct* or lesio* or block* or harden* or stiffen* or obliter* or calcificat* or restrict* or narrow* or block* or insufficien* or sclerosis)).tw. (4337)

49 (peripheral adj3 (occlus* or reocclus* or "re-occlus*" or steno* or restenos* or obstruct* or lesio* or block* or harden* or stiffen* or obliter* or calcificat* or restrict* or narrow* or block* or insufficien* or sclerosis)).tw. (22852)

50 ("lower extremit*" adj3 (occlus* or reocclus* or "re-occlus*" or steno* or restenos* or obstruct* or lesio* or block* or harden* or stiffen* or obliter* or calcificat* or restrict* or narrow* or block* or insufficien* or sclerosis)).tw. (2707)

51 Gangrene/ (9763)

52 gangren*.tw. (18326)

53 1 or 2 or 3 or 4 or 5 or 6 or 7 or 8 (197708)

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54 9 or 10 or 11 or 12 or 13 or 14 or 15 or 16 or 17 or 18 or 19 or 20 or 21 or 22 or 23 or 24 (1048226)

55 25 or 26 or 27 or 28 or 29 or 30 or 31 or 32 or 33 or 34 or 35 or 36 or 37 or 38 or 39 or 40 or 41 or 42 or 43 or 44 or 45 or 46 or 47 or 48 or 49 or 50 or 51 or 52 (3903577)

56 epidemiolog*.mp. (1504866)

57 inciden*.mp. (1328417)

58 prevalen*.mp. (1170777)

59 frequenc*.mp. (1238143)

60 (rate* adj3 (occur* or population*)).mp. (43835)

61 exp epidemiology/ (3268751)

62 epidemiological data/ (31787)

63 health survey/ (188176)

64 56 or 57 or 58 or 59 or 60 or 61 or 62 or 63 (5759910)

65 exp United Kingdom/ (408634)

66 (national health service* or nhs*).ti,ab,in,ad. (331006)

67 (english not ((published or publication* or translat* or written or language* or speak* or literature or citation*) adj5 english)).ti,ab. (40875)

68 (gb or "g.b." or britain* or (british* not "british columbia") or uk or "u.k." or united kingdom* or (england* not "new england") or northern ireland* or northern irish* or scotland* or scottish* or ((wales or "south wales") not "new south wales") or welsh*).ti,ab,jx,in,ad. (3065651)

69 (bath or "bath's" or ((birmingham not alabama*) or ("birmingham's" not alabama*) or bradford or "bradford's" or brighton or "brighton's" or bristol or "bristol's" or carlisle* or "carlisle's" or (cambridge not (massachusetts* or boston* or harvard*)) or ("cambridge's" not (massachusetts* or boston* or harvard*)) or (canterbury not zealand*) or ("canterbury's" not zealand*) or chelmsford or "chelmsford's" or chester or "chester's" or chichester or "chichester's" or coventry or "coventry's" or derby or "derby's" or (durham not (carolina* or nc)) or ("durham's" not (carolina* or nc)) or ely or "ely's" or exeter or "exeter's" or gloucester or "gloucester's" or hereford or "hereford's" or hull or "hull's" or lancaster or "lancaster's" or leeds* or leicester or "leicester's" or (lincoln not nebraska*) or ("lincoln's" not nebraska*) or (liverpool not (new south wales* or nsw)) or ("liverpool's" not (new south wales* or nsw)) or ((london not (ontario* or ont or toronto*)) or ("london's" not (ontario* or ont or toronto*)) or manchester or "manchester's" or (newcastle not (new south wales* or nsw)) or ("newcastle's" not (new south wales* or nsw)) or norwich or "norwich's" or nottingham or "nottingham's" or oxford or "oxford's" or peterborough or "peterborough's" or plymouth or "plymouth's" or portsmouth or "portsmouth's" or preston or "preston's" or ripon or "ripon's" or salford or "salford's" or salisbury or "salisbury's" or sheffield or "sheffield's" or southampton or "southampton's" or st albans or stoke or "stoke's" or sunderland or "sunderland's" or truro or "truro's" or wakefield or "wakefield's" or wells or westminster or "westminster's" or winchester or "winchester's" or wolverhampton or "wolverhampton's" or (worcester not (massachusetts* or boston* or harvard*)) or ("worcester's" not (massachusetts* or boston* or harvard*)) or (york not ("new york*" or ny or ontario* or ont or toronto*)) or ("york's" not ("new york*" or ny or ontario* or ont or toronto*))))).ti,ab,in,ad. (2353624)

70 (bangor or "bangor's" or cardiff or "cardiff's" or newport or "newport's" or st asaph or "st asaph's" or st davids or swansea or "swansea's").ti,ab,in,ad. (95988)

71 (aberdeen or "aberdeen's" or dundee or "dundee's" or edinburgh or "edinburgh's" or glasgow or "glasgow's" or inverness or (perth not australia*) or ("perth's" not australia*) or stirling or "stirling's").ti,ab,in,ad. (325298)

72 (armagh or "armagh's" or belfast or "belfast's" or lisburn or "lisburn's" or londonderry or "londonderry's" or derry or "derry's" or newry or "newry's").ti,ab,in,ad. (43503)

73 65 or 66 or 67 or 68 or 69 or 70 or 71 or 72 (3734952)

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74 (exp "arctic and antarctic"/ or exp oceanic regions/ or exp western hemisphere/ or exp africa/ or exp asia/ or exp "australia and new zealand"/) not (exp united kingdom/ or europe/) (2982458)

75 73 not 74 (3529253)

76 exp experimental organism/ (680593)

77 animal tissue/ (1318352)

78 animal cell/ (1463534)

79 exp animal disease/ (266885)

80 exp carnivore disease/ (46201)

81 exp bird/ (220421)

82 exp experimental animal welfare/ (3098)

83 exp animal husbandry/ (49846)

84 animal behavior/ (81143)

85 exp animal cell culture/ (12292)

86 exp mammalian disease/ (159889)

87 exp mammal/ (23800499)

88 exp marine species/ (6729)

89 nonhuman/ (5993421)

90 animal.hw. (5252369)

91 76 or 77 or 78 or 79 or 80 or 81 or 82 or 83 or 84 or 85 or 86 or 87 or 88 or 89 or 90 (26618565)

92 91 not human/ (6463716)

93 53 and 54 and 55 and 64 and 75 (900)

94 93 not 92 (896)

95 limit 94 to yr="2009 - 2020" (585)

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Database: CINAHL

Search Strategy:

#	Query	Limiters/Expanders	Last Run Via	Results
S1	TI amput* OR AB amput*	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	10,891
S2	TI disarticulat* OR AB disarticulat*	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	335

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S3	TI prosthe* OR AB prosthe*	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	19,654
S4	TI (artificial W1 (limb* OR leg*)) OR AB (artificial W1 (limb* OR leg*))	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	100
S5	(MH "Amputees")	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	2,691
S6	(MH "LIMB PROSTHESIS")	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	2,183
S7	(MH "Amputation+")	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	7,424
S8	TI LEA OR AB LEA	Expanders - Apply equivalent subjects	Interface - EBSCOhost Research Databases Search Screen - Advanced	386

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		Search modes - Boolean/Phrase	Search Database - CINAHL	
S9	S1 OR S2 OR S3 OR S4 OR S5 OR S6 OR S7 OR S8	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	31,971
S10	TI "lower limb*" OR AB "lower limb*"	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	12,988
S11	TI "lower extremit*" OR AB "lower extremit*"	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	15,723
S12	TI leg* OR AB leg*	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	96,668
S13	TI ((above OR below OR through) N3 knee*) OR AB ((above OR below OR through) N3 knee*)	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	1,899
S14	TI "trans-tibial" OR AB "trans-tibial"	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	107
S15	TI transtibial OR AB transtibial	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	1,189
S16	TI knee* OR AB knee*	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	56,084
S17	TI hip* OR AB hip*	Expanders - Apply equivalent subjects	Interface - EBSCOhost Research Databases	66,438

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		Search modes - Boolean/Phrase	Search Screen - Advanced Search Database - CINAHL	
S18	TI thigh* OR AB thigh*	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	5,640
S19	TI "trans-femoral" OR AB "trans-femoral"	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	86
S20	TI transfemoral OR AB transfemoral	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	1,408
S21	TI "trans-pelvic" OR AB "trans-pelvic"	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	0
S22	TI transpelvic OR AB transpelvic	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	9
S23	(MH "Hip/SU")	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	47
S24	(MH "Knee/SU")	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	0
S25	(MH "Leg/SU")	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	485
S26	(MH "Thigh/SU")	Expanders - Apply	Interface - EBSCOhost	133

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		equivalent subjects Search modes - Boolean/Phrase	Research Databases Search Screen - Advanced Search Database - CINAHL	
S27	(MH "Lower Extremity/SU")	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	822
S28	S10 OR S11 OR S12 OR S13 OR S14 OR S15 OR S16 OR S17 OR S18 OR S19 OR S20 OR S21 OR S22 OR S23 OR S24 OR S25 OR S26 OR S27	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	225,981
S29	TI vascular* OR AB vascular*	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	69,127
S30	TI dysvascular* OR AB dysvascular*	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	139
S31	TI (peripheral N3 ("arter* disease*" OR "arter* disorder*" OR arteriopath*)) OR AB (peripheral N3 ("arter* disease*" OR "arter* disorder*" OR arteriopath*))	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	3,943
S32	TI (limb* N1 threat*) OR AB (limb* N1 threat*)	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	371
S33	TI (((leg* OR limb* OR peripheral) N3 ischemi*) OR (leg* OR limb* OR peripheral) N3 ischaemi*)) OR AB (((leg* OR limb* OR peripheral) N3 ischemi*)	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	2,280

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	OR (leg* OR limb* OR peripheral) N3 ischaemi*))			
S34	TI PVD OR AB PVD	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	402
S35	TI PAOD OR AB PAOD	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	103
S36	TI (peripheral N3 ("vascul* disorder*" OR "vascul* disease*")) OR AB (peripheral N3 ("vascul* disorder*" OR "vascul* disease*"))	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	1,962
S37	TI CLI OR AB CLI	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	396
S38	TI (ulcer* N3 (arter* OR vascular*)) OR AB (ulcer* N3 (arter* OR vascular*))	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	511
S39	TI ALI OR AB ALI	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	1,512
S40	TI ulcer* OR AB ulcer*	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	32,901
S41	TI diabet* OR AB diabet*	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	173,481

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S42	TI atherosclero* OR AB atherosclero*	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	20,830
S43	TI arteriosclero* OR AB arteriosclero*	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	679
S44	TI claudic* OR AB claudic*	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	1,750
S45	TI ((arter* OR vascular) N3 (occlus* OR reocclus* OR "re-occlus*" OR steno* OR restenos* OR obstruct* OR lesio* OR block* OR harden* OR stiffen* OR obliter* OR calcificat* OR restrict* OR narrow* OR block* OR insufficien* OR sclerosis)) OR AB ((arter* OR vascular) N3 (occlus* OR reocclus* OR "re-occlus*" OR steno* OR restenos* OR obstruct* OR lesio* OR block* OR harden* OR stiffen* OR obliter* OR calcificat* OR restrict* OR narrow* OR block* OR insufficien* OR sclerosis))	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	20,456
S46	TI (limb* N3 (occlus* OR reocclus* OR "re-occlus*" OR steno* OR restenos* OR obstruct* OR lesio* OR block* OR harden* OR stiffen* OR obliter* OR calcificat* OR restrict* OR narrow* OR block* OR insufficien* OR sclerosis)) OR AB (limb*	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	762

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	N3 (occlus* OR reocclus* OR "re-occlus*" OR steno* OR restenos* OR obstruct* OR lesio* OR block* OR harden* OR stiffen* OR obliter* OR calcificat* OR restrict* OR narrow* OR block* OR insufficien* OR sclerosis))			
S47	TI (leg* N3 (occlus* OR reocclus* OR "re-occlus*" OR steno* OR restenos* OR obstruct* OR lesio* OR block* OR harden* OR stiffen* OR obliter* OR calcificat* OR restrict* OR narrow* OR block* OR insufficien* OR sclerosis)) OR AB (leg* N3 (occlus* OR reocclus* OR "re-occlus*" OR steno* OR restenos* OR obstruct* OR lesio* OR block* OR harden* OR stiffen* OR obliter* OR calcificat* OR restrict* OR narrow* OR block* OR insufficien* OR sclerosis))	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	994
S48	TI (peripheral N3 (occlus* OR reocclus* OR "re-occlus*" OR steno* OR restenos* OR obstruct* OR lesio* OR block* OR harden* OR stiffen* OR obliter* OR calcificat* OR restrict* OR narrow* OR block* OR insufficien* OR sclerosis)) OR AB (peripheral N3 (occlus* OR reocclus* OR "re-occlus*" OR steno* OR restenos* OR obstruct* OR lesio* OR block* OR harden* OR stiffen* OR obliter* OR calcificat* OR	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	2,700

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	restrict* OR narrow* OR block* OR insufficien* OR sclerosis))			
S49	TI ("lower extremit*" N3 (occlus* OR reocclus* OR "re-occlus*" OR steno* OR restenos* OR obstruct* OR lesio* OR block* OR harden* OR stiffen* OR obliter* OR calcificat* OR restrict* OR narrow* OR block* OR insufficien* OR sclerosis)) OR AB ("lower extremit*" N3 (occlus* OR reocclus* OR "reocclus*" OR steno* OR restenos* OR obstruct* OR lesio* OR block* OR harden* OR stiffen* OR obliter* OR calcificat* OR restrict* OR narrow* OR block* OR insufficien* OR sclerosis))	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	346
S50	TI gangren* OR AB gangren*	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	2,044
S51	(MH "VASCULAR SURGERY+")	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	26,829
S52	(MH "VASCULAR DISEASES+")	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	349,701
S53	(MH "ISCHEMIA")	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	6,485
S54	(MH "GANGRENE")	Expanders - Apply	Interface - EBSCOhost	701

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		equivalent subjects Search modes - Boolean/Phrase	Research Databases Search Screen - Advanced Search Database - CINAHL	
S55	S29 OR S30 OR S31 OR S32 OR S33 OR S34 OR S35 OR S36 OR S37 OR S38 OR S39 OR S40 OR S41 OR S42 OR S43 OR S44 OR S45 OR S46 OR S47 OR S48 OR S49 OR S50 OR S51 OR S52 OR S53 OR S54	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	575,685
S56	TX epidemiolog*	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	542,813
S57	TX inciden*	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	210,598
S58	TX prevalen*	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	220,384
S59	TX frequenc*	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	132,099
S60	TX (rate* N3 (occur* OR population*))	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	10,573
S61	(MH "EPIDEMIOLOGY+")	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	635,204
S62	(MH "INCIDENCE")	Expanders - Apply equivalent subjects	Interface - EBSCOhost Research Databases	62,878

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		Search modes - Boolean/Phrase	Search Screen - Advanced Search Database - CINAHL	
S63	(MH "PREVALENCE")	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	83,560
S64	(MH "EPIDEMIOLOGICAL RESEARCH+")	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	27,901
S65	S56 OR S57 OR S58 OR S59 OR S60 OR S61 OR S62 OR S63 OR S64	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	1,192,649
S66	S9 AND S28 AND S55 AND S65	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	1,031
S67	MH animals+	Limiters - Published Date: 20090101-20201231 Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	58,194
S68	MH (animal studies)	Limiters - Published Date: 20090101-20201231 Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	86,943
S69	TI (animal model*)	Limiters - Published Date: 20090101-20201231 Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	2,141
S70	S67 OR S68 OR S69	Expanders - Apply	Interface - EBSCOhost	141,713

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		equivalent subjects Search modes - Boolean/Phrase	Research Databases Search Screen - Advanced Search Database - CINAHL	
S71	S66 NOT S70	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	1,025
S72	S66 NOT S70	Limiters - Published Date: 20090101-20201231 Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	690

Database: Cochrane Library

Search Strategy:

ID	Search	Hits
#1	amput*	3156
#2	disarticulat*	39
#3	prosthe*	14698
#4	artificial NEXT (limb* or leg*)	162
#5	LEA	415
#6	MeSH descriptor: [Amputees] this term only	107
#7	MeSH descriptor: [Amputation] explode all trees	422
#8	MeSH descriptor: [Artificial Limbs] this term only	141
#9	[OR #1-#8]	17711
#10	"lower limb" OR "lower limbs"	8801
#11	"lower extremity" OR "lower extremities"	7011
#12	leg*	35011
#13	((above or below or through) NEAR/3 knee*)	1160
#14	"trans-tibial"	49
#15	transtibial	263

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#16 knee* 28100

#17 hip* 30315

#18 thigh* 3998

#19 "trans-femoral" 48

#20 transfemoral 592

#21 "trans-pelvic" 0

#22 transpelvic 3

#23 MeSH descriptor: [Hip] this term only 403

#24 MeSH descriptor: [Knee] this term only 765

#25 MeSH descriptor: [Leg] this term only 2886

#26 MeSH descriptor: [Thigh] this term only 460

#27 MeSH descriptor: [Lower Extremity] explode all trees 6900

#28 [OR #10-#27] 93386

#29 vascular* 52059

#30 dysvascular* 74

#31 MeSH descriptor: [Vascular Surgical Procedures] explode all trees 14772

#32 MeSH descriptor: [Vascular Diseases] explode all trees 74765

#33 peripheral NEAR/3 ("artery disease" OR "arterial disease" OR "artery disorder" OR "arterial disorder" OR arteriopath*) 3523

#34 "limb threat" 2

#35 (leg* or limb* or peripheral) NEAR/3 isch?emi* 2318

#36 PVD 13028

#37 PAOD 227

#38 peripheral NEAR/3 ("vascular disorder" OR "vascular disorders" OR "vascular disease" OR "vascular diseases") 2377

#39 CLI 535

#40 ulcer* NEAR/3 (arter* OR vascular*) 292

#41 MeSH descriptor: [Ischemia] this term only 1713

#42 ALI 5055

#43 ulcer* 26852

#44 diabet* 92003

#45 atherosclero* 13572

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#46 arteriosclero* 2802

#47 claudic* 2542

#48 (arter* or vascular) NEAR/3 (occlus* or reocclus* or steno* or restenos* or obstruct* or lesio* or block* or harden* or stiffen* or obliter* or calcificat* or restrict* or narrow* or block* or insufficien* or sclerosis) 13751

#49 limb* NEAR/3 (occlus* or reocclus* or steno* or restenos* or obstruct* or lesio* or block* or harden* or stiffen* or obliter* or calcificat* or restrict* or narrow* or block* or insufficien* or sclerosis) 723

#50 leg* NEAR/3 (occlus* or reocclus* or steno* or restenos* or obstruct* or lesio* or block* or harden* or stiffen* or obliter* or calcificat* or restrict* or narrow* or block* or insufficien* or sclerosis) 551

#51 peripheral NEAR/3 (occlus* or reocclus* or steno* or restenos* or obstruct* or lesio* or block* or harden* or stiffen* or obliter* or calcificat* or restrict* or narrow* or block* or insufficien* or sclerosis) 3610

#52 ("lower extremity" OR "lower extremities") NEAR/3 (occlus* or reocclus* or steno* or restenos* or obstruct* or lesio* or block* or harden* or stiffen* or obliter* or calcificat* or restrict* or narrow* or block* or insufficien* or sclerosis) 268

#53 gangren* 769

#54 MeSH descriptor: [Gangrene] this term only 70

#55 [OR #29 - #54] 240320

#56 epidemiolog* 72616

#57 inciden* 122718

#58 prevalen* 47943

#59 frequenc* 80831

#60 rate* NEAR/3 (occur* or population*) 4023

#61 MeSH descriptor: [Epidemiology] explode all trees 47

#62 MeSH descriptor: [Incidence] this term only 9858

#63 MeSH descriptor: [Prevalence] this term only 4841

#64 MeSH descriptor: [Epidemiologic Methods] explode all trees 405014

#65 MeSH descriptor: [Epidemiologic Studies] explode all trees 151650

#66 [OR #56 - #65] 576460

#67 #9 AND #28 AND #55 AND #66 with Publication Year from 2009 to 2019, with Cochrane Library publication date Between Jan 2009 and Jan 2019, in Trials 361

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Database: AMED

Search Strategy:

#	Database	Search term	Results
1	AMED	AMPUTATION/	2197
2	AMED	"AMPUTATION STUMPS"/	123
3	AMED	exp PROSTHESIS/	1955
4	AMED	(amput*).ti,ab	2641
5	AMED	(disarticulat*).ti,ab	138
6	AMED	(prosthe*).ti,ab	2759
7	AMED	(artificial ADJ (limb* OR leg*)).ti,ab	43
8	AMED	(LEA).ti,ab	20
9	AMED	(1 OR 2 OR 3 OR 4 OR 5 OR 6 OR 7 OR 8)	5260
10	AMED	("lower limb*").ti,ab	3143
11	AMED	("lower extremit*").ti,ab	3441
12	AMED	(leg*).ti,ab	8679
13	AMED	((above OR below OR through) ADJ3 knee*).ti,ab	525
14	AMED	("trans-tibial").ti,ab	159
15	AMED	(transtibial).ti,ab	446
16	AMED	(knee*).ti,ab	10846
17	AMED	(hip*).ti,ab	6708
18	AMED	(thigh*).ti,ab	856
19	AMED	("trans-femoral").ti,ab	69
20	AMED	(transfemoral).ti,ab	270

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21	AMED	("trans-pelvic").ti,ab	0
22	AMED	(transpelvic).ti,ab	1
23	AMED	THIGH/	207
24	AMED	KNEE/	1599
25	AMED	HIP/	909
26	AMED	LEG/	5036
27	AMED	(10 OR 11 OR 12 OR 13 OR 14 OR 15 OR 16 OR 17 OR 18 OR 19 OR 20 OR 21 OR 22 OR 23 OR 24 OR 25 OR 26)	28326
28	AMED	(vascular*).ti,ab	2118
29	AMED	(dysvascular*).ti,ab	63
30	AMED	"VASCULAR SURGERY"/	20
31	AMED	exp "VASCULAR DISEASE"/	13067
32	AMED	(peripheral ADJ3 ("arter* disease*" OR "arter* disorder*" OR arteriopath*)).ti,ab	133
33	AMED	("limb* threat*").ti,ab	32
34	AMED	((leg* OR limb* OR peripheral) ADJ3 isch?emi*).ti,ab	72
35	AMED	(PVD).ti,ab	13
36	AMED	(PAOD).ti,ab	10
37	AMED	(peripheral ADJ3 ("vascul* disorder*" OR "vascul* disease*")).ti,ab	152
38	AMED	(CLI).ti,ab	1
39	AMED	(ulcer* ADJ3 (arter* OR vascular*)).ti,ab	14
40	AMED	(ALI).ti,ab	52
41	AMED	(ulcer*).ti,ab	2267

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42	AMED	(diabet*).ti,ab	5359
43	AMED	(atherosclero*).ti,ab	471
44	AMED	(arteriosclero*).ti,ab	68
45	AMED	(claudic*).ti,ab	160
46	AMED	((arter* OR vascular) ADJ3 (occlus* OR reocclus* OR "re-occlus*" OR steno* OR restenos* OR obstruct* OR lesio* OR block* OR harden* OR stiffen* OR obliter* OR calcificat* OR restrict* OR narrow* OR block* OR insufficien* OR sclerosis)).ti,ab	455
47	AMED	(limb* ADJ3 (occlus* OR reocclus* OR "re-occlus*" OR steno* OR restenos* OR obstruct* OR lesio* OR block* OR harden* OR stiffen* OR obliter* OR calcificat* OR restrict* OR narrow* OR block* OR insufficien* OR sclerosis)).ti,ab	81
48	AMED	(leg* ADJ3 (occlus* OR reocclus* OR "re-occlus*" OR steno* OR restenos* OR obstruct* OR lesio* OR block* OR harden* OR stiffen* OR obliter* OR calcificat* OR restrict* OR narrow* OR block* OR insufficien* OR sclerosis)).ti,ab	78
49	AMED	(peripheral ADJ3 (occlus* OR reocclus* OR "re-occlus*" OR steno* OR restenos* OR obstruct* OR lesio* OR block* OR harden* OR stiffen* OR obliter* OR calcificat* OR restrict* OR narrow* OR block* OR insufficien* OR sclerosis)).ti,ab	128
50	AMED	("lower extremit*" ADJ3 (occlus* OR reocclus* OR "re-occlus*" OR steno* OR restenos* OR obstruct* OR lesio* OR block* OR harden* OR stiffen* OR obliter* OR calcificat* OR restrict* OR narrow* OR block* OR insufficien* OR sclerosis)).ti,ab	45
51	AMED	(gangren*).ti,ab	92
52	AMED	(28 OR 29 OR 30 OR 31 OR 32 OR 33 OR 34 OR 35 OR 36 OR 37 OR 38 OR	21019

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		39 OR 40 OR 41 OR 42 OR 43 OR 44 OR 45 OR 46 OR 47 OR 48 OR 49 OR 50 OR 51)	
53	AMED	(epidemiolog*).af	6512
54	AMED	(inciden*).af	4727
55	AMED	(prevalen*).af	5486
56	AMED	(frequenc*).af	7273
57	AMED	(rate* ADJ3 (occur* OR population*)).af	364
58	AMED	EPIDEMIOLOGY/	4880
59	AMED	(53 OR 54 OR 55 OR 56 OR 57 OR 58)	21072
60	AMED	(9 AND 27 AND 52 AND 59)	86
61	AMED	60 [DT FROM 2009]	25

Database: Scopus

Search Strategy:

List version:

1 amput*

2 disarticulat*

3 prosthe*

4 (artificial w/ (limb* or leg*))

5 LEA

6 #1 OR #2 OR #3 OR #4 OR #5

7 "lower limb*"

8 "lower extremit*"

9 leg*

10 ((above OR below OR through) w/3 knee*)

11 "trans-tibial"

12 transtibial

13 knee*

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14 hip*

15 thigh*

16 "trans-femoral"

17 transfemoral

18 "trans-pelvic"

19 transpelvic

20 #7 OR #8 OR #9 OR #10 OR #11 OR #12 OR #13 OR #14 OR #15 OR #16 OR #17 OR #18 OR #19

21 vascular*

22 dysvascular*

23 (peripheral w/3 ("arter* disease*" or "arter* disorder*" or arteriopath*))

24 "limb* threat*"

25 ((leg* or limb* or peripheral) w/3 isch?emi*)

26 PVD

27 PAOD

28 (peripheral w/3 ("vascul* disorder*" or "vascul* disease*"))

29 CLI

30 (ulcer* w/3 (arter* or vascular*))

31 ALI

32 ulcer*

33 diabet*

34 atherosclero*

35 claudic*

36 ((arter* or vascular) w/3 (occlus* or reocclus* or "re-occlus*" or steno* or restenos* or obstruct* or lesio* or block* or harden* or stiffen* or obliter* or calcificat* or restrict* or narrow* or block* or insufficien* or sclerosis))

37 (limb* w/3 (occlus* or reocclus* or "re-occlus*" or steno* or restenos* or obstruct* or lesio* or block* or harden* or stiffen* or obliter* or calcificat* or restrict* or narrow* or block* or insufficien* or sclerosis))

38 (leg* w/3 (occlus* or reocclus* or "re-occlus*" or steno* or restenos* or obstruct* or lesio* or block* or harden* or stiffen* or obliter* or calcificat* or restrict* or narrow* or block* or insufficien* or sclerosis))

39 (peripheral w/3 (occlus* or reocclus* or "re-occlus*" or steno* or restenos* or obstruct* or lesio* or block* or harden* or stiffen* or obliter* or calcificat* or restrict* or narrow* or block* or insufficien* or sclerosis))

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40 ("lower extremit*" w/3 (occlus* or reocclus* or "re-occlus*" or steno* or restenos* or obstruct* or lesio* or block* or harden* or stiffen* or obliter* or calcificat* or restrict* or narrow* or block* or insufficien* or sclerosis))

41 gangren*

42 #21 OR #22 OR #23 OR #24 OR #25 OR #26 OR #27 OR #28 OR #29 OR #30 OR #31 OR #32 OR #33 OR #34 OR #35 OR #36 OR #37 OR #38 OR #39 OR #40 OR #41

43 epidemiolog*

44 inciden*

45 prevalen*

46 frequenc*

47 (rate* w/3 (occur* or population*))

48 #43 OR #44 OR #45 OR #46 OR #47

49 #6 AND #20 AND #42 AND #49

50 #49 PUBYEAR > 2008 AND (LIMIT-TO (LANGUAGE, ”ENGLISH”))

String version:

((TITLE-ABS-KEY (amput*)) OR (TITLE-ABS-KEY (disarticulat*)) OR (TITLE-ABS-KEY (prosthe*)) OR (TITLE-ABS-KEY ((artificial W/ (limb* OR leg*)))) OR (TITLE-ABS-KEY (lea)))

AND

((TITLE-ABS-KEY ("lower limb*")) OR (TITLE-ABS-KEY ("lower extremit*")) OR (TITLE-ABS-KEY (leg*)) OR (TITLE-ABS-KEY (((above OR below OR through) W/3 knee*))) OR (TITLE-ABS-KEY ("trans-tibial")) OR (TITLE-ABS-KEY (transtibial)) OR (TITLE-ABS-KEY (knee*)) OR (TITLE-ABS-KEY (hip*)) OR (TITLE-ABS-KEY (thigh*)) OR (TITLE-ABS-KEY ("trans-femoral")) OR (TITLE-ABS-KEY (transfemoral)) OR (TITLE-ABS-KEY ("trans-pelvic")) OR (TITLE-ABS-KEY (transpelvic)))

AND

((TITLE-ABS-KEY (vascular*)) OR (TITLE-ABS-KEY (dysvascular*)) OR (TITLE-ABS-KEY ((peripheral W/3 ("arter* disease*" OR "arter* disorder*" OR arteriopath*)))) OR (TITLE-ABS-KEY ("limb* threat*")) OR (TITLE-ABS-KEY (((leg* OR limb* OR peripheral) W/3 isch?emi*))) OR (TITLE-ABS-KEY (pvd)) OR (TITLE-ABS-KEY (paod)) OR (TITLE-ABS-KEY ((peripheral W/3 ("vascul* disorder*" OR "vascul* disease*")))) OR (TITLE-ABS-KEY (cli)) OR (TITLE-ABS-KEY ((ulcer* W/3 (arter* OR vascular*)))) OR (TITLE-ABS-KEY (ali)) OR (TITLE-ABS-KEY (ulcer*)) OR (TITLE-ABS-KEY (diabet*)) OR (TITLE-ABS-KEY (atherosclero*)) OR (TITLE-ABS-KEY (claudic*)) OR (TITLE-ABS-KEY (((arter* OR vascular) W/3 (occlus* OR reocclus* OR "re-occlus*" OR steno* OR restenos* OR obstruct* OR lesio* OR block* OR harden* OR stiffen*

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OR obliter* OR calcificat* OR restrict* OR narrow* OR block* OR insufficien* OR sclerosis)))) OR (TITLE-ABS-KEY ((limb* W/3 (occlus* OR reocclus* OR "re-occlus*" OR steno* OR restenos* OR obstruct* OR lesio* OR block* OR harden* OR stiffen* OR obliter* OR calcificat* OR restrict* OR narrow* OR block* OR insufficien* OR sclerosis)))) OR (TITLE-ABS-KEY ((leg* W/3 (occlus* OR reocclus* OR "re-occlus*" OR steno* OR restenos* OR obstruct* OR lesio* OR block* OR harden* OR stiffen* OR obliter* OR calcificat* OR restrict* OR narrow* OR block* OR insufficien* OR sclerosis)))) OR (TITLE-ABS-KEY ((peripheral W/3 (occlus* OR reocclus* OR "re-occlus*" OR steno* OR restenos* OR obstruct* OR lesio* OR block* OR harden* OR stiffen* OR obliter* OR calcificat* OR restrict* OR narrow* OR block* OR insufficien* OR sclerosis)))) OR (TITLE-ABS-KEY (("lower extremit*" W/3 (occlus* OR reocclus* OR "re-occlus*" OR steno* OR restenos* OR obstruct* OR lesio* OR block* OR harden* OR stiffen* OR obliter* OR calcificat* OR restrict* OR narrow* OR block* OR insufficien* OR sclerosis)))) OR (TITLE-ABS-KEY (gangren*)))

AND

((TITLE-ABS-KEY (epidemiolog*)) OR (TITLE-ABS-KEY (inciden*)) OR (TITLE-ABS-KEY (prevalen*)) OR (TITLE-ABS-KEY (frequenc*)) OR (TITLE-ABS-KEY ((rate* W/3 (occur* OR population*)))))

PUBYEAR > 2008 AND (LIMIT-TO (LANGUAGE , "English"))

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Critical Appraisal

Table 1 Outcome of JBI critical appraisal for studies reporting prevalence data

Article	Question Answer								
	1	2	3	4	5	6	7	8	9
Ahmad, N. 2014a.	Yes	Yes	Yes	No	No	No	Yes	No	Unclear
Ahmad, N. 2014b.	Yes	Yes	Yes	No	Unclear	No	Yes	No	Unclear
Ahmad, N. 2016.	Yes	Yes	Yes	Yes	Unclear	No	Yes	No	Unclear
Ambler, GK. 2019.	Yes	Yes	Yes	No	Unclear	Unclear	Unclear	Unclear	Unclear
Behrendt, CA. 2018.	No	Yes	Yes	No	Unclear	Yes	Yes	No	Unclear
Holman, N. 2012	Yes	Yes	Yes	No	Unclear	Yes	Yes	Yes	Unclear
Kennon, B. 2012	Yes	Yes	Yes	No	Unclear	Yes	Yes	Yes	Unclear
Moxey, PW. 2010	Yes	Yes	Yes	No	Unclear	Yes	Yes	No	Unclear
Moxey, PW. 2012	Yes	Yes	Yes	Yes	Unclear	Yes	Yes	No	Unclear
NDA 2009-2010	Yes	Yes	Yes	No	Unclear	Unclear	Unclear	No	Unclear
NDA 2010-2011	Yes	Yes	Yes	No	Unclear	Unclear	Unclear	No	Unclear
NDA 2010-2012	Yes	Yes	Yes	No	Unclear	Unclear	Unclear	No	Unclear
NDA 2012-2013	Yes	Yes	Yes	No	Unclear	Unclear	Unclear	No	Unclear
NDA 2015-2016	Yes	Yes	Yes	No	Unclear	Unclear	Unclear	No	Unclear
NDA 2017-2018	Yes	Yes	Yes	No	Unclear	Unclear	Unclear	No	Unclear
NVR Report 2015	Yes	Yes	Yes	Yes	Unclear	Yes	Yes	No	No
NVR Report 2016	Yes	Yes	Yes	Yes	Unclear	Yes	Yes	No	No
NVR Report 2017	Yes	Yes	Yes	Yes	Unclear	Yes	Yes	No	No
NVR Report 2018	Yes	Yes	Yes	Yes	Unclear	Yes	Yes	No	No
NVR Report 2019	Yes	Yes	Yes	Yes	Unclear	Yes	Yes	No	No

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PHE Fingertips - 2019	Yes	Yes	Yes	No	Unclear	Yes	Yes	No	Unclear
Vamos, EP. 2010a	Yes	Yes	Yes	Yes	Unclear	Yes	Yes	No	Unclear
Vamos, EP. 2010b	Yes	Yes	Yes	Yes	Unclear	Yes	Yes	No	Unclear

Questions:

1. Was the sample frame appropriate to address the target population?
2. Were study participants sampled in an appropriate way?
3. Was the sample size adequate?
4. Were the study subjects and the setting described in detail?
5. Was the data analysis conducted with sufficient coverage of the identified sample?
6. Were valid methods used for the identification of the condition?
7. Was the condition measured in a standard, reliable way for all participants?
8. Was there appropriate statistical analysis?
9. Was the response rate adequate, and if not, was the low response rate managed appropriately?

Table 2 Outcome of JBI critical appraisal for cohort studies

	Question Answer										
Article	1	2	3	4	5	6	7	8	9	10	11
Nickinson, A. 2019	N/A	N/A	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Unclear	Yes

Questions:

1. Were the two groups similar and recruited from the same population?
2. Were the exposures measured similarly to assign people to both exposed and unexposed groups?
3. Was the exposure measured in a valid and reliable way?
4. Were confounding factors identified?
5. Were strategies to deal with confounding factors stated?
6. Were the groups/participants free of the outcome at the start of the study (or at the moment of exposure)?
7. Were the outcomes measured in a valid and reliable way?
8. Was the follow up time reported and sufficient to be long enough for outcomes to occur?
9. Was follow up complete, and if not, were the reasons to loss to follow up described and explored?
10. Were strategies to address incomplete follow up utilized?
11. Was appropriate statistical analysis used?

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MLEA codes

Table 1 Reported MLEA definition and MLEA database coding used by each included article

Article Author - Year	Ahmad 2014a	Ahmad 2014b	Ahmad 2016	Holman 2012	Moxey 2010	NDA Reports	Nickinson 2019	PHE Fingertips 2018	Vamos 2010a	Vamos 2010b	Behrendt 2018
Data source	HES	HES	HES	HES	HES	HES	HES*	HES	HES	HES	HES
MLEA definition	2	2	2	2	3	3(/5)	2	2	1	1	2
Code system	OPCS-4	OPCS-4	OPCS-4	OPCS-4	OPCS-4	OPCS-4	OPCS-4	OPCS-4	OPCS-4	OPCS-4	ICD-10-PCS
Codes	X09.1	X09.1	X09.1	X09.1			X09.1	X09.1			0Y62
	X09.2	X09.2	X09.2	X09.2			X09.2	X09.2	X09.2	X09.2	0Y63
	X09.3	X09.3	X09.3	X09.3	X09.3	X09.3	X09.3	X09.3	X09.3	X09.3	0Y64
	X09.4	X09.4	X09.4	X09.4	X09.4	X09.4	X09.4	X09.4	X09.4	X09.4	0Y67
	X09.5	X09.5	X09.5	X09.5	X09.5	X09.5	X09.5	X09.5	X09.5	X09.5	0Y68
	X09.8	X09.8	X09.8	X09.8	X09.8	X09.8	X09.8	X09.8	X09.9	X09.9	0Y6C
	X09.9	X09.9	X09.9	X09.9	X09.9	X09.9	X09.9	X09.9	X10.1	X10.1	0Y6D
									X10.2	X10.2	0Y6E
									X10.3	X10.3	0Y6F
									X10.4	X10.4	0Y6H
									X10.8	X10.8	0Y6J
									X10.9	X10.9	

*Denotes a study where CPRD was linked to HES

MLEA definition – the definition for MLEA used by each article:

1. Through or proximal to the ankle
2. Proximal to the ankle
3. BKA/TKA/AKA
4. BKA/AKA
5. 'Major Amputation' not further defined

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Diabetes definitions

Table 2 Case ascertainment method of people with diabetes for the reports that included those with diabetes only

Article author - Year	Population with diabetes case ascertainment method
Holman 2012	• Used ICD-10 codes in HES • Diagnostic codes E10-E14 • Code in any position of the discharge diagnosis fields
PHE Fingertips 2018	• Used ICD-10 codes in HES • Diagnostic codes E10-E14 • Only those with a code in the first or second position of the discharge diagnosis fields
NDA reports (all included)	• Those registered in the NDA database (in England and Wales by a GP on diagnosis)[1] • NDA does not cover all of those with diabetes in England and Wales as participation in the NDA is voluntary (NDA 2012-2013 reports coverage to be roughly 70%)
Kennon 2012	• Used Scottish National Diabetes Register Scottish Care Information – Diabetes Collaboration (SCI-DC)[2, 3] • Covers all people in Scotland with diabetes • Updated daily by GPs and hospital clinics

References

1. NHS Digital. National Diabetes Audit Programme. 2020. [updated 18/02/2020]. Available from: https://digital.nhs.uk/data-and-information/clinical-audits-and-registries/national-diabetes-audit].
2. NHS Scotland. Research Areas - Diabetes. 2020. [Available from: https://www.nhsresearchscotland.org.uk/research-areas/diabetes].
3. SCI-DC. SCI-Diabetes Features & Benefits. 2015. [Available from: https://www.sci-diabetes.scot.nhs.uk/features-benefits/].

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	Title page
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	Abstract page
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	4
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	4
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	4
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	4/5
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	4
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	Supplementary file 1
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	5
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	5
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	5
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	5/6
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	5/6
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	5/6

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	5/6
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	5/6
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	Figure 1
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	Table 1
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	Supplementary File 2
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	Table 1
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	Narrative
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	6
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	7-9 Figure 2-4
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	9
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	11
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	11
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	12

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

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Understanding variations in reported epidemiology of major lower extremity amputation in the UK: A systematic review

Journal:	*BMJ Open*
Manuscript ID	bmjopen-2021-053599.R1
Article Type:	Original research
Date Submitted by the Author:	04-Sep-2021
Complete List of Authors:	Meffen, Anna; University of Leicester, Health Sciences Houghton, John; University of Leicester Department of Cardiovascular Sciences Nickinson, Andrew; University of Leicester Department of Cardiovascular Sciences Pepper, Coral; University Hospitals of Leicester NHS Trust, Library and Information Services Sayers, Rob; University of Leicester Department of Cardiovascular Sciences Gray, Laura; University of Leicester Department of Health Sciences
Primary Subject Heading:	Epidemiology
Secondary Subject Heading:	Surgery
Keywords:	Vascular surgery < SURGERY, General diabetes < DIABETES & ENDOCRINOLOGY, EPIDEMIOLOGY

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Understanding variations in reported epidemiology of major lower extremity amputation in the UK: A systematic review

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Key words: Lower extremity amputation, systematic review, peripheral artery disease, diabetes, routinely collected health data

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Abstract

Objective

Estimate the prevalence/incidence/number of major lower extremity amputations (MLEA) in the UK, identify sources of routinely collected electronic health data used; assess time trends and regional variation and identify reasons for variation in reported incidence/prevalence of MLEA.

Design

Systematic review and narrative synthesis.

Data sources

Medline, EMBASE, EMcare, CINAHL, The Cochrane Library, AMED, Scopus and grey literature sources searched from 01/01/2009-12/08/2021.

Eligibility criteria for selecting studies

Reports which provided population based statistics, used routinely collected electronic health data, gave a measure of MLEA in adults in the general population or those with diabetes in the UK or constituent countries were included.

Data extraction and synthesis

Data extraction and quality assessment using the Joanna Briggs Institute Critical Appraisal Instruments were performed by two reviewers independently. Due to considerable differences in study populations and methodology, data pooling was not possible; data were tabulated, narratively synthesised and study differences discussed.

Results

Twenty-seven reports were included.

Incidence proportion for the general population ranged from 8.2 to 51.1 per 100,000 and from 70 to 291 per 100,000 for the population with diabetes. Evidence for trends over time was mixed, but there was no evidence of increasing incidence. Reports consistently found regional variation in England with incidence higher in the North. No studies reported prevalence.

Differences in database use, MLEA definition, calculation methods and multiple procedure inclusion which, together with identified inaccuracies, may account for the variation in incidence.

Conclusions

UK incidence and trends in MLEA remain unclear; estimates vary widely due to differences in methodology and inaccuracies. Reasons for regional variation also remain unexplained and prevalence un-investigated. International consensus on the definition of MLEA and medical code list is needed. Future research should recommend standards for the reporting of such outcomes and investigate further the potential to use primary care data in MLEA epidemiology.

Systematic review registration

PROSPERO CRD42020165592

Meffen A, Pepper CJ, Sayers RD, et al. Epidemiology of major lower limb amputation using routinely collected electronic health data in the UK: a systematic review protocol. BMJ Open 2020;10:e037053. doi: 10.1136/bmjopen-2020-037053

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Article Summary

Strengths and limitations of this study

- The study methods in the form of a protocol have been peer reviewed and published.
- Outcome measures investigated have been clearly defined and referenced.
- A comprehensive search strategy was employed in both peer-reviewed and grey literature searches.
- Included article methods and populations were not directly comparable and so it was not possible to pool data.
- This study only investigated data sources in the UK thus results may have limited use outside of this population.

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Introduction

Peripheral artery disease (PAD) is the leading cause of all major lower extremity amputations (MLEA) with diabetes, smoking, increasing age, hypertension increasing risk.[1, 2] Whilst diabetes is a risk factor of PAD, diabetes without PAD is also a cause of MLEA. With the global rise in diabetes prevalence and an aging population, incidence of MLEA has become a key indicator of health service performance and used for international comparisons.[3-6] Monitoring incidence globally, nationally and regionally is essential to determine the success of implemented prevention services.

In the UK the incidence and trends in incidence of MLEA are debated with variations reported.[7-9] Significant differences in regional estimates have also been reported.[8-10] The reasons for these disparities are unclear, although a recent review suggested differences in calculation methods may provide some explanation.[11]

Electronic health data are widely used for epidemiological research. Variation exists in the way individual databases collect, administer and report data and therefore differences between databases may also explain variation in reported MLEA statistics. There are many additional challenges to using such data, one notable, but improving issue is coding errors.[12-15] Including poor quality data with errors in incidence studies may affect the apparent trends in disease incidence.[15] Using different sets of MLEA codes and/or data may also account for some of the variation observed.

This systematic review aimed to ascertain the current UK incidence of MLEA; establish trends over time; report regional variation and review which routinely collected electronic health databases were used. The review also aimed to explain the reasons for variation in the reporting of MLEA epidemiology.

Methods

This review followed the Preferred Reporting Items for Systematic reviews and Meta-analyses (PRISMA) guidelines for systematic reviews.[16-18] A protocol was, published and registered in the PROSERO database (CRD42020165592).[19]

Search strategy

Searches were initially performed in December 2019 and re-run in August 2021.

Electronic databases Medline, EMBASE, EMcare, CINAHL, The Cochrane Library, AMED and Scopus were searched using keywords, thesaurus terms (indexing systems) and validated UK geographic filters where available.[20, 21] The search was developed in Medline and then adapted for other databases, Supplementary File 1.

Grey literature (non-peer-reviewed articles) was identified using www.opengrey.eu, openDoar.org, openAire.eu, base-search.net, eTHOS, OpenDisserations.org, www.gov.co.uk, www.parliament.uk, www.vvappg.com, www.digital.nhs.uk, www.QResearch.org and www.CPRD.com using the key word “amputation”/”amput*”.Inclusion criteria

Included reports presented: population based statistics; used routinely collected electronic health data; were written in English; and gave a measure (prevalence/incidence/number) of MLEA in the general population or in persons with diabetes in England, Scotland, Wales, Northern Ireland or the UK.

Studies published from 2009 onwards were included to ensure results were reflective of recent trends. This review aimed to focus on PAD and diabetes related MLEA as these may be preventable

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and only a small percentage of MLEA occur due to other aetiologies. [22] Analysis and any resulting care service improvements would therefore need to be based on vascular, diabetes and podiatric care. Studies of adults that included MLEA due to cancer and trauma were included in the study but this information is explicitly stated where available. Studies of children were excluded in order to exclude studies that did not focus on PAD and/or diabetes related amputations, however studies that included both adults and children were included.

No agreed definition of 'major' lower extremity amputation exists. Therefore, we did not define MLEA and all studies that reported MLEA were included with individual study definition of MLEA extracted.

For studies that reported for both minor and major lower extremity amputation, only data for major lower extremity amputation were included.

Screening and extraction

Identified reports were imported into EndNote X9 and duplicates removed. Initially, titles and abstracts were screened for inclusion by two reviewers independently (AM and JH). Of the selected reports, two reviewers independently screened the full texts with exclusion reasons recorded (AM plus JH or AN).

Data extraction was performed by two reviewers independently using a pre-tested tabulated form. Disagreements at any stage were resolved by discussion. Where data items were not reported, data were requested from authors.

Data extracted:

- Author, title, publication date
- Data source
- Epidemiological measure of MLEA - prevalence/incidence/number including any confidence interval (CI), standard error (SE) and variance presented
- Definition of MLEA
- Population studied (country, diabetic/non diabetic, age limits, date limits, comorbidities or reasons for amputation excluded e.g. cancer/trauma)
- Regional results
- Standardisation methods[23]
- MLEA code lists
- Reporting guidelines used

Quality assessment

The Joanna Briggs Institute (JBI) Critical Appraisal Instruments, either for Studies Reporting Prevalence Data or for Cohort Studies, were used to assess reporting quality.[24-26] Two reviewers (AM plus JH or AN) independently appraised included reports and resolved any disagreement. Additionally, use of appropriate reporting guidelines such as Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) and Reporting of Studies Conducted Using Observational Routinely-collected Data (RECORD) will be noteData synthesis

Data were converted, where possible, to provide consistent statistics across reports (e.g. percentages converted to per 100,000 population). Where multiple incidence values for an individual study were reported, the most recent figure was extracted.

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Extracted data were corrected where errors in terminology or calculations were identified using the following definitions (where populations refers to the population specified by the individual study).[27, 28]

$$\text{Incidence proportion} = \frac{\text{Number of new MLEA in specified time period}}{\text{Total population at risk at the start of time period}}$$

$$\text{Incidence rate} = \frac{\text{Number of new MLEA during specified time period}}{\text{Total population time at risk during specified time period}}$$

$$\text{Point prevalence} = \frac{\text{Number of new and pre existing MLEA at specified time point}}{\text{Population total at specified time point}}$$

$$\text{Period prevalence} = \frac{\text{Number of new and pre existing MLEA over specified time period}}{\text{Average or mid interval population for specified time period}}$$

Data were tabulated and synthesised narratively to investigate incidence/prevalence and time trends. Meta-analysis of incidence/prevalence and meta-regression of time trends using a random effects model were planned but not performed as extracted data were not suitable for pooling.[19]

Patient and public involvement

Patients or the public were not involved in the design, or conduct, or reporting, or dissemination plans of our research.

Ethics Approval

Ethics approval was not applicable for this study as it is a systematic review of published data.

Results

Search results

Titles and abstracts of 3722 peer-reviewed reports and 1728 grey literature reports were screened (Figure 1). Twenty-seven reports were included, 13 of which were identified through grey literature searches.[4, 7-10, 22, 29-49] Included grey literature consisted of reports from national healthcare quality improvement programmes, government healthcare department agencies, and government associated health care audits.[22, 32-36, 40, 41, 43, 46-49] Corresponding authors for nine reports were contacted for additional data, only one author responded.[44, 45]

Quality assessment

Overall reporting was poor to moderate with errors in key aspects such as statistic definition and calculations and poor reporting including the lack of descriptive statistics of their study population and denominator population descriptions. (Supplementary File 2). Six reports passed only 33% of the JBI prevalence study criteria whilst the remaining 19 passed between 33% and 67%. The two reports assessed by the JBI cohort study guidelines used more robust methodology and reporting and passed on 80-100% of the applicable criteria. Use of appropriate reporting guidelines such as STROBE and RECORD guidelines, where available, was unclear.[50, 51] Only one report stated the use of

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STROBE guidelines.[37] RECORD guidelines were published in 2015, none of the included reports published after this date stated the use of these guidelines.

Outcome measures and synthesis

A variety of MLEA statistics were reported by included reports (Table 1). Where available, standardised outcome measures were presented along with the CI where available, SE or variance were not provided by any included reports. Seventeen (63%) reports provided a calculated statistic,[4, 7-10, 29, 31-37, 41, 43-45] the remaining eight provided an absolute number.[22, 30, 38-40, 42, 46-49] Fourteen (52%) reported an incidence proportion.[4, 7, 10, 29, 32-38, 43-45] Two reported an incidence rate[8, 31]. The statistic presented for one report was unclear.[41] None presented prevalence.

Due to heterogeneity between study designs and reporting and the inability to accordingly recalculate outcomes homogeneously owing to a lack of available data (discussed further in the results and discussion sections), data pooling was not possible; results have been tabulated and narratively synthesised. An examination of the study method differences is explained in the 'Sources of outcome measure variation' section below.

Incidence proportion for the general population in England (not including reports for the population without diabetes) ranged from 8.2 to 51.1 per 100,000 population and from 70 to 291 per 100,000 for the population with diabetes in England.

Time trends

Time trends were investigated by six reports, (Figure 2).[4, 7, 37, 42, 44, 45] Differences in populations and methods mean that a direct comparison of trends could not be made.

One report investigated time trends in total number of amputations, they did not calculate a proportion or rate statistic. They reported a statistically significant 9.4% decrease in MLEA between 2000 and 2019 (5418 in 200 vs 4907 in 2019: 95% CI -49.6 to -12.5, p=0.003). However, as this does not take into account any changes in population and the methods used to account for multiple amputations per person were unclear, it was not possible to compare this trend with those of the other included reports.

For those that reported an incidence proportion, results were mixed, with some reports finding non-significant decreasing trends, however there was some evidence of a statistically significant decrease in MLEA in England over time for both the general population and for those with type 1 diabetes.[44, 45] One report found evidence of a statistically significant increase in MLEA for those with type 2 diabetes.[45] One possible explanation for the extreme differences in values seen in Figure 2 is the application of standardisation methods, reports that used these methods are labelled.

The report for Scotland reported a statistically significant (p<0.001) decrease in the incidence of MLEA for those with diabetes. [37]

For the studies that reported on a frequent basis, i.e. the National Diabetes Audit (NDA) and National Vascular Registry (NVR) reports, trends in incidence of MLEA over time could not be assessed as the NDA reports differ in frequency, study length and calculation methods and the NVR reports have overlapping study periods.

Regional trends

Four reports provide regional data.[8-10, 41] Public Health England (PHE) Fingertips provided regional estimates for those with diabetes by the 207 Clinical Commissioning Groups.[41] For the period of

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01/04/2017-31/03/2020 the regions with the highest and lowest incidence proportion per 100,000 population were Bradford City (In Northern England) (270/100,000) and Lewisham (In Greater London) (34/100,000). The report also gives values by the 42 Sustainability and Transformation Plan areas where incidence proportions per 100,000 population were highest in Cornwall and the Isles of Scilly (In South Western England) (117/100,000) and lowest in South East London (82/100,000), however, the time period these values cover is unclear. Holman et al studied MLEA incidence rates in those with and without diabetes by commissioning area (with different boundaries to those in the PHE fingertips report) but did not provide any extractable area level data.[8] They observed a 16-fold variation in incidence rate between areas for the population without diabetes and a 10-fold variation in incidence rate between areas for those with diabetes.

The two reports providing comparable data showed regional variation was statistically significant ($p<0.001$).[9, 10] Incidence was highest in the North and lowest in the South of England (Figure 3).

Sources of outcome measure variation

Database differences

Healthcare database use differed by country. All but one report ascertained cases of MLEA using only secondary care data with the remaining report using a combination of both primary and secondary care data.[31] Details of these databases are summarised in Table 2. Importantly, case ascertainment of MLEA in the NVR compared to HES is low in the NVR compared to HES (48-74% between 2014-2018).[52] Additionally, Clinical Practice Research Datalink (CPRD) primary care data does not cover the whole UK population.

Population differences

Study populations varied widely between reports with differences in country, exclusion criteria and MLEA definition and study period (Table 1).

Reports varied in how multiple procedures per person were dealt with. Nine reports made no reference to this.[4, 7, 9, 10, 29, 30, 38, 41, 42] Four used the highest level of amputation.[8, 37, 44, 45] Reports varied in how contralateral amputation during the same admission were counted. Two reports, the cohort studies, included the first procedure.[31, 39] The NDA reports implied each person was only included once no matter how many operations were performed although they do not specify which (first or highest) was included.[32-36, 43] The NVR 2015 report looked at unilateral amputations, no other information is given.[46] The NVR 2016-2019 reports excluded bilateral amputations and those associated with a bypass.[22, 47-49] The NVR 2020 makes the same exclusions as the NVR 2016-2019 reports but does, however, report the numbers of these excluded amputations.[40] Repeated amputations were not mentioned. None of the reports included revision specific procedures.

Study periods ranged from one to 19 years between the years 1996-2019 (Figure 4). No two reports with comparable populations covered the same period. Reports providing whole study estimates of incidence over a number of years generally reported higher incidence than those that reported for yearly incidence, as would be expected. This can be seen in the NDA reports in Table 1.

Coding differences

Seventeen reports provided code lists (Supplementary File 3, Table 1), all of which ascertained cases in HES or HES linked CPRD and used OPCS-4 codes apart from one which used ICD-10-PCS codes.[4, 7-10, 29, 31-36, 39, 41, 43-45] One report ascertained cases in CPRD in addition to HES using CPRD GOLD medcodes, these are listed under Table 1 in Supplementary File 3.[31] There were inconsistencies between the definition of MLEA used and the code lists and one report included an ICD-10-PCS code which does not exist whilst not including an expected code.[4]

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For those that reported for the population with diabetes only, differences were found in the case ascertainment methods of those with diabetes (Supplementary file 3, Table 2).

Outcome measure calculation differences

Statistics were inconsistently reported. Some reports used 'prevalence rate' or 'period prevalence' when actually stating an incidence proportion.[7, 10, 29, 32-36] Five repots used the terms 'incidence' and 'amputation rate' when reporting incidence proportion.[4, 9, 37, 44, 45] The term 'amputation rate' was also used by one other study, however, in this study an incidence rate was calculated.[31] One report did not clearly define the statistic used.[41]

Six reports used standardisation methods and only presented standardised results, these reports did not present enough information to be able to calculate crude (unstandardised) outcome measures.[7, 10, 29, 32, 41, 45] These reports varied in method and variables used to perform standardisation. Three reports standardised by age only;[7, 10, 29] two by age and sex[32, 45] and one by age and ethnicity.[41] Only one included report provided both standardised and crude outcome measures.[32] In this case, the effect of age and sex standardisation to reflect the national population was a greater than threefold increase in incidence proportion (crude: 87/100,000, standardised: 290/100,000).

Denominators were poorly reported with few reports providing the inclusion/exclusion criteria and calculation methods for incidence rate denominators were not described. One report stated the population country as the UK, however, the cases were ascertained using HES which only covers England and the denominator population was taken from Office for National Statistics population statistics for England and Wales.[4] Another report misreported incidence proportion as number of MLEAs per 100,000 population, the calculated figures were actually reported per 10,000 population.[9] As the population data were available to do so, correct recalculated statistics were presented in this review.

Discussion

Principle findings

Reported values of incidence proportion for the general population and population with diabetes varied with approximately a six and fourfold difference respectively.

Despite mixed reports of a decreasing trend in MLEA in England, it is encouraging that there is no evidence for an increasing trend in the general population considering the increase in prevalence of type 2 diabetes.[49] Some reports found evidence of an increase in the incidence of minor lower limb amputations which may partly explain this.[7] However, there is evidence of a statistically significant rise in the number of amputations in those with type 2 diabetes from one report.[45] This increasing trend was seen over the 1996 to 2005 period; with no more recent studies, it is unclear as to whether this trend is continuing.

Although two reports agree that incidence proportion is higher in the North of England compared with the South, this difference remains unexplained by demographic (including deprivation) or risk factors.[8-10] Additionally, another report found incidence to be highest in Cornwall and the Isles of Scilly in the South West, contrary to the trend of being highest in Northern areas.[41] It is unclear as to whether this report agrees with the others that, on the whole, incidence is higher in the North of England. All reports on regional trends used different geographic scales making comparisons difficult. Differences in geographic scale use may have an effect on epidemiology outcome measures.[53] One suggested explanation of regional variation in incidence is the difference in care

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provided, for example, areas where multidisciplinary preventative care services are available may have reduced regional incidence compared to other areas.[8, 54]

Two main sources of variation were found in reported MLEA incidence between reports. Firstly, there were considerable differences in the methodology used across reports including differences in country, study period, case and population (denominator) exclusions, MLEA definition (and coding), data source, standardisation, and statistic. For example, with Moxey et al 2010 and Ahmad et al 2014a, the slightly different study period and differences in age exclusions, standardisation methods, denominator populations and MLEA codes contributed to a twofold difference in reported incidence proportion. Secondly, there were reporting quality issues including inconsistent terminology and poorly reported denominators. It was not possible to further determine reasons for or the effect of the variation as reports did not include enough information to be able to clarify this. Adherence to reporting guidelines would have aided in clarifying variation sources and effects, however, adherence was low within included reports. Specifically, reports often failed to include; both crude (unstandardised and/or adjusted) and standardised whole study population outcome estimates; raw numerator and denominator values; both case and population (denominator) inclusion/exclusion criteria; referencing and detailing standardisation methods and data source. In addition to including details listed in RECORD guidelines, it would be of particular use in this case for reports to provide a breakdown of how many times an individual is included in the analysis to aid in interpretation and understanding of multiple procedure incidence.

Database use has an effect on MLEA epidemiology reporting. All studies ascertained cases using secondary care databases such as HES and the NVR which contain event based data and so only events within the study period can be analysed. Historical MLEA performed outside of the study period cannot be gained from these examples of event based databases. This explains the finding that prevalence of MLEA has not been investigated. It is also not possible to obtain population (/control) data for analysis from these databases, this would limit the ability to explain the regional variation found in some reports which is currently unexplained by the available population demographic (including deprivation) and risk factor data sourced externally to secondary care databases. Another option would be to use primary care data, which contains the whole medical history of each registered patient and so prevalence could potentially be calculated and more detailed control data obtained for use in analysis. However, current primary care databases in the UK do not contain data on the whole population and are therefore only a population sample.[55-57] One report did use primary care data from CPRD alongside HES to ascertain cases of MLEA, however, it was not clear how many cases were ascertained using primary care data. In addition to the loss of events due to population coverage of CPRD GOLD, this report excluded those with previous amputations and only counted one (the first) amputation per included individual further reducing the number of MLEA events counted. Another example of the effect database choice has on MLEA reporting is seen when considering the low case ascertainment of the NVR compared to HES. Although none of the reports using NVR calculated incidence proportion, using this database may underestimate MLEA incidence and could infer a potentially false increasing trends as case ascertainment improves. The lack of established linkages to the NVR and limited amount of patient data compared to HES may mean it could also be of limited use in further analysis.

Some reports ascertained cases in secondary care data and linked this to patient data in primary care records.[32-36, 39, 43] This obtains additional case patient information but does not aid in calculating prevalence or gaining population/control data.

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Comparison with other studies

A systematic review by Davies et al found many similar issues regarding denominator populations, MLEA coding and definition and general method differences.[11] They state that "prevalence of MLEA ranges from 0.7 to 332.4 per 100,000 in the diabetic population and 3.0 to 76.1 per 100,000 in the general population" a 480 and 25 fold difference respectively. This review differed to Davies et al in inclusion criteria by excluding reports that included 'minor' amputations; including only studies that use routinely collected electronic health data; excluding studies only reporting regional data and by excluding subgroup data from the main review outcome. These review method differences resulted in a considerable reduction in the magnitude of variation. However, the magnitude is still large and this review found additional sources of variation within the inclusion/exclusion criteria of reports; in multiple procedure counting; in standardisation techniques and data source. This review also included grey literature, investigated regional variation and discussed the effect of database use on MLEA epidemiology reporting.

As lower extremity amputation is a key health care quality indicator globally, national incidence of lower extremity amputation is an increasing topic of research in many countries.[53, 58-60] Systematic reviews and reports on the national and international incidence of lower extremity amputation have found similar problems with report comparisons as found in this review, i.e. population differences, lower extremity amputation definitions, database characteristics, and calculations methods.[4, 5, 53, 58-60]

Implications for clinical practice and further research

Being unable to accurately estimate the UK incidence of MLEA has important implications for clinicians and policy makers who will be unable to gauge trends and allocate resources appropriately. This will impact on national service configuration of vascular, diabetes and rehabilitation services. Additionally, unexplained regional differences in incidence mean that any resources allocated to specific areas may not be directly targeted at the cause. It also leads to inequalities in healthcare access and outcomes; inefficiency in resource allocation and uncertainty of the effect of preventative measures.

As MLEA definitions varied and were often not consistent with medical code lists, to aid in comparisons of future research, an internationally agreed definition of MLEA, alongside a corresponding medical coding list, should be agreed upon. Additionally, the appropriate reporting guidelines (STROBE, and more specifically, RECORD) should be strictly adhered to. Reports (particularly peer-reviewed) will, by nature, differ in research question, but by adhering to reporting guidelines, the data should be able to be unpicked so that sources of variation and their effect are more easily identified.

The use of primary care databases for ascertaining case and population data should be explored further to establish whether it is possible to estimate MLEA prevalence and explain the reasons for regional variation. Furthermore, current trends in MLEA in those with type 2 diabetes need to be investigated to ascertain whether the evidence of an increase trend found by one report continues.

Strengths and limitations

The main strength of this review is the extensive search strategy including a comprehensive grey literature search. Although not peer-reviewed, the included grey literature reports were found to be of no lesser quality, on the whole, than peer-review research and so were analysed alongside peer-review reports. Other strengths include, the clarification of epidemiological calculations and

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population definitions, the adherence to the PRISMA reporting guidelines, and the peer reviewed protocol.[19]

This systematic review was limited to reports on MLEA; so reports that did not distinguish between major and minor amputations were excluded. Only reports published since January 2009 were included. Outcome measures from these reports prior to 2009 were included in the narrative synthesis so may not be reflective of current trends. Some reports published prior to 2009 with data available from 1996 may be available but excluded due to publication date. Although UK grey literature sources were searched thoroughly, international sources (e.g. World Health Organisation (WHO) websites etc.) were not and thus there may be additional international reports fitting the criteria that were missed.

Moreover, this review was limited by the lack of reports with comparable methods, which led to the inability to statistically pool data. It was also limited by data availability. Notably, the majority of the data is from England, some reports included cases from Wales and Northern Ireland, however, none focused on these countries specifically. Additionally, the event based nature of the databases used by the included reports meant that determining and defining multiple amputations per patient and therefore also prevalence was problematic. This limitation is not restricted to the UK and has been a finding of reports globally.[4, 5, 53, 58-60]

Ranges in reported incidence were sensitive to the inclusion of particular reports, for example, removing the report that analysed only PAD related MLEA (excluding many who experienced MLEA related to diabetes who had no PAD associated diagnoses) would reduce this to a twofold difference in the general population.[4] Removing the report where the application of standardisation methods inflated incidence proportion by over three times the crude value would result in a reduction to a twofold fold difference in reported MLEA incidence proportion in the population with diabetes.[32]

Conclusions

The UK incidence and trends in MLEA remain unclear with estimates varying widely. There has been no research into the prevalence of MLEA. Encouragingly, there is no evidence of an increase in MLEA over time in the general population despite the national increase in diabetes type 2 diagnoses. There is however, dated evidence of an increase in MLEA in those with type 2 diabetes. Variation in reporting is due to differences in populations, methods and MLEA definitions, inaccurate calculations and terminology. There is evidence for regional variation in incidence, however, the reasons behind this are unexplained. This review highlights the need for improved MLEA epidemiology reporting with adherence to RECORD reporting guidelines. Importantly, an internationally agreed MLEA definition and code list is required. It also highlights the restrictive event based nature of secondary care data on analysis and the potential for further research of MLEA epidemiology using primary care data which could provide a prevalence estimate and the reasons behind the regional variation in MLEA in the UK.

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Acknowledgements

We would like to thank George Davies and the George Davies Charitable Trust for the charitable donation which help fund this work. This project was supported by the National Institute for Health Research (NIHR) Applied Research Collaboration East Midlands (ARC EM). The views expressed are those of the author(s) and not necessarily those of the NIHR or the Department of Health and Social Care.

Footnotes

Author Contributions: AM, RS, LG conceived and designed the study. CP, AM created and performed the search. AM, JH, AN screened, extracted data and undertook quality assessment. AM analysed the data, supervised by LG, RS. AM drafted and revised the manuscript which was reviewed by LG, all authors. Authors approved the final draft. The corresponding author attests that all listed authors meet authorship criteria and that no others meeting the criteria have been omitted.

Funding statement: AM, JH, AN are funded through the George Davies Charitable Trust (Registered Charity Number 1024818). RS is part funded through this Trust. The funders had no role in the design and conduct of the study, the collection, analysis, and interpretation of the data, or the preparation, review, or approval of the manuscript

Competing interests: All authors have completed the ICMJE uniform disclosure form at www.icmje.org/coi_disclosure.pdf and declare: RS reports grants from George Davies Charitable Trust during the conduct of the study and is Chair of the Vascular CRG for NHS England. All other authors declare no competing interests, or activities that could appear to have influenced the submitted work.

Data Sharing Statement: Statistical code, and dataset available from the corresponding author.

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Tables

Table 1: Table showing the method details and outcomes extracted from the included articles

Article	Country	**Case data source**	Exclusions	MLEA definition	Standardisation	**Statistic**	Study period	**Statistic value** (95% CI where available)
General population								
Ahmad, N. 2014a[29]	England	**HES**	Age<50 & >84	Proximal to the ankle	Age	**Incidence proportion**	2003 – 2008*	**26**
Ahmad, N. 2014b[10]	England	**HES**	Age<50 & >84	Proximal to the ankle	Age	**Incidence proportion**	2003 – 2008*	**26.3** (26, 26.6)
Ahmad, N. 2016[7]	England	**HES**	Age<50 & >84	Proximal to the ankle	Age	**Incidence proportion**	2003* 2004* 2005* 2006* 2007* 2008* 2009* 2010* 2011* 2012*	**27.7** **27.7** **26.6** **25.3** **26.0** **26.8** **25.3** **25.1** **23.7** **22.9**
Ambler, GK. 2020[30]	UK	**NVR**	None	BKA/TKA /AKA/ hip disarticulation/ hind quarter	N/A	**Number**	2014 – 2016	**9549**
Behrendt, CA. 2018[4]	England	**HES**	Trauma, cancer	Proximal to the ankle	None	**Incidence proportion** (PAD related amputations)	2010 2011 2012 2013 2014	**9.5** **9.1** **8.6** **8.7** **8.2**
Kennon, B. 2012[37]	Scotland	**SMR01**	Trauma, cancer	Through/ proximal to the ankle	None	**Incidence proportion** (Diabetes related amputations)	2004 2005 2006 2007 2008	**6.73** (6.06, 7.49) **5.06** (4.48, 5.72) **5.28** (4.68, 5.94) **5.05** (4.48, 5.71) **4.43** (3.89, 5.04)
Moxey, PW. 2010[9]	England	**HES**	None	BKA/TKA /AKA	None	**Incidence proportion**	2003 – 2007*	**51.1**
Moxey, PW. 2012[38]	England	**HES**	Age<50 & >84, trauma, cancer	BKA/TKA /AKA	N/A	**Number**	2002 – 2005*	**14168**
Nickinson, A. 2020[39]	England	**HES linked CPRD**	Age<17, trauma, cancer	Proximal to the ankle	N/A	**Number**	2000 – 2016	**3260**
NVR report 2015[46]	UK	**NVR**	Bilateral	BKA/AKA	N/A	**Number**	2014	**2465**
NVR report 2016[47]	UK	**NVR**	Trauma, bilateral MLEA	BKA/AKA	N/A	**Number**	2014 – 2015	**5318**
NVR report 2017[48]	UK	**NVR**	Trauma, bilateral MLEA	BKA/AKA	N/A	**Number**	2014 – 2016	**8866**

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NVR report 2018[49]	UK	**NVR**	Trauma, bilateral MLEA	BKA/AKA	N/A	**Number**	2015 – 2017	**9293**
NVR report 2019[22]	UK	**NVR**	Trauma, bilateral MLEA	BKA/AKA	N/A	**Number**	2016 – 2018	**9508**
NVR report 2020[40]	UK	**NVR**	Trauma, bilateral MLEA	BKA/TKA /AKA	N/A	**Number**	2017 – 2019	**10022**
Staniszewska, A. 2021[42]	England	**HES**	None	'Major Amputation' not further defined	N/A	**Number**	2000 2001 2002 2003 2004 2005 2006 2007 2008 2009 2010 2011 2012 2013 2014 2015 2016 2017 2018	**5418** **5334** **5208** **4864** **4881** **4804** **4542** **4829** **4973** **4994** **5004** **4638** **4608** **4526** **4473** **4573** **4735** **4742** **4907**
Vamos, EP. 2010a[44]	England	**HES**	Trauma	Through/ proximal to the ankle	None	**Incidence proportion** (people without diabetes†)	2004* 2005* 2006* 2007* 2008*	**7.7** **7.5** **6.8** **7.1** **6.9**
Vamos, EP. 2010b[45]	England	**HES**	Trauma	Through/ proximal to the ankle	Age, sex	**Incidence proportion** (people without diabetes†)	1996* 1997* 1998* 1999* 2000* 2001* 2002* 2003* 2004* 2005*	**7.0** **6.8** **6.8** **6.7** **6.2** **6.2** **5.7** **5.4** **5.1** **4.9**
Population with diabetes								
Gunn, LH, 2021[31]	England	**HES, CPRD**	Age <18, previous LEA, non-type 2 diabetes, censored prior to 01/04/2011	'Major amputation' defined by code list	None	**Incidence rate**	2010 – 2017*	**Type 2 diabetes 96**
Holman, N. 2012[8]	England	**HES**	Age <17	Proximal to the ankle	None	**Incidence rate**	2007 – 2009*	**99**
Kennon, B. 2012[37]	Scotland	**SMR01**	Trauma, cancer	Through/ proximal to the ankle	None	**Incidence proportion**	2004 2005 2006 2007 2008	**187** (168, 207) **134** (119, 151) **135** (120, 152) **126** (112, 142) **111** (99, 126)
NDA 2009-2010[43]	England	**HES**	None	'Major Amputation' not further defined	None	**Incidence proportion**	2009*	**70**

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NDA 2010-2011[32]	England & Wales	**HES, PEDW**	None	BKA/TKA/AKA	Age, sex	**Incidence proportion**	2010*	**291**
NDA 2011-2012[33]	England & Wales	**HES, PEDW**	None	BKA/TKA/AKA	None	**Incidence proportion**	2010 – 2011*	**170**
NDA 2012-2013[34]	England & Wales	**HES, PEDW**	None	BKA/TKA/AKA	None	**Incidence proportion**	2012*	**70**
NDA 2015-2016[36]	England & Wales	**HES, PEDW**	None	'Major Amputation' not further defined	None	**Incidence proportion**	2015*	**75**
NDA 2017-2018[35]	England & Wales	**HES, PEDW**	None	'Major Amputation' not further defined	None	**Incidence proportion**	2017*	**73**
PHE fingertips 2020[41]	England	**HES**	Age <17	Proximal to the ankle	Age, ethnicity	**Incidence proportion (/rate)‡**	2017 – 2019*	**82** (81, 84)
Vamos, EP. 2010a[44]	England	**HES**	Trauma	Through/ proximal to the ankle	None	**Incidence proportion**	2004* 2005* 2006* 2007* 2008*	**118** **106** **97** **100** **102**
Vamos, EP. 2010b[45]	England	**HES**	Trauma	Through/ proximal to the ankle	Age, sex	**Incidence proportion**		**Type 1 diabetes** / **Type 2 diabetes**
							1996*	**1.3** / **2.0**
							1997*	**1.4** / **2.1**
							1998*	**1.5** / **2.1**
							1999*	**1.4** / **2.3**
							2000*	**1.1** / **2.3**
							2001*	**1.2** / **2.4**
							2002*	**1.1** / **2.6**
							2003*	**0.9** / **2.6**
							2004*	**0.9** / **2.7**
							2005*	**0.7** / **2.7**

Statistic value is presented as incidence proportion per 100,000 population, incidence rate per 100,000 person years or total number where a calculated statistic was not reported. If both standardised and crude values were available, the standardised value has been presented. Where outcome measures were given for the population with diabetes by type, these are specified in the table. AKA = Above knee amputation, BKA = Below knee amputation, CI = Confidence interval, CPRD = Clinical Practice Research Datalink, HES = Hospital Episode Statistics, NDA = National Diabetes Audit, NVR = National Vascular Registry, LEA = Lower extremity amputation (minor & major), PEDW = Patient Episode Database for Wales, PHE = Public Health England, SMR01 = Scottish Morbidity Records, TKA = Through knee amputation.

* Study period conducted over the financial year (April-March) rather than the calendar year
† Incidence proportion of people without diabetes rather than the general population (which includes those with diabetes)
‡ Statistic definition unclear in report

Table 2: Characteristics of databases used to ascertain cases of MLEA by included reports

Database	Coverage	Linkage	Coding	Variables
Hospital Episode Statistics (HES)[61, 62] • England secondary care event based data	Whole England population	• Linkage using NHS number, postcode, date of birth, sex & postcode	• ICD-10 for diagnoses • OPCS-4 for procedures	• No information on prescriptions • No information on primary • care

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• Records work done so hospital can get paid		• Well established linkages: ○ Mortality data ○ Cancer registry ○ National longitudinal studies ○ Primary care data		• Over 400 variables including: ○ Episode (event) information ○ Diagnoses ○ Demographic characteristics
Patient Episode Database for Wales (PEDW)[63, 64] • Wales secondary care event based data • Records work done so hospital can get paid	Whole Wales population	• Linkage using NHS number, postcode, date of birth, sex & postcode • Well established linkages: ○ Mortality data ○ Cancer registry ○ National longitudinal studies ○ Primary care data	• ICD-10 for diagnoses • OPCS-4 for procedures	• No information on prescriptions • No information on primary care • Over 400 variables including: ○ Episode (event) information ○ Diagnoses ○ Demographic characteristics
Scottish Morbidity Record (SMR01)[65, 66] • Scotland secondary care event based data • Records work done so hospital can get paid	Whole Scotland population	• Scottish patient IDs include date of birth for ease of linkage • Well established linkages: ○ Mortality data ○ Cancer registry ○ National longitudinal studies Primary care data	• ICD-10 for diagnoses • OPCS-4 for procedures	• No information on prescriptions • No information on primary care • Similar amount of variables to HES and PEDW with extras geographical information including: ○ Census geographical information ○ Electoral ward and parliamentary constituency
National Vascular Registry (NVR)[52, 67] • UK data on 5 main vascular procedures (inc major and minor lower	UK data but procedure recording only mandatory for hospitals in England Contains 48% to 74% of MLEA cases over the years	• No established linkages	• No Internationally classified code use • Surgeon recorded procedure	• Limited tick box of pre-operative medications • No information on primary care • Detailed procedure information

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limb amputation)	2014-2018 compared to HES			• Comordbities recorded by limited tick boxes • Minimal patient characteristics
Clinical Practice Research Datalink (CPRD)[14, 55-57] • UK General Practice (GP) primary care data recorded by GPs for care purposes • Retrieved from practices, de-identified and transformed by CPRD • Two longitudinal databases, GOLD and Aurum from practices using the Vision and EMIS software respectively	• UK data ○ Practices and patients need to voluntarily opt in to CPRD data collection ○ Includes data on 60 million patients however only 16 million are currently registered and active (24% of UK population) ○ Coverage of the population varies by region, country and database ○ Aurum covers a greater proportion of the population in England than GOLD ○ GOLD covers a greater proportion of the wider UK population than Aurum	• Consent for linkages to other databases are optional on practice regisration to CPRD • Linkages performed by NHS digital and only available for patients and practices registered in England who have opted in to linkage • Four times more patients in the Aurum database have available linkage compared to GOLD • Well established linkages: ○ Mortality data ○ Cancer registry ○ HES data ○ Small area level data (deprivation and rural-urban classification)	• Uses the nationally classified Read codes which are converted to Aurum and GOLD specific medical codes • SNOMED CT, an internationally used coding system, has recently been introduced as a new coding system in GP practices, however, this system is not currently available in CPRD data	• Secondary care events are recorded on GP receipt of letter from secdonary care consultant, recording quality is variable • Wide range of primary care data: ○ Demographic characteristics ○ Diagnoses and symptoms ○ Prescriptions and drug exposure ○ Vaccination history ○ Labarorty tests ○ Referrals to hospital and specialist care

ICD-10 = International Classification of Diseases version 10, MLEA = Major lower extremity, NHS = National Health Service, amputation, OPCS-4 = Operation Procedure Codes version 4.

Figure and supplement captions

Figures

Figure 1: PRISMA flow diagram of the screening process.

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Figure 2: Reported time trends in incidence proportion of MLEA for (A) the general population and (B) the population with diabetes.

Figure 3: Reported regional incidence proportion of MLEA by (A) Moxey et al 2010 and (B) Ahmad et al 2014b.

Figure 4: Included article study periods by population type.

Supplementary files

Supplementary file 1: Database search strategy

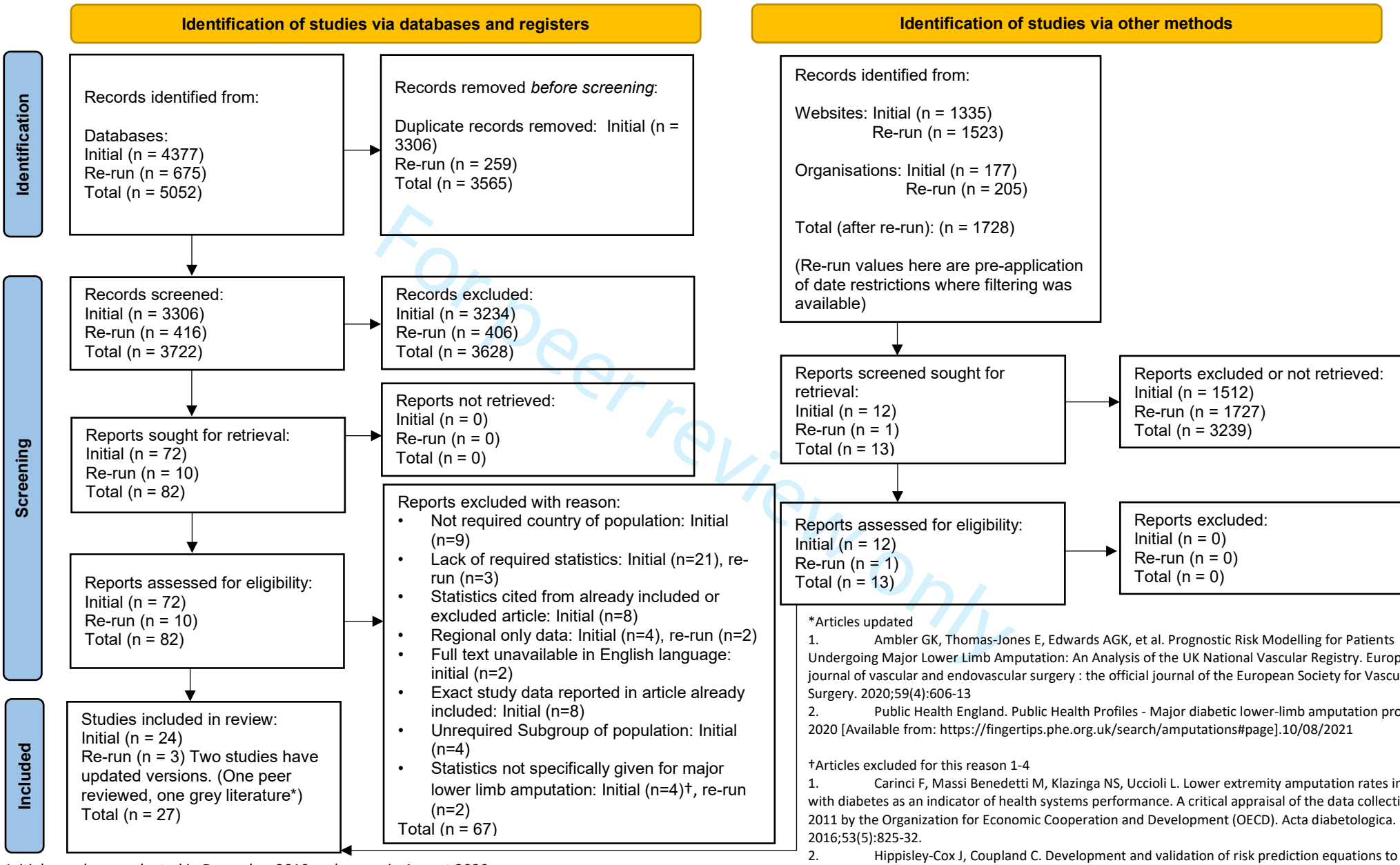
Supplementary file 2: Critical appraisal Table 1: Outcome of JBI critical appraisal for studies reporting prevalence data, Table 2: Outcome of JBI critical appraisal for cohort studies

Supplementary file 3: Table 1 - MLEA codes, Table 2 - Diabetes definitions

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PRISMA 2020 flow diagram for new systematic reviews which included searches of databases, registers and other sources



Initial searches conducted in December 2019 and re-run in August 2020.
Re-run n values are for additional reports unless otherwise specified.

*Articles updated

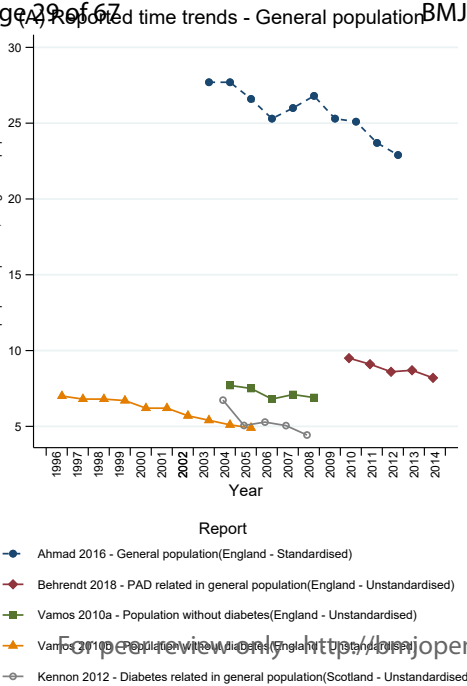
1. Ambler GK, Thomas-Jones E, Edwards AGK, et al. Prognostic Risk Modelling for Patients Undergoing Major Lower Limb Amputation: An Analysis of the UK National Vascular Registry. European journal of vascular and endovascular surgery : the official journal of the European Society for Vascular Surgery. 2020;59(4):606-13
2. Public Health England. Public Health Profiles - Major diabetic lower-limb amputation procedures 2020 [Available from: https://fingertips.phe.org.uk/search/amputations#page].10/08/2021

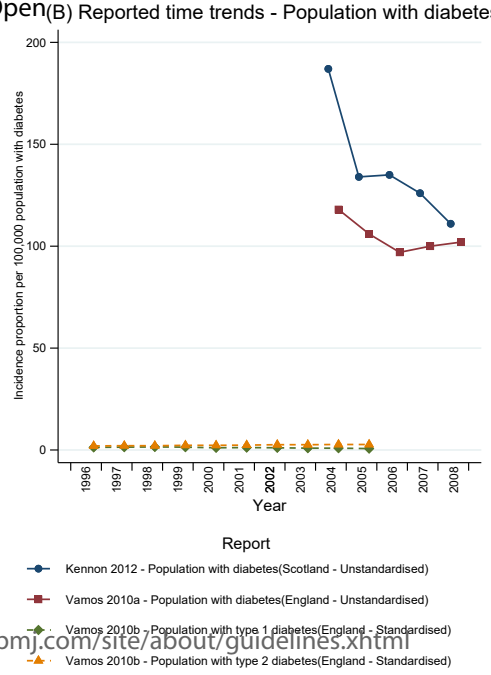
†Articles excluded for this reason 1-4

1. Carinci F, Massi Benedetti M, Klazinga NS, Uccioli L. Lower extremity amputation rates in people with diabetes as an indicator of health systems performance. A critical appraisal of the data collection 2000-2011 by the Organization for Economic Cooperation and Development (OECD). Acta diabetologica. 2016;53(5):825-32.
2. Hippisley-Cox J, Coupland C. Development and validation of risk prediction equations to estimate future risk of blindness and lower limb amputation in patients with diabetes. British Medical Journal. 2015;351(8033).
3. Hippisley-Cox J, Coupland C. Diabetes treatments and risk of amputation, blindness, severe kidney failure, hyperglycaemia, and hypoglycaemia: open cohort study in primary care. BMJ. 2016;352:i1450.
4. Hoffstad O, Mitra N, Walsh J, Margolis DJ. Diabetes, Lower-Extremity amputation, and death. Diabetes Care. 2015;38(10):1852-7.

From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71. For more information, visit: http://www.prisma-statement.org/

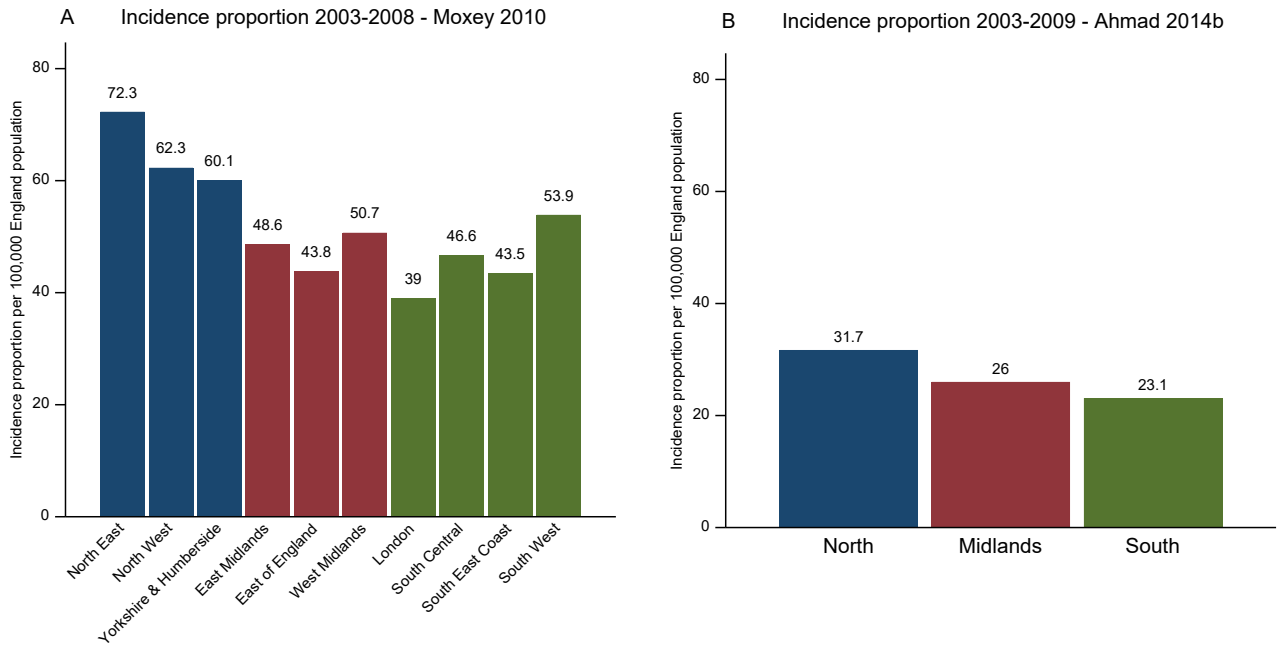
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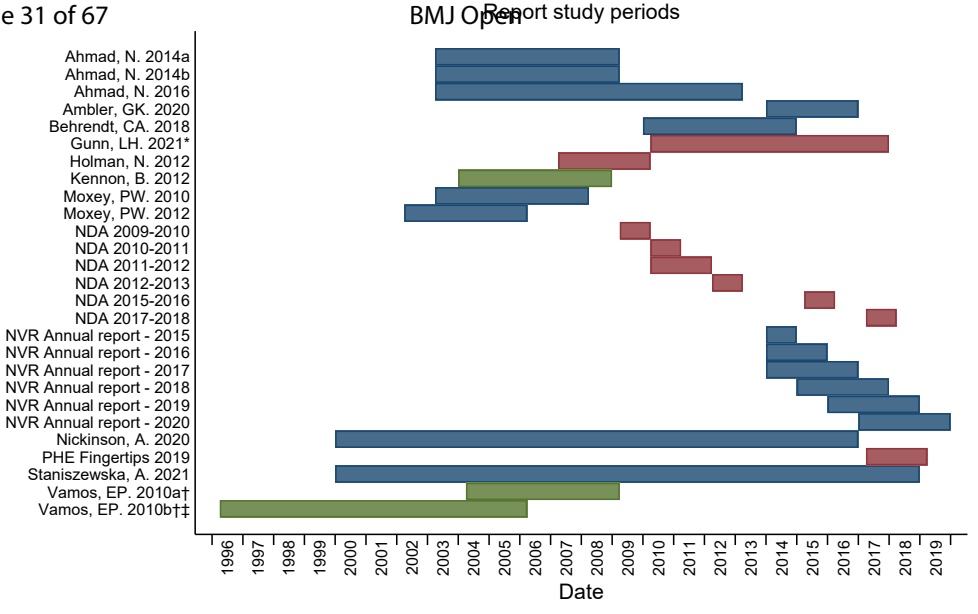


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Figure 3: Reported regional incidence proportion of MLEA by (A) Moxey et al 2010 and (B) Ahmad et al 2014b. Colours represent the North, Midlands and South of England in both plots.



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Population with type 2 diabetes

† General population excludes those with diabetes

‡ Populations with type 1 and type 2 diabetes reported separately

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Supplementary file 1 – Database search strategy

Database: Medline

Database: Ovid MEDLINE(R) and Epub Ahead of Print, In-Process & Other Non-Indexed Citations and Daily <1946 to December 12, 2019>

Search Strategy:

---

1 Amputees/ (3376)

2 exp Amputation/ (20930)

3 amput*.tw. (42609)

4 disarticulat*.tw. (1606)

5 prosthe*.tw. (118947)

6 (artificial adj (limb* or leg*)).tw. (536)

7 Artificial Limbs/ (6665)

8 LEA.tw. (2347)

9 1 or 2 or 3 or 4 or 5 or 6 or 7 or 8 (166117)

10 "lower limb*".tw. (47509)

11 "lower extremit*".tw. (52565)

12 leg*.tw. (293270)

13 ((above or below or through) adj3 knee*).tw. (7081)

14 "trans-tibial".tw. (358)

15 transtibial.tw. (1733)

16 knee*.tw. (141203)

17 hip*.tw. (325565)

18 thigh*.tw. (28640)

19 "trans-femoral".tw. (350)

20 transfemoral.tw. (5026)

21 "trans-pelvic".tw. (6)

22 transpelvic.tw. (82)

23 hip/su (2871)

24 knee/su (2650)

25 leg/su (5107)

26 thigh/su (1447)

27 lower extremity/su (1856)

28 10 or 11 or 12 or 13 or 14 or 15 or 16 or 17 or 18 or 19 or 20 or 21 or 22 or 23 or 24 or 25 or 26 or 27 (818087)

29 vascular*.tw. (589135)

30 dysvascular*.tw. (239)

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31 exp Vascular Surgical Procedures/ (246355)

32 exp Vascular Diseases/ (1626902)

33 (peripheral adj3 ("arter* disease*" or "arter* disorder*" or arteriopath*)).tw. (14144)

34 "limb* threat*".tw. (1439)

35 ((leg* or limb* or peripheral) adj3 isch?emi*).tw. (14235)

36 PVD.tw. (2376)

37 PAOD.tw. (758)

38 (peripheral adj3 ("vascul* disorder*" or "vascul* disease*")).tw. (10062)

39 CLI.tw. (2465)

40 (ulcer* adj3 (arter* or vascular*)).tw. (1442)

41 Ischemia/ (49164)

42 ALI.tw. (6610)

43 ulcer*.tw. (198843)

44 diabet*.tw. (603319)

45 atherosclero*.tw. (144792)

46 arteriosclero*.tw. (15481)

47 claudic*.tw. (10402)

48 ((arter* or vascular) adj3 (occlus* or reocclus* or "re-occlus*" or steno* or restenos* or obstruct* or lesio* or block* or harden* or stiffen* or obliter* or calcificat* or restrict* or narrow* or block* or insufficien* or sclerosis)).tw. (135311)

49 (limb* adj3 (occlus* or reocclus* or "re-occlus*" or steno* or restenos* or obstruct* or lesio* or block* or harden* or stiffen* or obliter* or calcificat* or restrict* or narrow* or block* or insufficien* or sclerosis)).tw. (3792)

50 (leg* adj3 (occlus* or reocclus* or "re-occlus*" or steno* or restenos* or obstruct* or lesio* or block* or harden* or stiffen* or obliter* or calcificat* or restrict* or narrow* or block* or insufficien* or sclerosis)).tw. (3356)

51 (peripheral adj3 (occlus* or reocclus* or "re-occlus*" or steno* or restenos* or obstruct* or lesio* or block* or harden* or stiffen* or obliter* or calcificat* or restrict* or narrow* or block* or insufficien* or sclerosis)).tw. (16324)

52 ("lower extremit*" adj3 (occlus* or reocclus* or "re-occlus*" or steno* or restenos* or obstruct* or lesio* or block* or harden* or stiffen* or obliter* or calcificat* or restrict* or narrow* or block* or insufficien* or sclerosis)).tw. (2057)

53 Gangrene/ (8076)

54 gangren*.tw. (16525)

55 29 or 30 or 31 or 32 or 33 or 34 or 35 or 36 or 37 or 38 or 39 or 40 or 41 or 42 or 43 or 44 or 45 or 46 or 47 or 48 or 49 or 50 or 51 or 52 or 53 or 54 (2865492)

56 Epidemiology/ (12267)

57 incidence/ (252140)

58 prevalence/ (279362)

59 exp Epidemiologic Methods/ (6001928)

60 exp epidemiologic studies/ (2404737)

61 Population Surveillance/ (57836)

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62 epidemiolog*.mp. (1828006)

63 inciden*.mp. (951016)

64 prevalen*.mp. (796766)

65 frequenc*.mp. (942710)

66 (rate* adj3 (occur* or population*)).mp. (34809)

67 56 or 57 or 58 or 59 or 60 or 61 or 62 or 63 or 64 or 65 or 66 (7703017)

68 exp Great Britain/ (358725)

69 (national health service* or nhs*).ti,ab,in. (182826)

70 (english not ((published or publication* or translat* or written or language* or speak* or literature or citation*) adj5 english)).ti,ab. (93210)

71 (gb or "g.b." or britain* or (british* not "british columbia") or uk or "u.k." or united kingdom* or (england* not "new england") or northern ireland* or northern irish* or scotland* or scottish* or ((wales or "south wales") not "new south wales") or welsh*).ti,ab,jw,in. (1990860)

72 (bath or "bath's" or ((birmingham not alabama*) or ("birmingham's" not alabama*) or bradford or "bradford's" or brighton or "brighton's" or bristol or "bristol's" or carlisle* or "carlisle's" or (cambridge not (massachusetts* or boston* or harvard*)) or ("cambridge's" not (massachusetts* or boston* or harvard*)) or (canterbury not zealand*) or ("canterbury's" not zealand*) or chelmsford or "chelmsford's" or chester or "chester's" or chichester or "chichester's" or coventry or "coventry's" or derby or "derby's" or (durham not (carolina* or nc)) or ("durham's" not (carolina* or nc)) or ely or "ely's" or exeter or "exeter's" or gloucester or "gloucester's" or hereford or "hereford's" or hull or "hull's" or lancaster or "lancaster's" or leeds* or leicester or "leicester's" or (lincoln not nebraska*) or ("lincoln's" not nebraska*) or (liverpool not (new south wales* or nsw)) or ("liverpool's" not (new south wales* or nsw)) or ((london not (ontario* or ont or toronto*)) or ("london's" not (ontario* or ont or toronto*)) or manchester or "manchester's" or (newcastle not (new south wales* or nsw)) or ("newcastle's" not (new south wales* or nsw)) or norwich or "norwich's" or nottingham or "nottingham's" or oxford or "oxford's" or peterborough or "peterborough's" or plymouth or "plymouth's" or portsmouth or "portsmouth's" or preston or "preston's" or ripon or "ripon's" or salford or "salford's" or salisbury or "salisbury's" or sheffield or "sheffield's" or southampton or "southampton's" or st albans or stoke or "stoke's" or sunderland or "sunderland's" or truro or "truro's" or wakefield or "wakefield's" or wells or westminster or "westminster's" or winchester or "winchester's" or wolverhampton or "wolverhampton's" or (worcester not (massachusetts* or boston* or harvard*)) or ("worcester's" not (massachusetts* or boston* or harvard*)) or (york not ("new york*" or ny or ontario* or ont or toronto*)) or ("york's" not ("new york*" or ny or ontario* or ont or toronto*))))).ti,ab,in. (1341900)

73 (bangor or "bangor's" or cardiff or "cardiff's" or newport or "newport's" or st asaph or "st asaph's" or st davids or swansea or "swansea's").ti,ab,in. (52438)

74 (aberdeen or "aberdeen's" or dundee or "dundee's" or edinburgh or "edinburgh's" or glasgow or "glasgow's" or inverness or (perth not australia*) or ("perth's" not australia*) or stirling or "stirling's").ti,ab,in. (199885)

75 (armagh or "armagh's" or belfast or "belfast's" or lisburn or "lisburn's" or londonderry or "londonderry's" or derry or "derry's" or newry or "newry's").ti,ab,in. (24719)

76 68 or 69 or 70 or 71 or 72 or 73 or 74 or 75 (2563344)

77 (exp africa/ or exp americas/ or exp antarctic regions/ or exp arctic regions/ or exp asia/ or exp oceania/) not (exp great britain/ or europe/) (2783662)

78 76 not 77 (2422301)

79 9 and 28 and 55 and 67 and 78 (704)

80 exp animals/ not humans.sh. (4651026)

81 79 not 80 (701)

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82 limit 81 to yr="2009-2020" (341)

****

- **Lines 1-9** are amputation terms
- **Lines 10-28** are lower extremity terms
- **Lines 29-55** are vascular terms (including diabetes)

The aim of combining these groups is to pick up only major lower limb amputation for vascular conditions.

Filters used:

- **Lines 68-74 is the UK filter** for Medline[22].
  This has been adapted for EMBASE[23].
  Validated UK filters for the other databases were not available.
- **Lines 76-77 is a human filter**, taken from the end of the Cochrane Handbook filter for RCTs[32].
- **Lines 56-67 is a study type filter[33]**.

Database: EMBASE

Search Strategy:

---

1 Amputee/ (933)

2 exp Amputation/ (44320)

3 amput*.tw. (51765)

4 disarticulat*.tw. (1845)

5 prosthe*.tw. (137059)

6 (artificial adj (limb* or leg*)).tw. (555)

7 exp "limb prosthesis"/ (6486)

8 LEA.tw. (2728)

9 "lower limb*".tw. (69278)

10 "lower extremit*".tw. (70925)

11 leg*.tw. (367108)

12 ((above or below or through) adj3 knee*).tw. (9584)

13 "trans-tibial".tw. (445)

14 transtibial.tw. (2073)

15 knee*.tw. (182897)

16 hip*.tw. (416631)

17 thigh*.tw. (39584)

18 "trans-femoral".tw. (840)

19 transfemoral.tw. (8600)

20 "trans-pelvic".tw. (13)

21 transpelvic.tw. (88)

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22 hip/su (1527)

23 exp leg/su (6845)

24 "lower limb"/ (15078)

25 vascular*.tw. (798977)

26 dysvascular*.tw. (270)

27 exp "Vascular Surgery"/ (442769)

28 exp "Vascular Disease"/ (2359700)

29 (peripheral adj3 ("arter* disease*" or "arter* disorder*" or arteriopath*)).tw. (22290)

30 "limb* threat*".tw. (1723)

31 ((leg* or limb* or peripheral) adj3 isch?emi*).tw. (20995)

32 PVD.tw. (4002)

33 PAOD.tw. (1230)

34 (peripheral adj3 ("vascul* disorder*" or "vascul* disease*")).tw. (14328)

35 CLI.tw. (4204)

36 (ulcer* adj3 (arter* or vascular*)).tw. (2062)

37 Ischemia/ (77155)

38 "Peripheral Ischemia"/ (934)

39 "Leg Ischemia"/ (6107)

40 ALI.tw. (11904)

41 ulcer*.tw. (254945)

42 diabet*.tw. (893260)

43 atherosclero*.tw. (199411)

44 arteriosclero*.tw. (15582)

45 claudic*.tw. (14191)

46 ((arter* or vascular) adj3 (occlus* or reocclus* or "re-occlus*" or steno* or restenos* or obstruct* or lesio* or block* or harden* or stiffen* or obliter* or calcificat* or restrict* or narrow* or block* or insufficien* or sclerosis)).tw. (184353)

47 (limb* adj3 (occlus* or reocclus* or "re-occlus*" or steno* or restenos* or obstruct* or lesio* or block* or harden* or stiffen* or obliter* or calcificat* or restrict* or narrow* or block* or insufficien* or sclerosis)).tw. (5473)

48 (leg* adj3 (occlus* or reocclus* or "re-occlus*" or steno* or restenos* or obstruct* or lesio* or block* or harden* or stiffen* or obliter* or calcificat* or restrict* or narrow* or block* or insufficien* or sclerosis)).tw. (4337)

49 (peripheral adj3 (occlus* or reocclus* or "re-occlus*" or steno* or restenos* or obstruct* or lesio* or block* or harden* or stiffen* or obliter* or calcificat* or restrict* or narrow* or block* or insufficien* or sclerosis)).tw. (22852)

50 ("lower extremit*" adj3 (occlus* or reocclus* or "re-occlus*" or steno* or restenos* or obstruct* or lesio* or block* or harden* or stiffen* or obliter* or calcificat* or restrict* or narrow* or block* or insufficien* or sclerosis)).tw. (2707)

51 Gangrene/ (9763)

52 gangren*.tw. (18326)

53 1 or 2 or 3 or 4 or 5 or 6 or 7 or 8 (197708)

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54 9 or 10 or 11 or 12 or 13 or 14 or 15 or 16 or 17 or 18 or 19 or 20 or 21 or 22 or 23 or 24 (1048226)

55 25 or 26 or 27 or 28 or 29 or 30 or 31 or 32 or 33 or 34 or 35 or 36 or 37 or 38 or 39 or 40 or 41 or 42 or 43 or 44 or 45 or 46 or 47 or 48 or 49 or 50 or 51 or 52 (3903577)

56 epidemiolog*.mp. (1504866)

57 inciden*.mp. (1328417)

58 prevalen*.mp. (1170777)

59 frequenc*.mp. (1238143)

60 (rate* adj3 (occur* or population*)).mp. (43835)

61 exp epidemiology/ (3268751)

62 epidemiological data/ (31787)

63 health survey/ (188176)

64 56 or 57 or 58 or 59 or 60 or 61 or 62 or 63 (5759910)

65 exp United Kingdom/ (408634)

66 (national health service* or nhs*).ti,ab,in,ad. (331006)

67 (english not ((published or publication* or translat* or written or language* or speak* or literature or citation*) adj5 english)).ti,ab. (40875)

68 (gb or "g.b." or britain* or (british* not "british columbia") or uk or "u.k." or united kingdom* or (england* not "new england") or northern ireland* or northern irish* or scotland* or scottish* or ((wales or "south wales") not "new south wales") or welsh*).ti,ab,jx,in,ad. (3065651)

69 (bath or "bath's" or ((birmingham not alabama*) or ("birmingham's" not alabama*) or bradford or "bradford's" or brighton or "brighton's" or bristol or "bristol's" or carlisle* or "carlisle's" or (cambridge not (massachusetts* or boston* or harvard*)) or ("cambridge's" not (massachusetts* or boston* or harvard*)) or (canterbury not zealand*) or ("canterbury's" not zealand*) or chelmsford or "chelmsford's" or chester or "chester's" or chichester or "chichester's" or coventry or "coventry's" or derby or "derby's" or (durham not (carolina* or nc)) or ("durham's" not (carolina* or nc)) or ely or "ely's" or exeter or "exeter's" or gloucester or "gloucester's" or hereford or "hereford's" or hull or "hull's" or lancaster or "lancaster's" or leeds* or leicester or "leicester's" or (lincoln not nebraska*) or ("lincoln's" not nebraska*) or (liverpool not (new south wales* or nsw)) or ("liverpool's" not (new south wales* or nsw)) or ((london not (ontario* or ont or toronto*)) or ("london's" not (ontario* or ont or toronto*)) or manchester or "manchester's" or (newcastle not (new south wales* or nsw)) or ("newcastle's" not (new south wales* or nsw)) or norwich or "norwich's" or nottingham or "nottingham's" or oxford or "oxford's" or peterborough or "peterborough's" or plymouth or "plymouth's" or portsmouth or "portsmouth's" or preston or "preston's" or ripon or "ripon's" or salford or "salford's" or salisbury or "salisbury's" or sheffield or "sheffield's" or southampton or "southampton's" or st albans or stoke or "stoke's" or sunderland or "sunderland's" or truro or "truro's" or wakefield or "wakefield's" or wells or westminster or "westminster's" or winchester or "winchester's" or wolverhampton or "wolverhampton's" or (worcester not (massachusetts* or boston* or harvard*)) or ("worcester's" not (massachusetts* or boston* or harvard*)) or (york not ("new york*" or ny or ontario* or ont or toronto*)) or ("york's" not ("new york*" or ny or ontario* or ont or toronto*))))).ti,ab,in,ad. (2353624)

70 (bangor or "bangor's" or cardiff or "cardiff's" or newport or "newport's" or st asaph or "st asaph's" or st davids or swansea or "swansea's").ti,ab,in,ad. (95988)

71 (aberdeen or "aberdeen's" or dundee or "dundee's" or edinburgh or "edinburgh's" or glasgow or "glasgow's" or inverness or (perth not australia*) or ("perth's" not australia*) or stirling or "stirling's").ti,ab,in,ad. (325298)

72 (armagh or "armagh's" or belfast or "belfast's" or lisburn or "lisburn's" or londonderry or "londonderry's" or derry or "derry's" or newry or "newry's").ti,ab,in,ad. (43503)

73 65 or 66 or 67 or 68 or 69 or 70 or 71 or 72 (3734952)

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74 (exp "arctic and antarctic"/ or exp oceanic regions/ or exp western hemisphere/ or exp africa/ or exp asia/ or exp "australia and new zealand"/) not (exp united kingdom/ or europe/) (2982458)

75 73 not 74 (3529253)

76 exp experimental organism/ (680593)

77 animal tissue/ (1318352)

78 animal cell/ (1463534)

79 exp animal disease/ (266885)

80 exp carnivore disease/ (46201)

81 exp bird/ (220421)

82 exp experimental animal welfare/ (3098)

83 exp animal husbandry/ (49846)

84 animal behavior/ (81143)

85 exp animal cell culture/ (12292)

86 exp mammalian disease/ (159889)

87 exp mammal/ (23800499)

88 exp marine species/ (6729)

89 nonhuman/ (5993421)

90 animal.hw. (5252369)

91 76 or 77 or 78 or 79 or 80 or 81 or 82 or 83 or 84 or 85 or 86 or 87 or 88 or 89 or 90 (26618565)

92 91 not human/ (6463716)

93 53 and 54 and 55 and 64 and 75 (900)

94 93 not 92 (896)

95 limit 94 to yr="2009 - 2020" (585)

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Database: CINAHL

Search Strategy:

#	Query	Limiters/Expanders	Last Run Via	Results
S1	TI amput* OR AB amput*	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	10,891
S2	TI disarticulat* OR AB disarticulat*	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	335

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S3	TI prosthe* OR AB prosthe*	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	19,654
S4	TI (artificial W1 (limb* OR leg*)) OR AB (artificial W1 (limb* OR leg*))	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	100
S5	(MH "Amputees")	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	2,691
S6	(MH "LIMB PROSTHESIS")	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	2,183
S7	(MH "Amputation+")	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	7,424
S8	TI LEA OR AB LEA	Expanders - Apply equivalent subjects	Interface - EBSCOhost Research Databases Search Screen - Advanced	386

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		Search modes - Boolean/Phrase	Search Database - CINAHL	
S9	S1 OR S2 OR S3 OR S4 OR S5 OR S6 OR S7 OR S8	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	31,971
S10	TI "lower limb*" OR AB "lower limb*"	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	12,988
S11	TI "lower extremit*" OR AB "lower extremit*"	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	15,723
S12	TI leg* OR AB leg*	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	96,668
S13	TI ((above OR below OR through) N3 knee*) OR AB ((above OR below OR through) N3 knee*)	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	1,899
S14	TI "trans-tibial" OR AB "trans-tibial"	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	107
S15	TI transtibial OR AB transtibial	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	1,189
S16	TI knee* OR AB knee*	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	56,084
S17	TI hip* OR AB hip*	Expanders - Apply equivalent subjects	Interface - EBSCOhost Research Databases	66,438

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		Search modes - Boolean/Phrase	Search Screen - Advanced Search Database - CINAHL	
S18	TI thigh* OR AB thigh*	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	5,640
S19	TI "trans-femoral" OR AB "trans-femoral"	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	86
S20	TI transfemoral OR AB transfemoral	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	1,408
S21	TI "trans-pelvic" OR AB "trans-pelvic"	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	0
S22	TI transpelvic OR AB transpelvic	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	9
S23	(MH "Hip/SU")	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	47
S24	(MH "Knee/SU")	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	0
S25	(MH "Leg/SU")	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	485
S26	(MH "Thigh/SU")	Expanders - Apply	Interface - EBSCOhost	133

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		equivalent subjects Search modes - Boolean/Phrase	Research Databases Search Screen - Advanced Search Database - CINAHL	
S27	(MH "Lower Extremity/SU")	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	822
S28	S10 OR S11 OR S12 OR S13 OR S14 OR S15 OR S16 OR S17 OR S18 OR S19 OR S20 OR S21 OR S22 OR S23 OR S24 OR S25 OR S26 OR S27	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	225,981
S29	TI vascular* OR AB vascular*	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	69,127
S30	TI dysvascular* OR AB dysvascular*	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	139
S31	TI (peripheral N3 ("arter* disease*" OR "arter* disorder*" OR arteriopath*)) OR AB (peripheral N3 ("arter* disease*" OR "arter* disorder*" OR arteriopath*))	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	3,943
S32	TI (limb* N1 threat*) OR AB (limb* N1 threat*)	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	371
S33	TI (((leg* OR limb* OR peripheral) N3 ischemi*) OR (leg* OR limb* OR peripheral) N3 ischaemi*)) OR AB (((leg* OR limb* OR peripheral) N3 ischemi*)	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	2,280

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	OR (leg* OR limb* OR peripheral) N3 ischaemi*))			
S34	TI PVD OR AB PVD	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	402
S35	TI PAOD OR AB PAOD	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	103
S36	TI (peripheral N3 ("vascul* disorder*" OR "vascul* disease*")) OR AB (peripheral N3 ("vascul* disorder*" OR "vascul* disease*"))	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	1,962
S37	TI CLI OR AB CLI	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	396
S38	TI (ulcer* N3 (arter* OR vascular*)) OR AB (ulcer* N3 (arter* OR vascular*))	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	511
S39	TI ALI OR AB ALI	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	1,512
S40	TI ulcer* OR AB ulcer*	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	32,901
S41	TI diabet* OR AB diabet*	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	173,481

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S42	TI atherosclero* OR AB atherosclero*	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	20,830
S43	TI arteriosclero* OR AB arteriosclero*	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	679
S44	TI claudic* OR AB claudic*	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	1,750
S45	TI ((arter* OR vascular) N3 (occlus* OR reocclus* OR "re-occlus*" OR steno* OR restenos* OR obstruct* OR lesio* OR block* OR harden* OR stiffen* OR obliter* OR calcificat* OR restrict* OR narrow* OR block* OR insufficien* OR sclerosis)) OR AB ((arter* OR vascular) N3 (occlus* OR reocclus* OR "re-occlus*" OR steno* OR restenos* OR obstruct* OR lesio* OR block* OR harden* OR stiffen* OR obliter* OR calcificat* OR restrict* OR narrow* OR block* OR insufficien* OR sclerosis))	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	20,456
S46	TI (limb* N3 (occlus* OR reocclus* OR "re-occlus*" OR steno* OR restenos* OR obstruct* OR lesio* OR block* OR harden* OR stiffen* OR obliter* OR calcificat* OR restrict* OR narrow* OR block* OR insufficien* OR sclerosis)) OR AB (limb*	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	762

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	N3 (occlus* OR reocclus* OR "re-occlus*" OR steno* OR restenos* OR obstruct* OR lesio* OR block* OR harden* OR stiffen* OR obliter* OR calcificat* OR restrict* OR narrow* OR block* OR insufficien* OR sclerosis))			
S47	TI (leg* N3 (occlus* OR reocclus* OR "re-occlus*" OR steno* OR restenos* OR obstruct* OR lesio* OR block* OR harden* OR stiffen* OR obliter* OR calcificat* OR restrict* OR narrow* OR block* OR insufficien* OR sclerosis)) OR AB (leg* N3 (occlus* OR reocclus* OR "re-occlus*" OR steno* OR restenos* OR obstruct* OR lesio* OR block* OR harden* OR stiffen* OR obliter* OR calcificat* OR restrict* OR narrow* OR block* OR insufficien* OR sclerosis))	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	994
S48	TI (peripheral N3 (occlus* OR reocclus* OR "re-occlus*" OR steno* OR restenos* OR obstruct* OR lesio* OR block* OR harden* OR stiffen* OR obliter* OR calcificat* OR restrict* OR narrow* OR block* OR insufficien* OR sclerosis)) OR AB (peripheral N3 (occlus* OR reocclus* OR "re-occlus*" OR steno* OR restenos* OR obstruct* OR lesio* OR block* OR harden* OR stiffen* OR obliter* OR calcificat* OR	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	2,700

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	restrict* OR narrow* OR block* OR insufficien* OR sclerosis))			
S49	TI ("lower extremit*" N3 (occlus* OR reocclus* OR "re-occlus*" OR steno* OR restenos* OR obstruct* OR lesio* OR block* OR harden* OR stiffen* OR obliter* OR calcificat* OR restrict* OR narrow* OR block* OR insufficien* OR sclerosis)) OR AB ("lower extremit*" N3 (occlus* OR reocclus* OR "reocclus*" OR steno* OR restenos* OR obstruct* OR lesio* OR block* OR harden* OR stiffen* OR obliter* OR calcificat* OR restrict* OR narrow* OR block* OR insufficien* OR sclerosis))	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	346
S50	TI gangren* OR AB gangren*	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	2,044
S51	(MH "VASCULAR SURGERY+")	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	26,829
S52	(MH "VASCULAR DISEASES+")	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	349,701
S53	(MH "ISCHEMIA")	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	6,485
S54	(MH "GANGRENE")	Expanders - Apply	Interface - EBSCOhost	701

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		equivalent subjects Search modes - Boolean/Phrase	Research Databases Search Screen - Advanced Search Database - CINAHL	
S55	S29 OR S30 OR S31 OR S32 OR S33 OR S34 OR S35 OR S36 OR S37 OR S38 OR S39 OR S40 OR S41 OR S42 OR S43 OR S44 OR S45 OR S46 OR S47 OR S48 OR S49 OR S50 OR S51 OR S52 OR S53 OR S54	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	575,685
S56	TX epidemiolog*	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	542,813
S57	TX inciden*	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	210,598
S58	TX prevalen*	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	220,384
S59	TX frequenc*	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	132,099
S60	TX (rate* N3 (occur* OR population*))	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	10,573
S61	(MH "EPIDEMIOLOGY+")	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	635,204
S62	(MH "INCIDENCE")	Expanders - Apply equivalent subjects	Interface - EBSCOhost Research Databases	62,878

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		Search modes - Boolean/Phrase	Search Screen - Advanced Search Database - CINAHL	
S63	(MH "PREVALENCE")	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	83,560
S64	(MH "EPIDEMIOLOGICAL RESEARCH+")	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	27,901
S65	S56 OR S57 OR S58 OR S59 OR S60 OR S61 OR S62 OR S63 OR S64	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	1,192,649
S66	S9 AND S28 AND S55 AND S65	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	1,031
S67	MH animals+	Limiters - Published Date: 20090101-20201231 Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	58,194
S68	MH (animal studies)	Limiters - Published Date: 20090101-20201231 Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	86,943
S69	TI (animal model*)	Limiters - Published Date: 20090101-20201231 Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	2,141
S70	S67 OR S68 OR S69	Expanders - Apply	Interface - EBSCOhost	141,713

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		equivalent subjects Search modes - Boolean/Phrase	Research Databases Search Screen - Advanced Search Database - CINAHL	
S71	S66 NOT S70	Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	1,025
S72	S66 NOT S70	Limiters - Published Date: 20090101-20201231 Expanders - Apply equivalent subjects Search modes - Boolean/Phrase	Interface - EBSCOhost Research Databases Search Screen - Advanced Search Database - CINAHL	690

Database: Cochrane Library

Search Strategy:

ID	Search	Hits
#1	amput*	3156
#2	disarticulat*	39
#3	prosthe*	14698
#4	artificial NEXT (limb* or leg*)	162
#5	LEA	415
#6	MeSH descriptor: [Amputees] this term only	107
#7	MeSH descriptor: [Amputation] explode all trees	422
#8	MeSH descriptor: [Artificial Limbs] this term only	141
#9	[OR #1-#8]	17711
#10	"lower limb" OR "lower limbs"	8801
#11	"lower extremity" OR "lower extremities"	7011
#12	leg*	35011
#13	((above or below or through) NEAR/3 knee*)	1160
#14	"trans-tibial"	49
#15	transtibial	263

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#16 knee* 28100

#17 hip* 30315

#18 thigh* 3998

#19 "trans-femoral" 48

#20 transfemoral 592

#21 "trans-pelvic" 0

#22 transpelvic 3

#23 MeSH descriptor: [Hip] this term only 403

#24 MeSH descriptor: [Knee] this term only 765

#25 MeSH descriptor: [Leg] this term only 2886

#26 MeSH descriptor: [Thigh] this term only 460

#27 MeSH descriptor: [Lower Extremity] explode all trees 6900

#28 [OR #10-#27] 93386

#29 vascular* 52059

#30 dysvascular* 74

#31 MeSH descriptor: [Vascular Surgical Procedures] explode all trees 14772

#32 MeSH descriptor: [Vascular Diseases] explode all trees 74765

#33 peripheral NEAR/3 ("artery disease" OR "arterial disease" OR "artery disorder" OR "arterial disorder" OR arteriopath*) 3523

#34 "limb threat" 2

#35 (leg* or limb* or peripheral) NEAR/3 isch?emi* 2318

#36 PVD 13028

#37 PAOD 227

#38 peripheral NEAR/3 ("vascular disorder" OR "vascular disorders" OR "vascular disease" OR "vascular diseases") 2377

#39 CLI 535

#40 ulcer* NEAR/3 (arter* OR vascular*) 292

#41 MeSH descriptor: [Ischemia] this term only 1713

#42 ALI 5055

#43 ulcer* 26852

#44 diabet* 92003

#45 atherosclero* 13572

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#46 arteriosclero* 2802

#47 claudic* 2542

#48 (arter* or vascular) NEAR/3 (occlus* or reocclus* or steno* or restenos* or obstruct* or lesio* or block* or harden* or stiffen* or obliter* or calcificat* or restrict* or narrow* or block* or insufficien* or sclerosis) 13751

#49 limb* NEAR/3 (occlus* or reocclus* or steno* or restenos* or obstruct* or lesio* or block* or harden* or stiffen* or obliter* or calcificat* or restrict* or narrow* or block* or insufficien* or sclerosis) 723

#50 leg* NEAR/3 (occlus* or reocclus* or steno* or restenos* or obstruct* or lesio* or block* or harden* or stiffen* or obliter* or calcificat* or restrict* or narrow* or block* or insufficien* or sclerosis) 551

#51 peripheral NEAR/3 (occlus* or reocclus* or steno* or restenos* or obstruct* or lesio* or block* or harden* or stiffen* or obliter* or calcificat* or restrict* or narrow* or block* or insufficien* or sclerosis) 3610

#52 ("lower extremity" OR "lower extremities") NEAR/3 (occlus* or reocclus* or steno* or restenos* or obstruct* or lesio* or block* or harden* or stiffen* or obliter* or calcificat* or restrict* or narrow* or block* or insufficien* or sclerosis) 268

#53 gangren* 769

#54 MeSH descriptor: [Gangrene] this term only 70

#55 [OR #29 - #54] 240320

#56 epidemiolog* 72616

#57 inciden* 122718

#58 prevalen* 47943

#59 frequenc* 80831

#60 rate* NEAR/3 (occur* or population*) 4023

#61 MeSH descriptor: [Epidemiology] explode all trees 47

#62 MeSH descriptor: [Incidence] this term only 9858

#63 MeSH descriptor: [Prevalence] this term only 4841

#64 MeSH descriptor: [Epidemiologic Methods] explode all trees 405014

#65 MeSH descriptor: [Epidemiologic Studies] explode all trees 151650

#66 [OR #56 - #65] 576460

#67 #9 AND #28 AND #55 AND #66 with Publication Year from 2009 to 2019, with Cochrane Library publication date Between Jan 2009 and Jan 2019, in Trials 361

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Database: AMED

Search Strategy:

#	Database	Search term	Results
1	AMED	AMPUTATION/	2197
2	AMED	"AMPUTATION STUMPS"/	123
3	AMED	exp PROSTHESIS/	1955
4	AMED	(amput*).ti,ab	2641
5	AMED	(disarticulat*).ti,ab	138
6	AMED	(prosthe*).ti,ab	2759
7	AMED	(artificial ADJ (limb* OR leg*)).ti,ab	43
8	AMED	(LEA).ti,ab	20
9	AMED	(1 OR 2 OR 3 OR 4 OR 5 OR 6 OR 7 OR 8)	5260
10	AMED	("lower limb*").ti,ab	3143
11	AMED	("lower extremit*").ti,ab	3441
12	AMED	(leg*).ti,ab	8679
13	AMED	((above OR below OR through) ADJ3 knee*).ti,ab	525
14	AMED	("trans-tibial").ti,ab	159
15	AMED	(transtibial).ti,ab	446
16	AMED	(knee*).ti,ab	10846
17	AMED	(hip*).ti,ab	6708
18	AMED	(thigh*).ti,ab	856
19	AMED	("trans-femoral").ti,ab	69
20	AMED	(transfemoral).ti,ab	270

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21	AMED	("trans-pelvic").ti,ab	0
22	AMED	(transpelvic).ti,ab	1
23	AMED	THIGH/	207
24	AMED	KNEE/	1599
25	AMED	HIP/	909
26	AMED	LEG/	5036
27	AMED	(10 OR 11 OR 12 OR 13 OR 14 OR 15 OR 16 OR 17 OR 18 OR 19 OR 20 OR 21 OR 22 OR 23 OR 24 OR 25 OR 26)	28326
28	AMED	(vascular*).ti,ab	2118
29	AMED	(dysvascular*).ti,ab	63
30	AMED	"VASCULAR SURGERY"/	20
31	AMED	exp "VASCULAR DISEASE"/	13067
32	AMED	(peripheral ADJ3 ("arter* disease*" OR "arter* disorder*" OR arteriopath*)).ti,ab	133
33	AMED	("limb* threat*").ti,ab	32
34	AMED	((leg* OR limb* OR peripheral) ADJ3 isch?emi*).ti,ab	72
35	AMED	(PVD).ti,ab	13
36	AMED	(PAOD).ti,ab	10
37	AMED	(peripheral ADJ3 ("vascul* disorder*" OR "vascul* disease*")).ti,ab	152
38	AMED	(CLI).ti,ab	1
39	AMED	(ulcer* ADJ3 (arter* OR vascular*)).ti,ab	14
40	AMED	(ALI).ti,ab	52
41	AMED	(ulcer*).ti,ab	2267

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42	AMED	(diabet*).ti,ab	5359
43	AMED	(atherosclero*).ti,ab	471
44	AMED	(arteriosclero*).ti,ab	68
45	AMED	(claudic*).ti,ab	160
46	AMED	((arter* OR vascular) ADJ3 (occlus* OR reocclus* OR "re-occlus*" OR steno* OR restenos* OR obstruct* OR lesio* OR block* OR harden* OR stiffen* OR obliter* OR calcificat* OR restrict* OR narrow* OR block* OR insufficien* OR sclerosis)).ti,ab	455
47	AMED	(limb* ADJ3 (occlus* OR reocclus* OR "re-occlus*" OR steno* OR restenos* OR obstruct* OR lesio* OR block* OR harden* OR stiffen* OR obliter* OR calcificat* OR restrict* OR narrow* OR block* OR insufficien* OR sclerosis)).ti,ab	81
48	AMED	(leg* ADJ3 (occlus* OR reocclus* OR "re-occlus*" OR steno* OR restenos* OR obstruct* OR lesio* OR block* OR harden* OR stiffen* OR obliter* OR calcificat* OR restrict* OR narrow* OR block* OR insufficien* OR sclerosis)).ti,ab	78
49	AMED	(peripheral ADJ3 (occlus* OR reocclus* OR "re-occlus*" OR steno* OR restenos* OR obstruct* OR lesio* OR block* OR harden* OR stiffen* OR obliter* OR calcificat* OR restrict* OR narrow* OR block* OR insufficien* OR sclerosis)).ti,ab	128
50	AMED	("lower extremit*" ADJ3 (occlus* OR reocclus* OR "re-occlus*" OR steno* OR restenos* OR obstruct* OR lesio* OR block* OR harden* OR stiffen* OR obliter* OR calcificat* OR restrict* OR narrow* OR block* OR insufficien* OR sclerosis)).ti,ab	45
51	AMED	(gangren*).ti,ab	92
52	AMED	(28 OR 29 OR 30 OR 31 OR 32 OR 33 OR 34 OR 35 OR 36 OR 37 OR 38 OR	21019

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		39 OR 40 OR 41 OR 42 OR 43 OR 44 OR 45 OR 46 OR 47 OR 48 OR 49 OR 50 OR 51)	
53	AMED	(epidemiolog*).af	6512
54	AMED	(inciden*).af	4727
55	AMED	(prevalen*).af	5486
56	AMED	(frequenc*).af	7273
57	AMED	(rate* ADJ3 (occur* OR population*)).af	364
58	AMED	EPIDEMIOLOGY/	4880
59	AMED	(53 OR 54 OR 55 OR 56 OR 57 OR 58)	21072
60	AMED	(9 AND 27 AND 52 AND 59)	86
61	AMED	60 [DT FROM 2009]	25

Database: Scopus

Search Strategy:

List version:

1 amput*

2 disarticulat*

3 prosthe*

4 (artificial w/ (limb* or leg*))

5 LEA

6 #1 OR #2 OR #3 OR #4 OR #5

7 "lower limb*"

8 "lower extremit*"

9 leg*

10 ((above OR below OR through) w/3 knee*)

11 "trans-tibial"

12 transtibial

13 knee*

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14 hip*

15 thigh*

16 "trans-femoral"

17 transfemoral

18 "trans-pelvic"

19 transpelvic

20 #7 OR #8 OR #9 OR #10 OR #11 OR #12 OR #13 OR #14 OR #15 OR #16 OR #17 OR #18 OR #19

21 vascular*

22 dysvascular*

23 (peripheral w/3 ("arter* disease*" or "arter* disorder*" or arteriopath*))

24 "limb* threat*"

25 ((leg* or limb* or peripheral) w/3 isch?emi*)

26 PVD

27 PAOD

28 (peripheral w/3 ("vascul* disorder*" or "vascul* disease*"))

29 CLI

30 (ulcer* w/3 (arter* or vascular*))

31 ALI

32 ulcer*

33 diabet*

34 atherosclero*

35 claudic*

36 ((arter* or vascular) w/3 (occlus* or reocclus* or "re-occlus*" or steno* or restenos* or obstruct* or lesio* or block* or harden* or stiffen* or obliter* or calcificat* or restrict* or narrow* or block* or insufficien* or sclerosis))

37 (limb* w/3 (occlus* or reocclus* or "re-occlus*" or steno* or restenos* or obstruct* or lesio* or block* or harden* or stiffen* or obliter* or calcificat* or restrict* or narrow* or block* or insufficien* or sclerosis))

38 (leg* w/3 (occlus* or reocclus* or "re-occlus*" or steno* or restenos* or obstruct* or lesio* or block* or harden* or stiffen* or obliter* or calcificat* or restrict* or narrow* or block* or insufficien* or sclerosis))

39 (peripheral w/3 (occlus* or reocclus* or "re-occlus*" or steno* or restenos* or obstruct* or lesio* or block* or harden* or stiffen* or obliter* or calcificat* or restrict* or narrow* or block* or insufficien* or sclerosis))

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40 ("lower extremit*" w/3 (occlus* or reocclus* or "re-occlus*" or steno* or restenos* or obstruct* or lesio* or block* or harden* or stiffen* or obliter* or calcificat* or restrict* or narrow* or block* or insufficien* or sclerosis))

41 gangren*

42 #21 OR #22 OR #23 OR #24 OR #25 OR #26 OR #27 OR #28 OR #29 OR #30 OR #31 OR #32 OR #33 OR #34 OR #35 OR #36 OR #37 OR #38 OR #39 OR #40 OR #41

43 epidemiolog*

44 inciden*

45 prevalen*

46 frequenc*

47 (rate* w/3 (occur* or population*))

48 #43 OR #44 OR #45 OR #46 OR #47

49 #6 AND #20 AND #42 AND #49

50 #49 PUBYEAR > 2008 AND (LIMIT-TO (LANGUAGE, ”ENGLISH”))

String version:

((TITLE-ABS-KEY (amput*)) OR (TITLE-ABS-KEY (disarticulat*)) OR (TITLE-ABS-KEY (prosthe*)) OR (TITLE-ABS-KEY ((artificial W/ (limb* OR leg*)))) OR (TITLE-ABS-KEY (lea)))

AND

((TITLE-ABS-KEY ("lower limb*")) OR (TITLE-ABS-KEY ("lower extremit*")) OR (TITLE-ABS-KEY (leg*)) OR (TITLE-ABS-KEY (((above OR below OR through) W/3 knee*))) OR (TITLE-ABS-KEY ("trans-tibial")) OR (TITLE-ABS-KEY (transtibial)) OR (TITLE-ABS-KEY (knee*)) OR (TITLE-ABS-KEY (hip*)) OR (TITLE-ABS-KEY (thigh*)) OR (TITLE-ABS-KEY ("trans-femoral")) OR (TITLE-ABS-KEY (transfemoral)) OR (TITLE-ABS-KEY ("trans-pelvic")) OR (TITLE-ABS-KEY (transpelvic)))

AND

((TITLE-ABS-KEY (vascular*)) OR (TITLE-ABS-KEY (dysvascular*)) OR (TITLE-ABS-KEY ((peripheral W/3 ("arter* disease*" OR "arter* disorder*" OR arteriopath*)))) OR (TITLE-ABS-KEY ("limb* threat*")) OR (TITLE-ABS-KEY (((leg* OR limb* OR peripheral) W/3 isch?emi*))) OR (TITLE-ABS-KEY (pvd)) OR (TITLE-ABS-KEY (paod)) OR (TITLE-ABS-KEY ((peripheral W/3 ("vascul* disorder*" OR "vascul* disease*")))) OR (TITLE-ABS-KEY (cli)) OR (TITLE-ABS-KEY ((ulcer* W/3 (arter* OR vascular*)))) OR (TITLE-ABS-KEY (ali)) OR (TITLE-ABS-KEY (ulcer*)) OR (TITLE-ABS-KEY (diabet*)) OR (TITLE-ABS-KEY (atherosclero*)) OR (TITLE-ABS-KEY (claudic*)) OR (TITLE-ABS-KEY (((arter* OR vascular) W/3 (occlus* OR reocclus* OR "re-occlus*" OR steno* OR restenos* OR obstruct* OR lesio* OR block* OR harden* OR stiffen*

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OR obliter* OR calcificat* OR restrict* OR narrow* OR block* OR insufficien* OR sclerosis)))) OR (TITLE-ABS-KEY ((limb* W/3 (occlus* OR reocclus* OR "re-occlus*" OR steno* OR restenos* OR obstruct* OR lesio* OR block* OR harden* OR stiffen* OR obliter* OR calcificat* OR restrict* OR narrow* OR block* OR insufficien* OR sclerosis)))) OR (TITLE-ABS-KEY ((leg* W/3 (occlus* OR reocclus* OR "re-occlus*" OR steno* OR restenos* OR obstruct* OR lesio* OR block* OR harden* OR stiffen* OR obliter* OR calcificat* OR restrict* OR narrow* OR block* OR insufficien* OR sclerosis)))) OR (TITLE-ABS-KEY ((peripheral W/3 (occlus* OR reocclus* OR "re-occlus*" OR steno* OR restenos* OR obstruct* OR lesio* OR block* OR harden* OR stiffen* OR obliter* OR calcificat* OR restrict* OR narrow* OR block* OR insufficien* OR sclerosis)))) OR (TITLE-ABS-KEY (("lower extremit*" W/3 (occlus* OR reocclus* OR "re-occlus*" OR steno* OR restenos* OR obstruct* OR lesio* OR block* OR harden* OR stiffen* OR obliter* OR calcificat* OR restrict* OR narrow* OR block* OR insufficien* OR sclerosis)))) OR (TITLE-ABS-KEY (gangren*)))

AND

((TITLE-ABS-KEY (epidemiolog*)) OR (TITLE-ABS-KEY (inciden*)) OR (TITLE-ABS-KEY (prevalen*)) OR (TITLE-ABS-KEY (frequenc*)) OR (TITLE-ABS-KEY ((rate* W/3 (occur* OR population*)))))

PUBYEAR > 2008 AND (LIMIT-TO (LANGUAGE , "English"))

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Critical Appraisal

Table 1 Outcome of JBI critical appraisal for studies reporting prevalence data

Article	Question Answer								
	1	2	3	4	5	6	7	8	9
Ahmad, N. 2014a.	Yes	Yes	Yes	No	No	No	Yes	No	Unclear
Ahmad, N. 2014b.	Yes	Yes	Yes	No	Unclear	No	Yes	No	Unclear
Ahmad, N. 2016.	Yes	Yes	Yes	Yes	Unclear	No	Yes	No	Unclear
Ambler, GK. 2020.	Yes	Yes	Yes	Yes	Unclear	Yes	Yes	No	Unclear
Behrendt, CA. 2018.	No	Yes	Yes	No	Unclear	Yes	Yes	No	Unclear
Holman, N. 2012	Yes	Yes	Yes	No	Unclear	Yes	Yes	Yes	Unclear
Kennon, B. 2012	Yes	Yes	Yes	No	Unclear	Yes	Yes	Yes	Unclear
Moxey, PW. 2010	Yes	Yes	Yes	No	Unclear	Yes	Yes	No	Unclear
Moxey, PW. 2012	Yes	Yes	Yes	Yes	Unclear	Yes	Yes	No	Unclear
NDA 2009-2010	Yes	Yes	Yes	No	Unclear	Unclear	Unclear	No	Unclear
NDA 2010-2011	Yes	Yes	Yes	No	Unclear	Unclear	Unclear	No	Unclear
NDA 2010-2012	Yes	Yes	Yes	No	Unclear	Unclear	Unclear	No	Unclear
NDA 2012-2013	Yes	Yes	Yes	No	Unclear	Unclear	Unclear	No	Unclear
NDA 2015-2016	Yes	Yes	Yes	No	Unclear	Unclear	Unclear	No	Unclear
NDA 2017-2018	Yes	Yes	Yes	No	Unclear	Unclear	Unclear	No	Unclear
NVR Report 2015	Yes	Yes	Yes	Yes	Unclear	Yes	Yes	No	No
NVR Report 2016	Yes	Yes	Yes	Yes	Unclear	Yes	Yes	No	No
NVR Report 2017	Yes	Yes	Yes	Yes	Unclear	Yes	Yes	No	No
NVR Report 2018	Yes	Yes	Yes	Yes	Unclear	Yes	Yes	No	No
NVR Report 2019	Yes	Yes	Yes	Yes	Unclear	Yes	Yes	No	No
NVR Report	Yes	Yes	Yes	Yes	Unclear	Yes	Yes	No	No

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2020									
PHE Fingertips 2020	Yes	Yes	Yes	No	Unclear	Yes	Yes	No	Unclear
Staniszewska, A. 2021	Yes	Yes	Yes	No	Unclear	Unclear	Yes	No	Unclear
Vamos, EP. 2010a	Yes	Yes	Yes	Yes	Unclear	Yes	Yes	No	Unclear
Vamos, EP. 2010b	Yes	Yes	Yes	Yes	Unclear	Yes	Yes	No	Unclear

Questions:

1. Was the sample frame appropriate to address the target population?
2. Were study participants sampled in an appropriate way?
3. Was the sample size adequate?
4. Were the study subjects and the setting described in detail?
5. Was the data analysis conducted with sufficient coverage of the identified sample?
6. Were valid methods used for the identification of the condition?
7. Was the condition measured in a standard, reliable way for all participants?
8. Was there appropriate statistical analysis?
9. Was the response rate adequate, and if not, was the low response rate managed appropriately?

Table 2 Outcome of JBI critical appraisal for cohort studies

	Question Answer										
Article	1	2	3	4	5	6	7	8	9	10	11
Gunn, LH. 2021	N/A	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Nickinson, A. 2019	N/A	N/A	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Unclear	Yes

Questions:

1. Were the two groups similar and recruited from the same population?
2. Were the exposures measured similarly to assign people to both exposed and unexposed groups?
3. Was the exposure measured in a valid and reliable way?
4. Were confounding factors identified?
5. Were strategies to deal with confounding factors stated?
6. Were the groups/participants free of the outcome at the start of the study (or at the moment of exposure)?
7. Were the outcomes measured in a valid and reliable way?
8. Was the follow up time reported and sufficient to be long enough for outcomes to occur?
9. Was follow up complete, and if not, were the reasons to loss to follow up described and explored?
10. Were strategies to address incomplete follow up utilized?
11. Was appropriate statistical analysis used?

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MLEA codes

Table 1 Reported MLEA definition and MLEA database coding used by each included article

Article Author - Year	Ahmad 2014a	Ahmad 2014b	Ahmad 2016	Gunn 2021	Holman 2012	Moxey 2010	NDA Reports	Nickinson 2019	PHE Fingertips 2020	Vamos 2010a	Vamos 2010b	Behrendt 2018
MLEA definition	2	2	2	6	2	3	3(/5)	2	2	1	1	2
Data source	HES	HES	HES	HES	HES	HES	HES	HES†	HES	HES	HES	HES
Code system	OPCS-4	OPCS-4	OPCS-4	OPCS-4*	OPCS-4	OPCS-4	OPCS-4	OPCS-4	OPCS-4	OPCS-4	OPCS-4	ICD-10-PCS
Codes	X09.1	X09.1	X09.1		X09.1			X09.1	X09.1			0Y62
	X09.2	X09.2	X09.2	X09.2	X09.2			X09.2	X09.2	X09.2	X09.2	0Y63
	X09.3	X09.3	X09.3	X09.3	X09.3	X09.3	X09.3	X09.3	X09.3	X09.3	X09.3	0Y64
	X09.4	X09.4	X09.4	X09.4	X09.4	X09.4	X09.4	X09.4	X09.4	X09.4	X09.4	0Y67
	X09.5	X09.5	X09.5	X09.5	X09.5	X09.5	X09.5	X09.5	X09.5	X09.5	X09.5	0Y68
	X09.8	X09.8	X09.8	X09.8	X09.8	X09.8	X09.8	X09.8	X09.8	X09.9	X09.9	0Y6C
	X09.9	X09.9	X09.9		X09.9	X09.9	X09.9	X09.9	X09.9	X10.1	X10.1	0Y6D
										X10.2	X10.2	0Y6E
										X10.3	X10.3	0Y6F
										X10.4	X10.4	0Y6H
										X10.8	X10.8	0Y6J
										X10.9	X10.9	

*CPRD GOLD medcodes also used: 6955, 15011, 32049, 69518, 12206, 25347, 12563, 21167, 11765, 11242, 3701, 27350, 47940, 50589, 52033, 111037, 18062, 3779, 100835, 71110, 2361, 97164, 62099, 18778, 39553, 100814, 104364, 50580, 3590, 97895, 54211, 108792, 51330, 15010

†Denotes a study where CPRD was linked to HES

MLEA definition – the definition for MLEA used by each article:

1. Through or proximal to the ankle
2. Proximal to the ankle
3. BKA/TKA/AKA
4. BKA/AKA
5. 'Major Amputation' not further defined
6. Defined by code list – no worded definition

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Diabetes definitions

Table 2 Case ascertainment method of people with diabetes for the reports that included those with diabetes only

Article author - Year	Population with diabetes case ascertainment method
Gunn 2021	• Used CPRD GOLD Medcodes • Diagnostic codes for type 2 diabetes prior to 01/04/2010 • Excluded other forms of diabetes by diagnostic medcode at any point • Excluded those with records for being prescribed insulin within 3 months of a diabetes diagnosis if aged ≥35 years, or within 6 months of a diagnosis made if aged <35 years, to exclude those with potentially inaccurate diagnoses • Excluded those with any level of lower limb amputation prior to 01/04/2011 • CPRD GOLD only covers registered practices using the Vision software, just less than half of these have available HES/ONS linkage[1]
Holman 2012	• Used ICD-10 codes in HES • Diagnostic codes E10-E14 • Code in any position of the discharge diagnosis fields
PHE Fingertips 2020	• Used ICD-10 codes in HES • Diagnostic codes E10-E14 • Only those with a code in the first or second position of the discharge diagnosis fields
NDA reports (all included)	• Those registered in the NDA database (in England and Wales by a GP on diagnosis)[2] • NDA does not cover all of those with diabetes in England and Wales as participation in the NDA is voluntary (NDA 2012-2013 reports coverage to be roughly 70%)
Kennon 2012	• Used Scottish National Diabetes Register Scottish Care Information – Diabetes Collaboration (SCI-DC)[3, 4] • Covers all people in Scotland with diabetes • Updated daily by GPs and hospital clinics

References

1. CPRD. Clinical practice research datalink 2020. [Available from: https://www.cprd.com.
2. NHS Digital. National Diabetes Audit Programme. 2020. [updated 18/02/2020]. Available from: https://digital.nhs.uk/data-and-information/clinical-audits-and-registries/national-diabetes-audit.
3. NHS Scotland. Research Areas - Diabetes. 2020. [Available from: https://www.nhsresearchscotland.org.uk/research-areas/diabetes.
4. SCI-DC. SCI-Diabetes Features & Benefits. 2015. [Available from: https://www.sci-diabetes.scot.nhs.uk/features-benefits/.

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PRISMA 2020 for Abstracts Checklist

Section and Topic	Item #	Checklist item	Reported (Yes/No)
TITLE			
Title	1	Identify the report as a systematic review.	YES
BACKGROUND			
Objectives	2	Provide an explicit statement of the main objective(s) or question(s) the review addresses.	YES
METHODS			
Eligibility criteria	3	Specify the inclusion and exclusion criteria for the review.	YES
Information sources	4	Specify the information sources (e.g. databases, registers) used to identify studies and the date when each was last searched.	YES
Risk of bias	5	Specify the methods used to assess risk of bias in the included studies.	YES
Synthesis of results	6	Specify the methods used to present and synthesise results.	YES
RESULTS			
Included studies	7	Give the total number of included studies and participants and summarise relevant characteristics of studies.	YES
Synthesis of results	8	Present results for main outcomes, preferably indicating the number of included studies and participants for each. If meta-analysis was done, report the summary estimate and confidence/credible interval. If comparing groups, indicate the direction of the effect (i.e. which group is favoured).	YES, where applicable
DISCUSSION			
Limitations of evidence	9	Provide a brief summary of the limitations of the evidence included in the review (e.g. study risk of bias, inconsistency and imprecision).	YES
Interpretation	10	Provide a general interpretation of the results and important implications.	YES
OTHER			
Funding	11	Specify the primary source of funding for the review.	N/A
Registration	12	Provide the register name and registration number.	YES

From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71

For more information, visit: http://www.prisma-statement.org/

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	Page 1
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	Page 2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	Page 5
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	Page 5
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	Page 5 - 6
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	Page 5
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	Supplementary File 1
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	Page 5 - 6
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	Page 5 - 6
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	Page 5 - 6
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	Page 5 - 6
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	Page 6 - 7
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	Page 6 - 7
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	Page 6 - 7
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	Page 6 - 7
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	Page 6 - 7
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	Page 6 - 7
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	Page 6 - 7
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	Page 6 - 7
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	Page 6 - 7
Certainty	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	N/A

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PRISMA 2020 Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
assessment			
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	Page 7
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	Figure 1
Study characteristics	17	Cite each included study and present its characteristics.	Table 1
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	Supplementary File 2
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	Table 1
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	Table 1
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	N/A
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	Pages 9 - 10
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	N/A
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	N/A
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	N/A
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	Pages 10 - 11
	23b	Discuss any limitations of the evidence included in the review.	Pages 12 - 13
	23c	Discuss any limitations of the review processes used.	Pages 12 - 13
	23d	Discuss implications of the results for practice, policy, and future research.	Page 12
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	Pages 1, 2 & 5
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	Pages 1, 2 & 5
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	Pages 7, 12, 13
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	Page 14
Competing interests	26	Declare any competing interests of review authors.	Page 14
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	Page 14

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PRISMA 2020 Checklist

From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71

For more information, visit: http://www.prisma-statement.org/

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Synthesis Without Meta-analysis (SWiM) reporting items

The citation for the Synthesis Without Meta-analysis explanation and elaboration article is: Campbell M, McKenzie JE, Sowden A, Katikireddi SV, Brennan SE, Ellis S, Hartmann-Boyce J, Ryan R, Shepperd S, Thomas J, Welch V, Thomson H. Synthesis without meta-analysis (SWiM) in systematic reviews: reporting guideline BMJ 2020;368:l6890 http://dx.doi.org/10.1136/bmj.l6890

SWiM is intended to complement and be used as an extension to PRISMA			
SWiM reporting item	**Item description**	**Page in manuscript where item is reported**	**Other***
Methods			
1 Grouping studies for synthesis	1a) Provide a description of, and rationale for, the groups used in the synthesis (e.g., groupings of populations, interventions, outcomes, study design)	N/A	
	1b) Detail and provide rationale for any changes made subsequent to the protocol in the groups used in the synthesis	N/A	
2 Describe the standardised metric and transformation methods used	Describe the standardised metric for each outcome. Explain why the metric(s) was chosen, and describe any methods used to transform the intervention effects, as reported in the study, to the standardised metric, citing any methodological guidance consulted	Pages 6 - 7	
3 Describe the synthesis methods	Describe and justify the methods used to synthesise the effects for each outcome when it was not possible to undertake a meta-analysis of effect estimates	Pages 7 - 10	
4 Criteria used to prioritise results for summary and synthesis	Where applicable, provide the criteria used, with supporting justification, to select the particular studies, or a particular study, for the main synthesis or to draw conclusions from the synthesis (e.g., based on study design, risk of bias assessments, directness in relation to the review question)	N/A	

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Synthesis Without Meta-analysis (SWiM) reporting items

SWiM reporting item	Item description	Page in manuscript where item is reported	Other*
5 Investigation of heterogeneity in reported effects	State the method(s) used to examine heterogeneity in reported effects when it was not possible to undertake a meta-analysis of effect estimates and its extensions to investigate heterogeneity	Pages 6, 9, 10	
6 Certainty of evidence	Describe the methods used to assess certainty of the synthesis findings	N/A	
7 Data presentation methods	Describe the graphical and tabular methods used to present the effects (e.g., tables, forest plots, harvest plots). Specify key study characteristics (e.g., study design, risk of bias) used to order the studies, in the text and any tables or graphs, clearly referencing the studies included	Pages 7, 8	
Results			
8 Reporting results	For each comparison and outcome, provide a description of the synthesised findings, and the certainty of the findings. Describe the result in language that is consistent with the question the synthesis addresses, and indicate which studies contribute to the synthesis	Pages 7 - 10, Table 1	
Discussion			
9 Limitations of the synthesis	Report the limitations of the synthesis methods used and/or the groupings used in the synthesis, and how these affect the conclusions that can be drawn in relation to the original review question	Pages 12 - 13	

PRISMA=Preferred Reporting Items for Systematic Reviews and Meta-Analyses.
*If the information is not provided in the systematic review, give details of where this information is available (e.g., protocol, other published papers (provide citation details), or website (provide the URL)).

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